# CFA® LEVEL I ESSENTIAL SUMMARIES BY ANALYSTPREP

*These essential summaries include most of the relevant concepts and formulas covered in the level I curriculum. They represent a good review right before exam time. However, we recommend reading either the Level I curriculum or AnalystPrep's study notes if you want more in-depth content.*

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# Quantitative Methods

## Reading 1 – The Time Value of Money

The Time Value of Money is an important concept for the level 1 exam. You will need to be comfortable discounting the **Present Value** and **Future Value** of cash flows for individual and ongoing payments in the exam. Your calculator has functions built in to make these calculations easier. Here are the calculations that you need to know:

$$FV = PV\left(1 + \frac{r}{n}\right)^{n \times t}$$
$$PV = \frac{FV}{\left(1 + \frac{r}{n}\right)^{n \times t}}$$

Where:

$FV = Future\ value$

$PV = Present\ value$

$r = Discount\ rate$

$n = Number\ or\ discounting\ periods\ per\ year$

$t = Number\ or\ years$

A simple example of this application is if you needed to determine how much you would pay today to receive \$1 in one year, assuming your required rate of return was 5%. The formula would look like this:

$$PV = \frac{\$1}{(1 + .05)} = \$0.95$$

Using this discount rate, \$1 one year from now is worth \$0.95 today.

You won't often have to do the formulas written out by hand but be familiar with them because the logic here applies to many other applications through this topic area. A similar issue that you'll have to deal with involves ongoing cash flows such as annuities and perpetuities. Both involve finding the current value of a series of cash flows, just that an annuity continues for a certain period of time and a perpetuity never ends. The formulas for these are as follows:

$$PV\ Annuity = \frac{(1 - V_n)}{r}$$

Where:

$r = Rate\ of\ return$

$n = Term\ of\ annuity$

$V_n = (1 + r)^{-n}$

$$PV\ Annuity\ Due = \frac{(1 - V_n)}{d}$$

Where:

$d = \frac{r}{(1+r)}$

$$PV\ Perpetuity = \frac{C}{r}$$

Where:

$C = \ Amount\ of\ each\ payment$

Your calculator has built-in functionality for PV and FV equations using the row of buttons labeled "N', "I/Y", "PV", "PMT", and "FV". To use these for a problem, you input the value of all the parameters that you know and then have the calculator compute the missing value. To input a value, you type in the number for the value and then press the corresponding value's button (i.e., to input 10,000 as the PV, you type in 10000, then press the "PV" button). Once you have input all figures you have, you press the CPT button and then the parameter button you want to calculate. Don't forget to input the interest amount in decimals, so that 5% is entered as 0.05.

# Reading 2 – Organizing, Visualizing, and Describing Data

Data refer to a collection of facts such as numbers, characters, measurements, observations, audios, and videos. Data can be classified as:

i. Numerical versus categorical data.

ii. Cross-sectional versus time-series versus panel data.

iii. Structured versus unstructured data.

**Numerical Data** represent values that can be measured or counted. Examples of numerical data include age, height, weight, or the number of shares in a portfolio.

**Categorical Data** (also called qualitative data) represent qualitative outcomes (i.e., quality or characteristic) of a group of observations. Examples of categorical data include investment style (i.e., growth vs. value stock), or investment-grade and junk bonds.

**Cross-Sectional Data** refer to a set of observations made at a point in time. A good example of cross-sectional data can be the stock returns earned by shareholders of Microsoft, IBM, and Samsung in the year ended, 31st December 2021.

**Time-Series Data** refer to a set of observations made over a given period of time at specific and equally spaced time intervals. An example of time-series data could be the daily or weekly closing price of a stock recorded over a period spanning 13 weeks.

**Panel Data** comprise a combination of time series data and cross-sectional data. An Example of panel data can be studying the GDP of three developing countries for a period spanning three years, from 2019 to 2021.

**Structured Data** are organized in any pre-defined manner (i.e., rows and columns), and there is a relationship between different rows and columns. Typical examples of structured company financial data include market data. For example, daily trading yields of a bond, daily closing stock prices, bond prices, and trading volumes.

**Unstructured Data** are not organized in any pre-defined manner. They can be textual, numbers or dates. Examples of unstructured data include financial news, posts on social media, company filings with the regulator, audio, or video recordings.

**Data Visualization** refers to the presentation of data in a pictorial or graphical format using different graphs. Examples include histograms, frequency polygons, bar charts, pareto charts, grouped bar charts, stacked bar charts, tree maps, word cloud, line charts, bubble line charts, scattered plots, scatter plot matrix and heat maps.

There are several methods of calculating the average value for a set of numbers as laid out in the curriculum. The formulas you need to know are:

$$Arithmetic\ Mean = \frac{\sum X_i}{N}$$

$$Weighted\ Mean = \sum_{i=1}^{N} X_i W_i$$

$$Geometric\ Mean = (X_1 \times X_2 \times \cdots \times X_n)^{\frac{1}{n}}$$

$$Harmonic\ Mean = \frac{N}{\sum_{i=1}^{N}\left(\frac{1}{X_i}\right)}$$

Where:

$N = Number\ of\ observations$

$W_i = Percentage\ weight$

***Examples***

The **Arithmetic Mean** of the numbers {1,2,3,4} is:

$$Arithmetic\ Mean = \frac{\sum X_i}{N} = \frac{1 + 2 + 3 + 4}{4} = 2.5$$

The **Weighted Mean** of a portfolio with the following assets is:

| | **Stocks** | **Bonds** |
|---|---|---|
| Portfolio Weight | 60% | 40% |
| Return | 10% | 6% |

$$Weighted\ Mean = \sum X_i W_i = (0.6 \times 0.1) + (0.4 \times 0.06) = 0.084$$

The **Harmonic Mean** of the numbers {1,2,3,4} is:

$$Harmonic\ Mean = \frac{N}{\sum_{i=1}^{N}\left(\frac{1}{X_i}\right)} = \frac{4}{\frac{1}{1}+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}} = 1.92$$

For the same set of numbers, it is always the case that:

Arithmetic Mean > Geometric Mean > Harmonic Mean

In addition to measures of average, there are several measurements of the distribution for datasets that you will need to know.

The **Variance** is the average of the squared deviations from the mean, and the formula to calculate this for a sample of data looks like this:

$$Variance = \sigma^2 = \frac{\sum_{i=1}^{N}(X_i - \bar{X})^2}{n-1}$$

Where:

$\bar{X} = Mean$

The **Standard Deviation** is simply the square root of the variance.

The **Mean Absolute Deviation** is a measure of the average of the absolute values of deviations from the mean in a data set. We must use the absolute values because a sum of the deviations from the mean in a data set is always 0. This formula is as follows:

$$MAD = \frac{\sum_{i=1}^{N} |X_i - \bar{X}|}{n-1}$$

Now we'll calculate these values for the following returns data:

| Year 1 | Year 2 | Year 3 | Mean |
|---|---|---|---|
| 10% | 7% | 5% | 7.3% |

$$Variance = \sigma^2 = \frac{(0.1 - 0.073)^2 + (0.07 - 0.073)^2 + (0.05 - 0.073)^2}{2}$$
$$= 0.00063$$

$$Standard\ Deviation = \sigma = \sqrt{0.00063} = 0.025$$

$$MAD = \frac{\sum_{i=1}^{N} |X_i - \bar{X}|}{n-1}$$
$$= \frac{|0.1 - 0.073| + |0.07 - 0.073| + |0.05 - 0.073|}{2} = 0.027$$

Another important formula related to the previous metrics is the **Sharpe Ratio**. It measures the risk-adjusted returns of a portfolio. The higher the Sharpe Ratio of a portfolio, the more excess return you are getting for each additional unit of risk. This formula shows up repeatedly in all levels of the CFA curriculum, so it's important to know it well.

$$Sharpe\ Ratio = \frac{(r_p - r_f)}{\sigma_p}$$

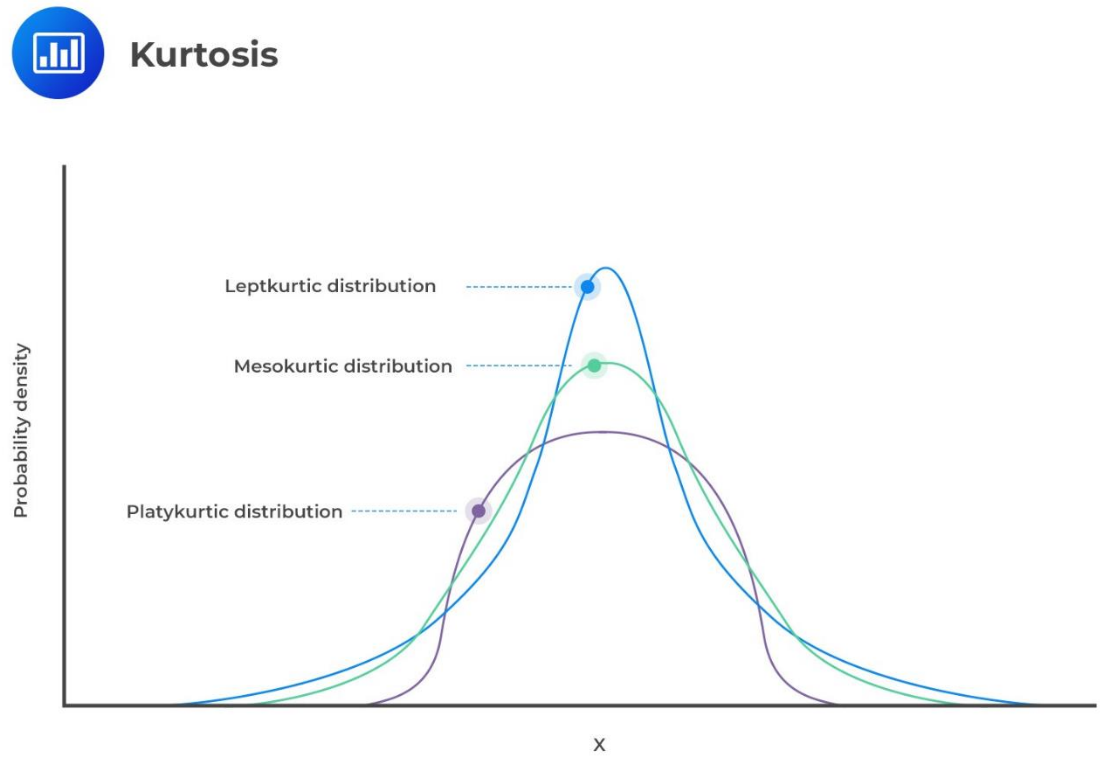
Where:

$r_p = Portfolio\ return$

$r_f = Risk\ free\ rate\ of\ return$

$\sigma_p = Standard\ deviation\ of\ portfolio\ returns$

When looking at data sets, there are different ways to categorize the distribution of the data. Two of these methods that you need to know are Kurtosis and Skewness. Kurtosis refers to the degree to which a data set is distributed, relative to the Normal Distribution. There are three categories of kurtosis, as illustrated in the graph below:




A data set would be described as **Mesokurtic** if it closely resembled a normal distribution, **Leptokurtic** if it was more tightly clustered around the mean, and **Platykurtic** if it is more widely distributed. The kurtosis of a normal distribution has a value of 3, so the excess Kurtosis is calculated as:

$$Excess\ Kurtosis = \left(\frac{\sum_{i=1}^{N}(X_i - X)^4}{s^4} \times \frac{1}{n}\right) - 3$$

Where:

$s = Standard\ deviation$

$n = Number\ of\ observations$

**Skewness** is another way to describe the distribution of a data set. It refers to how symmetrical a data's distribution is relative to a completely symmetrical normal distribution. A data set with negative skew has more values to the extreme left side of the distribution, and positive skew indicates the same occurring to the right side of the scale. Either of these conditions would be referred to as the distribution having "fat" tails.

$$Skewness = \frac{\sum_{i=1}^{N}(X_i - X)^3}{s^3} \times \frac{1}{n}$$

**Covariance** is a measure of how two variables move together. **Correlation** is a measure of the linear relationship between two variables.

**Limitations of Correlation Analysis**

- Two variables can have a very low correlation despite having a strong nonlinear relationship.
- Correlation can be an unreliable measure when outliers are present in the data.
- Correlation does not imply causation.

# Reading 3 – Probability Concepts

**Probability** is a measure of the likelihood that something will happen. Probabilities are expressed as percentages, from 0% (impossible to happen) to 100% (guaranteed).

A **random variable** is any quantity whose expected future value is not known in advance. An **outcome** is any possible value that a random variable can take. A single outcome or a set of outcomes is known as an **event**.

**Unconditional probability** (also known as marginal probability) is simply the probability that the occurrence of an event does not, in any way, depend on any other preceding events. **Conditional probability** is the probability of one event occurring with some relationship to one or more other events.

There are several concepts involving calculating probabilities you need to know. Most of these involve determining the probability of multiple events occurring, given the probability of each individual event.

The **Multiplication Rule** applies to the joint probability of multiple events occurring. The formula is as follows:

$$P(AB) = P(A|B)P(B)$$

The joint probability of both events happening (P(AB)) is the conditional probability of A, assuming B occurs (P(A|B)), times the probability of B (P(B)). Since the probability of both events happening is less than the odds of either one individually, the joint probability will always be lower than either input.

The **Addition Rule** applies to the probability of multiple events occurring, i.e., A and/or B occurs. That formula is:

$$P(A \text{ or } B) = P(A) + P(B) - P(AB)$$

Since both P(A) and P(B) include the possibility of both events occurring, we subtract the probability of both so that it's not double-counted.

The **Total Probability** rule is similar to the addition rule but deals with the combination of conditional probabilities to create the entire probability of an event occurring. The sum of all conditional probabilities for an event adds up to the non-conditional probability of that event if the conditions are mutually exclusive.

Two important concepts related to statistics in portfolio management are Correlation and Covariance. **Correlation** refers to the ratio of covariance between two variables and the product of their standard deviations. **Covariance** is the degree to which two variables move in sync with one another.

The formula to calculate the covariance of two assets (a and b) in a portfolio appears as follows.

$$\sigma_{R_a,R_b} = \sum P(R_a)[R_a - E(R_a)][R_b - E(R_b)]$$

It is important to understand the nature of covariance, but you should not have to calculate this formula by hand on the exam. You should expect, however, to be calculating correlations, which use the following formula:

$$Correlation(R_a, R_b) = \frac{Covariance(R_a, R_b)}{StdDev(R_a) \times StdDev(R_b)}$$

Be careful in exam questions involving these equations, because sometimes you will be given the portfolio risk expressed as the Standard Deviation or as the Variance. Don't forget that the Standard Deviation is the square root of the Variance.

The Correlation value ranges between -1 and 1.

- 1 means that the two variables move in the same direction.
- 0 means that they have no relationship.
- -1 means that they move exactly opposite.

**Bayes' Formula** is a method of calculating an updated probability of an event occurring, given a set of prior probabilities.

$$P(E_i|A) = \frac{P(E_i)P(A|E)_i}{\sum P\left(E_j\right)P\left(A\middle|E_j\right)}$$

Where:

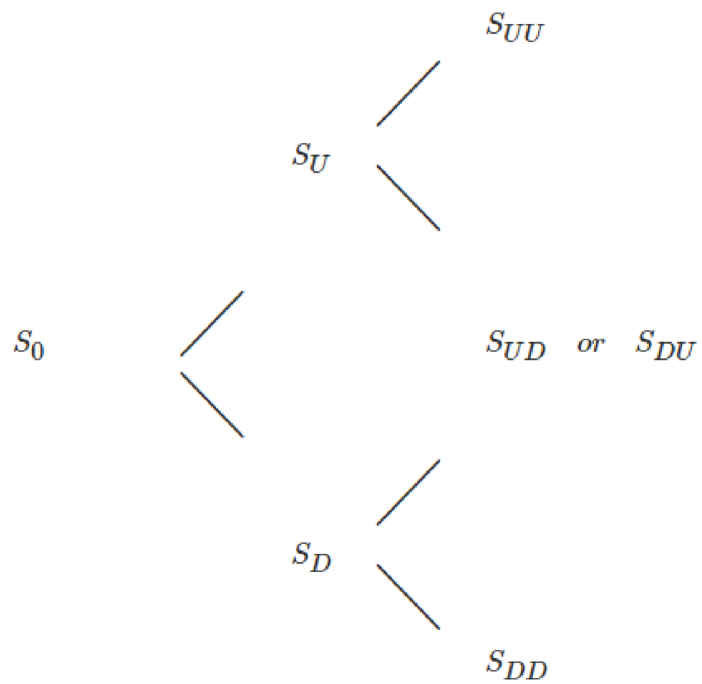
$$P\left(E_j\right) = Prior\ probabilities$$
$$A = An\ event\ known\ to\ have\ occured$$
$$P(E_i|A) = Posterior\ probability$$

# Reading 4 – Common Probability Distributions

There are several types of distributions you will need to understand, and this includes understanding a few common kinds of randomly distributed variables.

A **Discrete Uniform Random Variable** is one where the probabilities of all outcomes are equal, like the roll of a die. A **Bernoulli Random Variable** is one that can have only two outcomes. Similarly, a **Binomial Random Variable** is the number of specific outcomes in a Bernoulli situation that are independent of each other, such as the results of flipping a coin.

One way to calculate potential outcomes using a Bernoulli approach is known as a **Binomial Tree**:




In this case, each final outcome is seen as successive iterations of Bernoulli events. At each point in time, the stock goes up or down. The tree lists all possible outcomes for this stock, and the probability of each final outcome is the multiplied product of the probability at each step along the way. A good way to catch potential errors when calculating the values of a binomial tree is that each column (a specific point in time) should add up to a probability of 100%.

The curriculum specifies several methods of tracking and managing risk exposures using quantitative means. An important one is known as **Roy's Safety-First Ratio.** This is based on the principle that an optimal portfolio is one that minimizes the probability of a returning less than a given threshold level. Achieving a return less than a specified required return is known as **Shortfall Risk.** The Safety-First Ratio is expressed as the excess return of the portfolio (expected returns – threshold returns) divided by portfolio risk (standard deviation):

$$SF\ Ratio = \frac{E(R_p) - R_L}{\sigma_p}$$

Computing technology has introduced new methods of performance forecasting, including **Monte Carlo Simulation.** This method involves building a model of the portfolio and potential outcomes, then running a simulation of what could happen over time (factoring in all available probabilities and covariances of related variables) thousands of times. The output of this method is a distribution of the potential outcomes far more detailed than what a simple linear model could project. This is compared to the other common technique of **Historical Simulation**. In that approach, a projected portfolio is run through actual historical situations to see how it would have performed. This approach gives a much more limited set of outcomes than the Monte Carlo but has the benefit of being based on observed market events rather than projected simulations.

# Reading 5 – Sampling and Estimation

A core tenet of statistics is using samples of data to find information about the entire population of a data set. It's often impossible to gather all of the data of a population, so we rely on representative samples and try to make sure they properly represent the population from which they are drawn (like polls given to small groups of people to represent a whole state or country). One major assumption that underpins the validity of this approach is the **Central Limit Theorem**. This posits that a data sample will have a mean and variance that approach the mean and variance of the population it represents as the sample size becomes sufficiently large. It is widely accepted that a sample of size of at least 30 is usually enough to make the sample representative, but that can vary depending on the skewness of the sample distribution.

A common way (that will definitely appear multiple times on the exam) of analyzing the probability distribution of a data sample is to use the **Student's T-Distribution**. This distribution is appropriate for small samples when the population variance is unknown. The formula for calculating a t-statistic is:

$$t = \frac{x - \mu}{\frac{s}{\sqrt{n}}}$$

Where:

$x = Sample\ mean$

$\mu = Population\ mean$

$s = Sample\ standard\ deviation$

$n = Sample\ size$

The t-distribution has an important parameter known as the **Degrees of Freedom**. The degrees of freedom is calculated as n-1. The t-distribution is symmetrical about its mean and has thicker tails than the normal distribution. The shape of the distribution is dependent on the number of degrees of freedom. As the number of degrees of freedom increases, the distribution becomes more closely gathered around the mean.

A big part of the statistics part of the curriculum deals with the use of **Confidence Intervals**. A confidence interval is the range of values in which a statistician believes that a certain population parameter lies. They incorporate a trade-off of value range and confidence, such that the smaller the area in which the value is believed to exist, the lower the confidence that it is actually there. There are three primary scenarios for calculating the confidence intervals in the CFA curriculum.

1. A normal distribution with a known variance:

$$CI = X \pm z_{\frac{\alpha}{2}} \times \frac{\sigma}{\sqrt{n}}$$

2. A normal distribution with unknown variance:

$$CI = X \pm t_{\frac{\alpha}{2}} \times \frac{s}{\sqrt{n}}$$

3. A normal distribution with unknown variance and a sample size is large enough:

$$CI = X \pm t_{\frac{\alpha}{2}} \times \frac{s}{\sqrt{n}}$$

When the population variance is known, we can use the more accurate z-score to determine the confidence interval instead of the student's t-score, but the Central Limit Theorem allows us to also use the z-score when the sample size is large enough.

# Reading 6 – Hypothesis Testing

Hypothesis testing is a method of determining the statistical significance of a given data point in a distribution. A statistical test has to be performed to establish whether a hypothesis is correct or not.

The following procedure is performed whenever a statistical test is performed:

1. Statement of both the null and the alternative hypotheses.

2. Selection of the appropriate test statistic, i.e., what's being tested, e.g., the population mean, the difference between sample means, or variance.

3. Specification of the level of significance.
4. Clear statement of the decision rule to guide the choice of whether to reject or approve the null hypothesis.
5. Calculation of the sample statistic.
6. Arrival at a decision based on the sample results.

There are **One-Tailed Tests** that test the possibility of a change in one direction, and **Two-Tailed Tests** regarding changes in both positive and negative directions. This method uses calculated test statistics to determine whether a given hypothesis should be accepted or rejected.

The test statistic for the hypothesis testis is calculated as follows:

$$Test\ statistic = \frac{Sample\ statistic - Hypothesized\ value}{Standard\ error\ of\ the\ sample\ statistic}$$

There are two types of errors that could occur when running a hypothesis test. **Type 1** error occurs when we reject a true null hypothesis and **Type 2** error occurs when we fail to reject a false null hypothesis. The level of significance ($\alpha$) represents the probability of making a type 1 error.

The **Power of a Test** is the direct opposite of the level of significance. The power of a test gives us the probability of correctly discrediting and rejecting the null hypothesis when it is false i.e., it gives the likelihood of rejecting $H_0$. It is expressed as:

$$Power\ of\ a\ test\ =\ 1 - \ \beta\ =\ 1 - P(type\ II\ error)$$

**Statistical Significance vs. Economic Significance**

**Statistical Significance** refers to the use of a sample to carry out a statistical test meant to reveal any significant deviation from the stated null hypothesis. **Economic Significance** does not only entail the statistical significance but also the economic effect inherent in the decision made after data analysis and testing.

Another important component of hypothesis testing is the **p-Value**. If the calculated P-value is lower than the level of significance, then we can reject the null hypothesis.

For one-tailed tests, the $p$-value is given by the probability that lies below the calculated test statistic for left-tailed tests. In the case of right-tailed tests, the probability that lies above the test statistic gives the $p$-value. If the test is two-tailed, this value is given by the sum of the probabilities in the two tails.

The correct hypothesis test for a normal distribution of the random variable is the **Z-test**, which is calculated as:

$$z - statistic = \frac{X - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

Where:

$X = Sample\ mean$

$\sigma_0 = Hypothesized\ mean\ of\ the\ population$

$\mu = Standard\ deviation\ of\ the\ population$

$n = Sample\ size$

For hypothesis testing a population mean when the population variance is unknown and the sample size is large, a **T-test** is more appropriate:

$$t_{n-1} = \frac{X - \mu_0}{\frac{s}{\sqrt{n}}}$$

Where:

$X = Sample\ mean$

$\sigma_0 = Hypothesized\ mean\ of\ the\ population$

$s = Standard\ deviation\ of\ the\ sample$

$n = Sample\ size$

Your testing materials will include tables of significant values for t- and z-scores, so that you can compare to them in determining whether a calculated value exceeds the level necessary to reject a null hypothesis.

For hypothesis testing to show whether the population correlation coefficient equals zero (absence of linear relationship between two variables), we use the t-statistic:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

Where:

$r = Sample\ correlation$

$n = Sample\ size$

The decision rule is to reject the null hypothesis (ρ=0)/to uphold the alternative hypothesis (ρ≠0) if $t$ is greater than the critical value from the t-distribution table. Otherwise, we fail to reject the null hypothesis.

A **Chi-square Test** is used to establish whether a hypothesized value of variance is equal to, less than, or greater than the true population variance. The chi-square distribution is asymmetrical. However, the distribution approaches the "normal" one as the degrees of freedom increase. As a natural consequence, the chi-square distribution has no negative values and is bounded by zero. A chi-square statistic with (n – 1) degrees of freedom is computed as:

$$X_{n-1}^2 = \frac{(n-1)s^2}{\delta_0^2}$$

Where:

$n\ =\ Sample\ size$

$s^2\ =\ Sample\ variance$

$\sigma_0^2\ =\ Hypothesized\ population\ variance$

There are two classifications of statistical tests that you will need to know, **Parametric** and **Non-Parametric**. Parametric tests involve any testing where the given parameter follows a specific distribution. Non-parametric testing does not involve making any assumptions about the distribution of the parameter being studied. An instance where this could be necessary is when there are significant outliers that could cause the mean value of a dataset to be unrepresentative of the data. For this reason, if you wanted to run a test focused on the median of a dataset instead of the mean you would run a non-parametric test.

The **Spearman's Rank Correlation** Coefficient is a non-parametric statistical test used to examine whether there is a significant relationship between two sets of data. It can have any value between -1 and +1. A value of 0 indicates no relationship, and values of +1 indicate a perfectly positive correlation, while -1 indicates a perfectly negative correlation.

### Steps of Calculating Spearman's Rank Correlation Coefficient

i. Rank the values of variables X and Y in descending order.

ii. Find the difference (denoted as di) between the rank of the pair values of X and Y.

iii. Square the difference and calculate the sum of the difference, that is $\sum$di.

iv. Calculate the Spearman's rank correlation coefficient as follows:

$$r_s = 1 - \frac{6\sum d_i^2}{n(n^2-1)}$$

To test the relationship between two categorical variables, we use a contingency table and a test of independence based on a chi-square distribution. The test statistic is calculated as follows:

$$X^2 = \sum_{i=1}^{m} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Where:

$$E_{ij} = \frac{(\,Total\ row_i) \times \left(Total\ column_j\right)}{Overall\ total}$$

$m$ = Number of cells in the table, the number of groups in the first class, multiplied by the number of groups in the second class.

$O_{ij}$ = Number of observations in each cell of row *i* and column *j* (i.e., observed frequency).

$E_{ij}$ = Expected number of observations in each cell of row $i$ and column $j$, assuming independence (i.e., expected frequency).

# Reading 7 – Introduction to Linear regression

Linear regression forecasts the value of a dependent variable given the value of an independent variable. It assumes that there is a linear relationship between dependent and independent variables.

A **Dependent Variable** is predicted by an independent variable and is also known as the explained variable or endogenous variable. An **Independent Variable** explains the variation in the dependent variable.

The following is a simple linear regression equation:

$$Y = \beta_0 + \beta_1 X_1 + \varepsilon$$

Where:

$Y$ $= \textit{Dependent variable}$

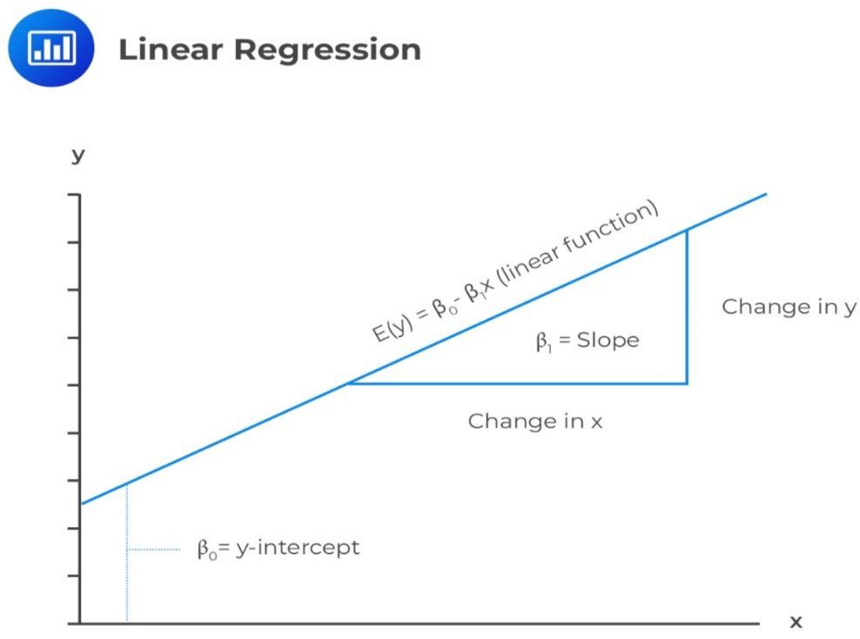
$\beta_0$ $= \textit{Intercept}$

$\beta_1$ $= \textit{Slope coefficient}$

$X$ $= \textit{Independent variable}$

$\epsilon$ $= \textit{Error term (Noise)}$

$\beta_0$ and $\beta_1$ are known as regression coefficients.




The **Slope Coefficient** is defined as the change in the dependent variable caused by a one-unit change in the value of the independent variable. The **Intercept** is the estimated value of the dependent variable when the independent variable is zero.

The normal linear regression model assumptions are:

- The relationship between the dependent variable, $Y$, and the independent variable, $X$, is linear.
- The expectation of the error terms is zero.
- The error terms (residuals) must be normally distributed.
- The variance of the error terms is constant across all observations.
- The error terms, $\epsilon$, must be uncorrelated across all observations.

**Sum of Squares Total** (total variation) is a measure of the total variation of the dependent variable. It is the sum of the squared differences of the *actual* y-value and *mean* of y-observations.

$$SST = \sum_{t=1}^{n} (Y_i - \bar{Y})^2$$

The Sum of Squares Total contains two parts:

i. Sum of Square Regression (SSR)

ii. Sum of Squares Error (SSE)

The **Sum of Square Regression** (SSR) is the measure of the *explained* variation in the dependent variable. It is given by the sum of the squared differences of the *predicted* y-value ($\hat{Y}$), and *mean* of y-observations, ($\bar{Y}$):

$$SSR = \sum_{i=1}^{n} \left(\hat{Y}_i - \bar{Y}\right)^2$$

The sum of squared errors is also called the residual sum of squares. It is defined as the variation of the dependent variable *unexplained* by the independent variable. SSE is given by the sum of the squared differences of the *actual* y-value ($Y_i$), and the *predicted* y-values, ($\hat{Y}_i$).

$$SSE = \sum_{i=1}^{n} \left(Y_i - \hat{Y}_i\right)^2$$

Therefore, the sum of squares total is given by:

$$\begin{aligned} Sum\ of\ Squares\ Total &= Explained\ Variation\ +\ Unexplained\ Variation \\ &= SSR +\ SSE \end{aligned}$$

The **Coefficient of Determination** ($R^2$) measures the proportion of the total variability of the dependent variable explained by the independent variable.

$$R^2 = \frac{Explained\ Variation}{Total\ Variation} = \frac{Sum\ of\ Squares\ Regression(SSR)}{Sum\ of\ Squares\ Total\ (SST)}$$

The sum of squares of a regression model is usually represented in the **Analysis of Variance** (ANOVA) table. The ANOVA table contains the sum of squares (SST, SSE, and SSR), the degrees of freedom, the mean squares (MSR and MSE), and F-statistics. The typical format of ANOVA is as shown below:

| Source of Variation | Sum of Squares | Degrees of Freedom | Mean Squares | F-Statistic |
|---|---|---|---|---|
| Regression (Explained) | SSR | 1 | $MSR = \frac{SSR}{1}$ | $F = \frac{MSR}{MSE}$ |
| Residual Error (Unexplained) | SSE | N – 2 | $MSE = \frac{SSE}{n-2}$ | |
| Total | SST | N – 1 | | |

**Standard Error of Estimate**, $S_e$ or $SEE$, is alternatively referred to as the root mean square error or standard error of regression. It measures the distance between the observed dependent variables and the dependent variables predicted by the regression model. The Standard Error of Estimate is easily calculated from the ANOVA table using the following formula:

$$Standard\ Error\ of\ Estimation\ (S_e) = \sqrt{MSE} = \sqrt{\frac{\sum_{i=1}^{n}(Y_i - \hat{Y})^2}{n-2}}$$

**Hypothesis Testing** is used to confirm if the estimated regression coefficients bear any statistical significance. Either the confidence interval approach or the t-test approach can be used in hypothesis testing. The following are the steps followed in the performance of the t-test:

i. Set the significance level for the test.

ii. Formulate the null and the alternative hypotheses.

iii. Calculate the t-statistic using the formula:

$$t = \frac{\hat{b}_1 - b_1}{S_{\hat{b}_1}}$$

Where:

$b_1 = \textit{True slope coefficient.}$

$\hat{b}_1 = \textit{Point estimate for } b_1$

$S_{\hat{b}_1} = \textit{Standard error of the regression coefficient.}$

i. Compare the absolute value of the t-statistic to the critical t-value ($t_c$). Reject the null hypothesis if the absolute value of the t-statistic is greater than the critical t-value i.e.,

$$t > + t_{critical} \text{ or } t < -t_{critical}$$

We calculate the predicted value of the dependent variable, Y, by inserting the estimated value of the independent variable, X, into the regression equation using the formula:

$$\hat{Y} = \hat{b}_0 + \hat{b}_1 X$$

Where:

$\hat{Y} = \textit{Predicted value of the dependent variable}$

$X = \textit{Estimated value of the independent variable}$

# Economics

## Reading 8 – Topics in Demand and Supply Analysis

You will need to be able to calculate and explain the importance of **Price Elasticity** on the exam. Elasticity measures the sensitivity of one variable to another. If something is highly elastic, it will experience a large price change in response to a small change in the related variable. Elasticity is calculated as follows:

$$E_{P_x}^d = \frac{\%\Delta Q_x^d}{\%\Delta P_x} = \left(\frac{\Delta Q_x^d}{\Delta P_x}\right) \times \left(\frac{P_x}{Q_x^d}\right)$$

Where:

$E_{P_x}^d = Price\ elasticity\ of\ demand\ of\ good\ X$

$Q = Quantity$

$P_x = Price\ of\ good\ X$

$\Delta P_x = Change\ in\ own\ price\ of\ good\ X$

$\Delta Q_x^d = Change\ in\ quantity\ demanded\ of\ good\ X$

Knowing both versions of the formula will help you whether the amounts in a problem are expressed as percentages or in whole numbers. The triangle symbol (∆) represents the change in a value. In addition to elasticity of prices (related to demand), there is also **Elasticity of Income** (related to supply). The formula is very similar to the price formula above:

$$E_I^d = \frac{\%\Delta Q_x^d}{\%\Delta I} = \left(\frac{\Delta Q_x^d}{\Delta I}\right) \times \left(\frac{I}{Q_x^d}\right)$$

Where:

$E_I^d = Income\ elasticity\ of\ demand$

$I = Income$

$\Delta I = Change\ in\ income$

There are two types of goods, as it relates to elasticity of income. **Normal Goods** have positive income elasticity (elasticity greater than 0), which means demand for them increases as consumer incomes increase. **Inferior Goods** have negative income elasticity (elasticity value less than 0) experience decreasing demand as income increases. A simple example is that a BMW is a normal good while a Kia is an inferior good, since people with higher incomes tend to buy more BMWs and fewer Kias.

There are also **Cross-Price Elasticities** among different goods.

$$E_{P_y}^d = \frac{\%\Delta Q_x^d}{\%\Delta P_y} = \left(\frac{\Delta Q_x^d}{\Delta P_y}\right) \times \left(\frac{P_y}{Q_x^d}\right)$$

Where:

$P_y = Own\ price\ of\ good\ Y$

$\Delta P_y = Change\ in\ own\ price\ of\ good\ Y$

**Substitutes** have positive cross-price elasticities since the increase in the price of one will increase demand of the other. This is common among different brands of the same product because people only need one or the other. **Complements** are goods with negative cross price elasticity. An increase in the price of a good will decrease demand for its complements because they tend to be purchased together, like peanut butter and jelly.

**The Law of Diminishing Marginal Returns** states that the marginal return from increasing input will decrease. One example would be that if you are hungry, eating an additional cookie will make you feel better. As you continue eating cookies, however, each new cookie will bring you less happiness.

**Economies of Scale** occur when increasing volume of production leads to decreasing average cost. You've probably seen this in action when you've received per-unit discounts for buying larger quantities of items, such as business cards.

# Reading 9 – The Firm and Market Structures

There are several common types of market structures you will need to know on the exam. In a **Monopoly**, one firm controls a market entirely. They set both the price and quantity of goods available and there are large barriers to entry preventing competition. In an **Oligopoly**, several firms control production in a market. There are high barriers to new entrants and the few controlling firms set prices to maximize profits. In **Perfect Competition**, many firms sell homogeneous goods to many buyers. No one seller can grow enough to influence the market. **Monopolistic Competition** involves several firms competing based on differentiating of products. Due to competition, no one firm can have out-sized profits over the long run.

There are different profit expectations for all four market types. Firms in a Perfectly Competitive market will experience virtually no economic profit over the long term. Any profit they earn will attract new entrants, which will increase supply to the point where Marginal Cost (MC) equals Marginal Revenue (MR) for additional production. Monopolistic Competition will draw new entrants into the market as profits increase, but firms can maintain some of their earnings through spending on marketing to differentiate their products from competitors'. In Oligopoly, firms can earn profits over the long run, but no one firm can earn enough to gain market dominance. As their profits increase, other firms will change their pricing strategy or new entrants will come in. Monopolist firms can earn economic profits over the long term. The firm can use its profit earnings to maintain the barriers to entry to prevent competition from eroding its market control. This is why it's important to regulate monopolies to keep a properly functioning market.

The four market types also have different optimal pricing strategies for firms. Perfect Competition firms are price takers, and the optimal price for their goods is the equilibrium price on the supply/demand curve. Firms in Monopolistic Competition have control over their own prices because they are able to differentiate their product to prevent consumers from substituting their goods with those from competitors. Oligopoly prices are determined by competitors. The few companies in the market can influence the price levels, but not control them. Monopolists set their own prices. Their only practical limit is not to raise their prices so high that it attracts new entrants to their market.

You will be expected to identify what type of market a firm operates in, given just a few pieces of information. Some common factors include the concentration ratio for the market, which ranges from 0 for Perfect Competition to 100 in a Monopoly. The level of price elasticity is also a good indicator. Monopoly firms have inelastic prices, while more competitive markets display much more elasticity.

# Reading 10 – Aggregate Output, Prices, and Economic Growth

**Gross Domestic Product** (GDP) is a measure of economic activity for a country. It can be measured as the total income earned by all households, companies, and the government (**Income Approach**) or as the total of all goods and services produced (**Expenditure/Output Approach**) over a given time. The formulas for the two approaches are:

**Income**

$$GDP = Total\ National\ Income + Sales\ Tax + Depreciation + Net\ Foreign\ Factor\ Income$$

**Expenditure**

$$GDP = Consumer\ Expenditures\ (C) + Business\ Investment\ (I) + Government\ Spending\ (G) + Exports\ (X) - Imports\ (I)$$

There are two ways to denominate GDP. These are **Nominal GDP**, which is how it is measured using unadjusted, observed values, and **Real GDP**, which involves adjusting the observations to account for inflation. The principle of real GDP is similar to that of finding a present or future value of the GDP amounts, using the inflation rate as the discount rate. One formula you will need to know for the exam is the **GDP Deflator**, which measures the level of inflation based on how it affects GDP values:

$$GDP\ Deflator = \frac{Nominal\ GDP}{Real\ GDP} \times 100$$

Economic growth factors are also an important part of the Econ curriculum. There are several measures for determining expected GDP growth and estimating the sustainability of that growth. Estimated GDP growth can be categorized into three categories: **Technology**, **Labor**, and **Capital**. The best one for estimating potential GDP growth is Labor. The two components of labor in GDP growth are in total hours worked and **Labor Productivity**.

$$Labor\ Productivity = \frac{Real\ GDP}{Aggregate\ Hours\ Worked}$$

Increases in labor productivity mean that more can be produced by the same number of labor hours. This is captured by the **Total Factor Productivity** component of GDP modeling. This figure is multiplied by the total labor hours to determine aggregate production. It represents the effects of technological improvements that increase the productivity of workers.

# Reading 11 – Understanding Business Cycles

The **Business Cycle** is the pattern of growth and decline that occurs as part of economic growth over time. While overall growth generally follows an upward trend, the business cycle represents the fluctuations around that trend line.

There are four phases of the business cycle that encompass each wave in the chart above:

1. **Expansion** features increasing growth, decreasing unemployment, and upward pressure on prices.

2. **Peak** occurs when inflationary pressure on prices and wages slows down further growth.

3. **Contraction** is when the economy slows down or declines. Prices start to decrease, and unemployment rises.

4. **Trough** is the lowest point in the cycle.

**Credit cycles** describe the changing availability and pricing of credit. In a strong economic environment, lenders are highly willing to extend credit and on favorable terms. in a weak economic environment, lenders' willingness to offer credit will be low while making the terms unfavorable, resulting in a decline in the value of assets causing more economic weakness and higher defaults.

There are many theories that attempt to explain the causes of business cycles, but the four primary ones in the curriculum are:

1. **Models with Money:** These models assume that money supply is a primary driver of business cycles, and that inflation is a result of these fluctuations.

2. **New Classical School:** This relies on the assumption of rational expectations for economic agents (buyers and sellers) in the market. Agents rely on rational beliefs about future inflation to influence their current day buying and selling

decisions. The economic results are the aggregation of these decisions made by each individual.

3. **Neoclassical and Austrian Schools:** These believe that the invisible hand of market forces will reach an equilibrium on its own unless that is prevented by government intervention. They posit that government intervention to increase growth in a recession is actually interrupting the natural flow of supply and demand that would bring growth back in line on its own.

4. **Keynesian and Monetarist School:** This school of thought believes that intervention is necessary to pull the economy out of a recession. Keynes supported fiscal policy as a tool to increase economic growth when spending is stuck at low rates.

Understanding **Inflation** is an important part of Economic analysis. Inflation is the increase in the prices of goods and services over time. Inflation reduces the purchasing power of consumers and can slow economic growth as a result. Some advantages of inflation include a decrease in the real amount of debt and an increase in unemployment due to the stickiness of nominal wages. An extreme case of inflation is known as **Hyperinflation**. This is when the supply and demand of money are so out of line that the value of money declines at astronomical rates. **Deflation** occurs when price levels decline, usually a result of a decrease in the money supply. Deflation causes a slowdown of economic activity because consumers defer purchases in expectation of lower prices in the future.

## Reading 12 – Monetary and Fiscal Policy

This reading focuses on the two primary mechanisms for market intervention by governments. **Fiscal Policy** involves decisions about government spending and taxation. **Monetary Policy** refers to actions by central banks to influence the quantity of money and credit throughout the economy. In trying to stimulate economic growth, monetary policy could be the Federal Reserve lowering interest rates through open market buying operations, while fiscal policy could be the government starting a stimulus program to increase aggregate spending.

Since economic growth is highly related to the supply of money, the CFA curriculum includes information about the nature of money. There are three important functions that money serves, including being a store of value, a unit of account, and a medium of exchange.

You need to be familiar with the **Quantity Theory of Money**, which is expressed by the formula:

$$M \times V = P \times Y$$

Where:

$M = Quantity\ of\ money$

$V = Velocity\ of\ money$

$P = Price\ level$

$Y = Real\ output$

Another important equation related to money is the Fisher relationship, which is the underlying principle of many central banking decision tools. The Fisher equation tells us:

$$R_N = R_R + \pi^e$$

Where:

$R_N = Nominal\ interest\ rate$

$R_R = Real\ interest\ rate$

$\pi^e = Expected\ inflation\ rate$

This model assumes that real rates are stable over time and so the primary determinant of nominal interest rates in the economy is the expected inflation rate. The equation itself is very simple but expect to see questions that try to mislead you into viewing it as more complex than it needs to be.

Central Banks play a major role in economic activity. They serve as the Controller of Credit through the use of monetary policy, where they use buying and selling of government securities to alter the supply of money in the banking system. They are also the custodian of cash reserves for commercial banks, in addition to being the "lender of last resort." In this role, they provide liquidity to banks through loans when market conditions are bad, and banks cannot get funding from other sources.

Central banks have several mechanisms through which they can apply **Monetary Policy** to influence economic activity. They set lending rates for banks by altering the rates at which they provide loans to commercial banks. The banks then alter their own lending rates to accommodate those changes while maintaining their earnings on funds. Central banks also use open market operations, wherein they buy or sell Treasury bonds to commercial banks to either decrease or increase, respectively, the total money in circulation.

Monetary policy can be either Expansionary or Contractionary, depending on whether the bank is trying to increase or decrease economic growth rates. **Expansionary** policies include increasing the money supply and lowering interest rates. These are intended to increase spending throughout the economy. **Contractionary** policies are activities in the opposite direction and are intended to decrease economic growth. A major indicator that central banks use to determine when they need to perform contractionary activities is when inflation rates begin to get exceedingly high.

Governments also utilize **Fiscal Policy** in order to influence economic growth and development. Fiscal policy entails spending and taxation programs by governments at different levels. Government spending can include **Capital Expenditures**, which are investments in infrastructure to increase the productive capacity of an economy, and **Current Spending**, which is directly spending on goods and services such as defense, health, and education. Another kind of spending is **Transfer Payments**. These are payments made to change the distribution of income throughout a society, such as welfare programs, Social Security, and pensions. These payments are not included in GDP calculations because they do not represent new production.

Like monetary policy, fiscal policy can be both expansionary or contractionary. Programs like tax cuts and spending increases are intended to spur economic growth, while tax increases and spending cuts do the opposite.

# Reading 13 – Introduction to Geopolitics

You will be expected to understand what geopolitics means and how geography influences national and international relations. Analysts in this field examine the people, groups, businesses, and national governments that engage in political, economic, and financial activities and their interactions with one another.

There are two main types of actors relevant to geopolitics risk:

i. **State Actors** often exercise authority over a nation's national security and resources. They do so through national governments, political groups, or country leaders. Examples of state actors are presidents, heads of government, and political parties.
ii. **Non-state Actors** engage in international political, economic, or financial activities without direct access to a nation's resources or security. Non-governmental organizations (NGOs), multinational corporations, charitable groups, and even influential people such as corporate executives or cultural celebrities are a few examples.

At the most fundamental level, interactions between nations or national governments (state actors) can either assume a **Cooperative** or **Competitive** form. A **Cooperative Nation** participates in and supports international accords on commerce, immigration, and regulation. In addition, such a nation embraces standardization of laws, tariff harmonization, and the free flow of information, including technology transfer. A **Non-cooperative Nation**, on the other hand, has erratic, often arbitrary policies and limits cross-border trade in capital, goods, and people. Aside from all these, a non-cooperative nation engages in retaliation and makes little contribution to technological exchange.

A country may cooperate with other countries for reasons such as national security/military interest, economic interest, geophysical resource endowment, standardization, and cultural considerations and "soft power".

Globalization is the interaction and integration of individuals, organizations, and governments on a global scale. There are various motivations for globalization, including increasing profits, access to resources and markets and intrinsic gain. It is important to note that there are some drawbacks to globalization, including unequal accrual of economic and financial gains, lower environmental, social and governance standards, political consequences, interdependence, autarky, hegemony, multilateralism, bilateralism.

**National Security** tools, **Economic Tools** and **Financial Tools** are geopolitical tools used by geopolitical actors to pitch their interests before others. These tools ultimately result in geopolitical risk.

There are three types of geopolitical risks:

**Event Risk:** is a type of geopolitical risks based on predetermined dates, such as elections, the enactment of new laws, or other date-based events, such as festivals or political anniversaries. Investor expectations frequently shift as a result of political events. These risks are mainly political timeline motivated.

**Exogenous Risk:** is a type of geopolitical risks involving an abrupt or unplanned concern that affects a nation's willingness to cooperate and non-state actors' capacity to go internationally. For example, an invasion, earthquake etc.

**Thematic Risk:** is a risk that concerns are well-known, and they change and grow with time. Examples include the persistent danger of terrorism, climate change, migratory patterns, the emergence of populist movements, and cyber threats.

An investor evaluates a geopolitical risk based on three categories:

i. Likelihood it will occur.

ii. Velocity of its impact.

iii. Size and nature of that impact.

## Reading 14 – International Trade and Capital Flows

You will be expected to understand the importance of trade between countries and how this affects economic growth within and between nations. International trade has many benefits, including increasing economic growth through specialization, improving the standard of living between countries, and new competition decreasing local monopoly power. There are also important limitations of trade, including increasing income inequality caused by international competition and loss of certain industries to international competitors who have competitive advantages.

There are two types of competitive advantage related to trading nations. A country has an **Absolute Advantage** when they can produce goods at a lower total cost than a trading partner. A **Comparative Advantage** is when a country can produce goods at a lower opportunity cost. Specializing in areas of comparative advantage allows countries to focus on what they do most efficiently and reduces average costs throughout the entire world economy through trade.

There are limits most countries' governments place on **Trade** flows in order to protect their own domestic interests. They can enact **Tariffs**, which are taxes on certain imported goods done to protect local industries from too much international competition. They can also require **Licenses** and set **Import Quotas** that limit the type and amount of goods that can be imported and sold to their citizens. Similar mechanisms can also be used for **Capital** controls that apply to the flow of money into and out of a country. Too few capital controls can make it difficult for a country (especially a smaller country with large trading partners) to enact the fiscal and monetary policies that it prefers.

These trading interactions are captured by the **Balance of Payments** between nations. It is a record of all economic transactions between different countries over a period of time and has three components. The **Current Account** tracks the inflow and outflow of goods or services. The **Capital Account** includes the moving of all non-financial and non-produced assets. The **Financial Account** is where all monetary assets are tracked.

There are several important international organizations to help promote and manage international trade. The **International Monetary Fund** lends foreign currencies to member states to aid them during financial difficulties in order to promote international monetary system stability. The **World Trade Organization** regulates cross-border trade relationships. The **World Bank Group** lends funding to projects in developing countries and helps create international economic infrastructure.

# Reading 15 – Currency Exchange Rates

Exchange rates represent the price of one currency in terms of another. A EUR/USD exchange rate of 0.75 means that 1 US dollar (USD) will buy 0.75 Euros (EUR). In this example, EUR is the price currency and USD is the base currency. This is known as a **Spot Exchange Rate** because it involves a transaction of one currency for another at the present time. There are also **Forward Exchange Rates**, which represent the exchange of one currency for another at a future date.

Forward rates are often expressed in terms of **Points**, which stands for basis points. This means that in order to convert them to decimals, you need to divide them by 10,000 (a basis point is 1/100th of a percentage). Here is an example:

- USD/EUR Spot Rate=1.5252
- 6-month forward points = 3bps

Therefore the 6-month forward exchange rate id

$$1.5252 + \frac{3}{10{,}000} = 1.5252 + .0003 = 1.5255$$

Forward rates can also be expressed as a **Premium** or **Discount** to the spot rate. This is based on the relative interest rates between the two countries whose currencies are involved.

$$F_{f/d} = S_{f/d}\left(\frac{1 + i_f}{1 + i_d}\right)$$

$S_{f/d} = \ Spot\ exchange\ rate$

$F_{f/d} = \ Forward\ exchange\ rate$

$i_d$ = *Domestic risk-free interest rate*

$i_f$ = *Foreign risk-free interest rate*

Suppose the investment horizon is a fraction, denoted as $\tau$ the forward exchange rate is given by:

$$F_{f/d} = S_{f/d}\left(\frac{1 + i_f\tau}{1 + i_d\tau}\right)$$

As an example, here is how to find the 90-day forward premium (or discount) based on the following:

- EUR/USD Spot Rate = 1.500
- EUR 1-year risk-free interest rate = 4%
- USD 1-year risk-free interest rate = 6%

The forward exchange rate is:

$$F = 1.5 \times \left(\frac{1 + 0.04 \times \frac{90}{360}}{1 + 0.06 \times \frac{90}{360}}\right) = 1.5 \times \left(\frac{1.44721}{1.49492}\right) = 1.45213$$

Exchange rates can also be expressed as **Cross Currency Rates**, which is the combination of several exchange rates to get a new rate in terms of two different currencies. This looks like:

$$\frac{EUR}{USD} \times \frac{USD}{CNY} = \frac{EUR}{CNY}$$

An important consideration when answering cross currency problems on the exam is to make sure that you are using each rate correctly. In the example above, you'll see that USD is the price currency in one equation and the base currency in the other. When you are translating these rates, it's important to make sure that your price and base currencies line up appropriately. It is very common for the exam to quote exchange rates in a different order than what you need for the equation, like in the example below:

$$EUR/USD = 0.75; \;\; CNY/USD = 7$$

$$\frac{EUR}{CNY} = \frac{EUR}{USD} \times \left(\frac{CNY}{USD}\right)^{-1} = \frac{0.75}{1} \times \frac{1}{7} = 0.107$$

According to the **Interest Rate Parity** theory, the currency of a nation with a lower interest rate should be at a forward premium compared to the currency of a nation with a higher rate.

# Financial Reporting and Analysis

There's no getting around it, financial reporting and analysis is a big part of the CFA level 1 curriculum, and there is a lot of material with which you'll need to become familiar. As one of the largest sections of the exam, you will want to be familiar with the FRA formula sheet information, as many of the questions will involve giving you a snippet of information from a financial statement and asking you to calculate a specific value from that. Knowing which data points go into specific formulas will be necessary to get a good score here.

## Reading 16 – An introduction to Financial Statement Analysis

There are four financial statements that are used to summarize a company's financial positions and performance. The statement of financial position (**Balance Sheet**) provides a description of the company's financial position at a particular point in time. The statement of comprehensive income (**Income Statement**) shows the company's financial results over a specified time period. The Statement of Changes in Equity provides information on the changes in owners' investments in the company over time, including the balances at the beginning and end of the reporting period and the changes that occurred during. The statement of cash flows (**Cash Flow Statement**) shows a company's sources and uses of cash during a specified period.

These statements are prepared according to specific standards based on where the company is located. US GAAP ("Generally Accepted Accounting Principles") applies to a company in the United States and IFRS ("International Financial Reporting Standards") is used everywhere else. In addition to the primary statements described above, there will usually be supplementary notes included in a company's financial statement disclosures. These notes include important information about risks the company faces, estimates used in preparing the financial statements, and performance of specific business units within the company.

The curriculum specifies a 6-step framework for financial statement analysis, which goes as follows:

1. Articulate the purpose and context of the analysis: Specify the target audience, timeframe, and main focus on the analysis to be done.
2. Collect input data: Gather the necessary data to answer the questions specified in step 1.
3. Process data: Perform calculations and create charts or other outputs to synthesize the information.
4. Analyze and interpret the data: Review the data and use it to reach a conclusion or recommendation.
5. Develop and communicate conclusions and recommendations.
6. Follow-up: Periodically update and review the data to determine if it still supports the original conclusion.

# Reading 17 – Financial Reporting Standards

The purpose of having standards for the preparation of financial statements is to provide consistency between different companies. As an analyst, you would be expected to utilize information from financial statements to influence investment decisions and having standards to which these statements must conform makes it easier to compare widely different companies. There are currently two sets of standards used throughout the world, US Generally Accepted Accounting Principles (**GAAP**) in the United States and International Financial Reporting Standards (**IFRS**). These have been converging over time to align more closely with IFRS, but companies in the US must still adhere to GAAP. It will be important to understand where these standards differ in their reporting requirements.

The Conceptual Framework outlines the primary characteristics behind IFRS, which are Comparability, Verifiability, Timeliness, and Understandability. Two primary assumptions that impact how financial statements are prepared are **Accrual Accounting** and "**Going Concern**". As covered earlier, accrual accounting means revenue and expense items must be recorded when recognized, and going concern is the assumption that a company will continue in business for the foreseeable future. IFRS also requires consistency of item classification from one reporting period to the next.

There are several important differences in how GAAP and IFRS define revenue and expense recognition. IFRS includes the word "probable" in its definition of when revenue should be recognized, specifically that it needs to be recognized when it is probable to provide a future economic benefit to the reporting company. GAAP does not include the word probable in its criteria for revenue recognition.

# Reading 18 – Understanding Income Statements

The income statement provides information on the financial performance of a company over a specified period of time. There are several required elements that must always appear on the income statement. **Revenue** is the top line on the income statement and is the amount of sales that the company took in. **Expenses** are all outflows and depletion of assets that occurred during the reporting period. **Gross Profit** is the amount of revenue left after subtracting the cost of goods. **Operating Profit** is sometimes referred to as Earnings Before Interest and Taxes (EBIT) and is the revenue left after subtracting all operating expenses. **Net Income**, also known as Profit or Loss, appears at the bottom of the income statement.

An important consideration in the preparation of financial statements is **Revenue Recognition**. IFRS and GAAP differ on the guidance for when revenue is to be recognized.

IFRS instructs that revenue should be recognized when the following conditions are satisfied:

- The significant risks and rewards of ownership of the goods have been transferred to the buyer.
- The company no longer retains effective control over the goods sold.
- Revenue can be reliably measured.
- The economic benefits of the transaction will probably flow to the company.
- Transaction costs have been incurred.

The US GAAP criteria for determining when revenue is to be realized are:

- Evidence exists of arrangement between buyer and seller.
- A product has been delivered or service rendered.
- Price is determined.
- There is reasonable assurance that the seller will collect money.

Revenue recognition can be calculated using the new converged standard. The new converged standard proposes that revenue should be recognized in a five-step framework:

1. Identify the contract(s) with a customer.

2. Identify the performance obligations in the contract.

3. Set the transaction price.

4. Allocate the transaction price to the performance obligations in the contract.

5. Recognize the revenue upon satisfaction of a performance obligation.

For recognition of expenses, both IFRS and US GAAP rely on the principle of **Matching.** This means that expenses associated with specific revenue are to be recognized in the same period as those revenues. There are several methods by which this can be applied. The **Specific Identification Method** is used when individual inventory items that are sold can have their exact costs matched to each item. This typically applies to items that are large and unique, such as expensive jewelry. More commonly, the **First in First Out** (FIFO) method is used to match expenses with sales. Both IFRS and GAAP allow for FIFO, which allocates the sales expenses of the oldest inventory item to each additional sale. In the **Weighted Average** cost method, the average cost of goods available for sale is allocated to each sale. Another method is the **Last in First Out** (LIFO) method, which is allowed only by the US GAAP. This assumes that the newest goods for sale are sold first. The problem with this method is that the older items might stay indefinitely in the inventory, therefore not accounting for inflation or obsolescence.

The use of different expense methods can have a big impact on a company's reported profits. In an environment of rising prices, LIFO will cause reported income to be lower (each new item in inventory was more expensive than the last, so Cost of Goods Sold will be higher), which will also cause tax expenses to be lower (less profit means less taxable income). You will definitely see questions about comparing FIFO and LIFO effects on financial statement values.

The **Earnings Per Share** (EPS) value calculated on the income statement is an important metric in a company's performance but can be complicated to determine if a company has a complex capital structure. The simple formula for EPS only includes the weighted average shares outstanding, but this needs to be adjusted for new shares or potential new shares if the company has stock options or convertible debt that could be used to create new shares. The formulas for each condition appear in your formulas sheet. These securities can have either dilutive or antidilutive effects, depending on whether the inclusion in the formula decreases or increases EPS, respectively. Securities that have an antidilutive effect cannot be included in the calculation of diluted EPS under IFRS or GAAP.

## Reading 19 – Understanding Balance Sheets

The balance sheet illustrates a company's total assets and sources of capital at a particular point in time. There are three elements that make up the balance sheet. **Assets** are the resources which a company owns or controls and will use to derive future economic benefits. **Liabilities** are anything that the company owes. **Equity** is the owners' residual interest in the company after deducting the liabilities. They make up the basic accounting equation:

$$Assets = Liabilities + Equity$$

Balance sheets are useful for highlighting a company's abilities to meet its operating liquidity needs, keep up with debt obligations, and make distributions to shareholders. There are some limitations to this reporting, however. Not all items on the balance sheet are measured in the same manner, so some items may reflect historical costs while others are at current market value. The values presented as current were only current at the time for which the balance sheet was prepared.

Both IFRS and GAAP require companies to report current and non-current assets and liabilities separately. Current assets are those held for trading or that are expected to be sold or used up within the next reporting period, including cash, trade receivables, and inventories. Non-current assets are expected to be held or used up over longer periods of time. These long-lived assets represent the infrastructure of the company, including property plant and equipment, goodwill, and long-maturity financial assets. Current liabilities are expected to be settled within one reporting cycle. These include trade payables and accrued expenses. IFRS requires some current liabilities (trade payables and some accruals) be considered part of working capital and must always be classified as current, regardless of the when they will be settled. Non-current liabilities will be settled in greater than a year or reporting cycle, such as deferred tax liabilities.

Current assets such as cash or marketable securities are typically valued at current market prices. Trade receivables are reported at their net realizable value, which includes an estimate of collectability. Inventories are valued at either the lower of cost and net realizable value (IFRS) or the lower of cost and market value (GAAP). Non-current assets are usually valued using the cost model (GAAP and IFRS) or the revaluation model (IFRS). For Property, Plant, and Equipment, the cost model is the historical cost of the item less accumulated depreciation, and the revaluation model is the fair value at the date of revaluation less subsequent accumulated depreciation. Investment Property (which is used to generate non-operating income) can be valued using the cost model or the fair value model, but under the fair value model, all gains or losses arising from a change in the fair value of the property are recognized on the income statement. Goodwill is capitalized when realized under both IFRS and GAAP. Rather than amortized over time, goodwill must be periodically tested for impairment. Impairment losses are charged against income when they occur. Liabilities are also reported base on whether they are current or non-current. Current assets include Trade Payables and Accrued Expenses that will be settled in the following reporting period. Non-current payables include long-term loans and deferred tax liabilities.

There are a number of liquidity and solvency ratios in your ratio sheet, downloadable in the AnalystPrep.com's platform, that you can use to evaluate a company's financial health based on information found in the balance sheet. Expect to see questions asking you to calculate those ratios and use them to rate the financial situation of an example company.

## Reading 20 – Understanding Cash Flow Statements

There are three primary components of the cash flow statement. **Cash Flow from Operations** is derived from the day-to-day activities of the company. **Cash Flow from Investing Activities** results from the purchase and sale of assets not related to the normal activities of the business. **Cash Flow from Financing Activities** results from capital financing activities.

There are a number of differences in cash flow classification between US GAAP and IFRS. IFRS allows interest paid or received to be classified as either operating or investing, but GAAP requires it to be under operating. GAAP requires dividends received to be classified as operating, but IFRS allows them to be either operating or investing. Dividends paid must be classified as financing activities under GAAP but can be either financing or operating under IFRS.

Cash Flow from Operations can be reported using the **Direct** or the **Indirect Method**, but both IFRS and GAAP encourage the direct method. Under the direct method, each cash inflow and outflow related to cash receipts or payments is shown. This eliminates the impact of accruals. The indirect method provides only the net results of receipts and payments. It uses net income and a series of adjustments to reconcile to the operational cash flow figures.

There is a multi-step process for converting cash flows from the indirect method to the direct method. Net income in disaggregated into total revenues and total expenses, then non-operating and non-cash items are removed from aggregated revenues and expenses into the relevant cash flow items, and then finally the revenue and expense accrual amounts are converted to cash flow amounts. Expect questions on the exam asking you to go through these steps.

Just like for the information regarding the other financial statements, there are a number of ratios using information from the cash flow statement to determine the financial health of the reporting company. This includes formulas like Free Cash Flow to the Firm and Free Cash Flow to Equity that will definitely appear on the exam. Be sure to familiarize yourself with these equations by doing practice questions.

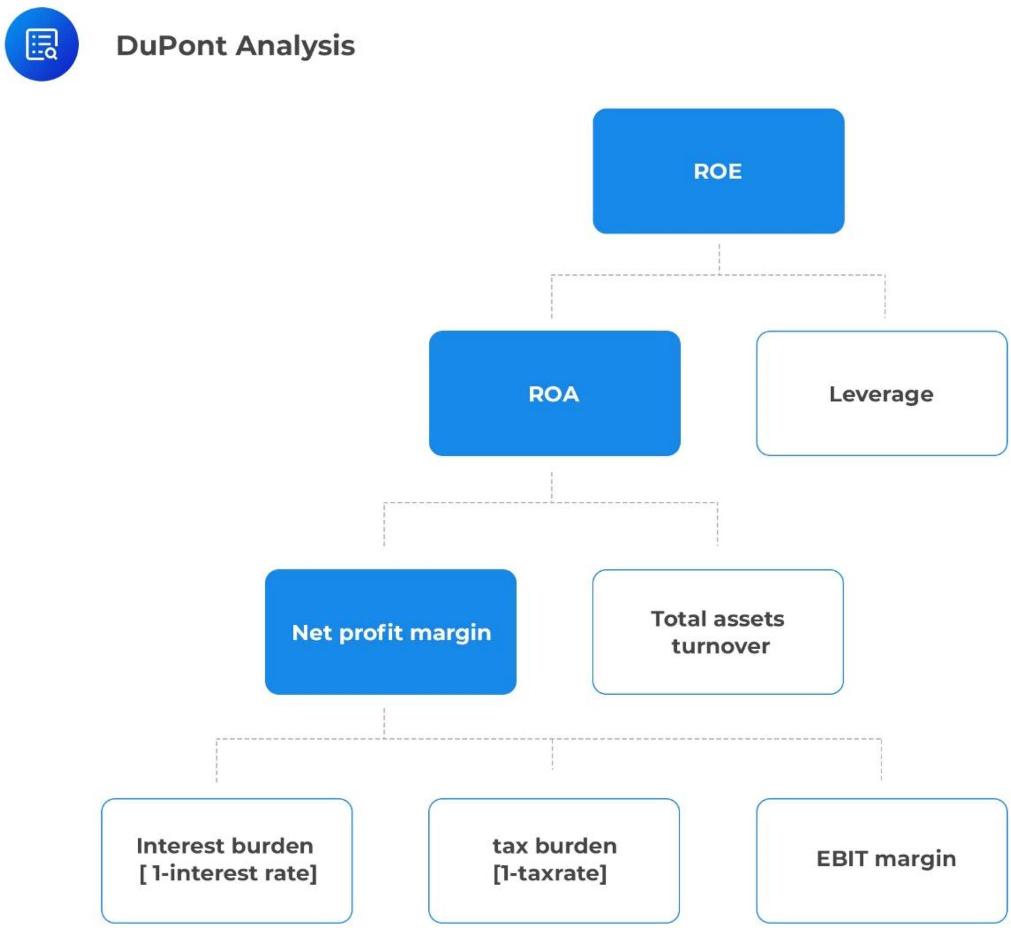
# Reading 21 – Financial Analysis Techniques

Two common techniques for analyzing company financial data are Ratio Analysis and Common-Size Analysis. **Ratio Analysis** evaluates a company's past performance using financial ratios. These can be used to project future results but are also limited by judgments made in the calculations, ratios that might provide conflicting results, and heterogeneity of a company's operating activities over time. **Common-size Analysis** involves calculating all financial statements inputs as a percentage of total revenue or assets to facilitate comparisons between different companies.

Several categories of ratios are used for performance analysis. **Activity Ratios** such as Inventory Turnover are used to measure how efficiently a company uses its resources to perform normal business functions. **Liquidity Ratios** measure a company's financial health and ability to meet short-term obligations. Common examples include the Current Ratio and Quick Ratio that compare liquid assets to short-term financial liabilities. **Solvency Ratios** are similar to liquidity ratios but are used to measure the ability to meet long-term obligations. **Profitability Ratios** measure a company's ability to generate profits from its resources. **Valuation Ratios** measure the quantity of an asset or cash flow that is associated with ownership of a specific capital claim. Examples of all of these appear in your formula sheet.

When calculating ratios that involve items from the balance sheet and income statement, there are a few concerns that must be addressed. Since the income statement represents activity over time and the balance sheet is a snapshot, there are certain best practices to ensure appropriate calculation methodology. For example, in a ratio where there is an income statement or cash flow statement item in the numerator and balance sheet item in the denominator, you should use an average value of the balance sheet item over the reporting period.

The **DuPont Analysis** method of decomposing Return on Equity is an important part of the financial analysis section. It is a way of breaking down the ROE into component parts in order to better analyze the drivers of returns. The specific formulas for the DuPont method appear in your formula sheet, but the concept is illustrated below:




Both IFRS and GAAP require the breaking out of results for individual business segments if certain requirements are met. This includes any business segment that constitutes more than 10% of total company revenue or profit. Companies are able to combine individual business segment results if the segments are small and share common factors that allow for sensible grouping, such as the type of business or sales locations.

# Reading 22 – Inventories

**Inventories** and **Cost of Goods Sold** are two important items contained within a company's financial statements. Amounts held in inventory represent assets for sale that will eventually show up on the income statement as Cost of Goods Sold when sales are made. There is a choice made by companies whether these items appear on the balance sheet as inventory in the first place, or just expensed as soon as they are acquired. By putting them on the balance sheet as assets until sales are made, companies can defer recognizing them as expenses until a later period. If costs are included in inventory that should have been expensed right away, it overstates profitability in the reporting period and inflates inventory values on the balance sheet.

Both IFRS and GAAP require costs to be included in inventories if they represent items such as the purchase price of goods, transportation costs, or other costs incurred to acquire these goods and make them available for sale. Both frameworks exclude costs from inventory (and require them to be expensed) if they represent abnormal costs related to product waste and administrative overhead.

There are four methods for valuing inventory as it relates to calculating the Cost of Goods Sold: Specific Identification, First-In First-Out (FIFO), Last-In First-Out (LIFO), and Weighted Average Cost. LIFO is only permitted under GAAP, but both systems permit the use of the other three. **Specific Identification** is for goods that are not interchangeable and are typically produced for individual projects. The cost of each of the goods is matched to the physical flow and sales for that good. **FIFO** assumes that the oldest items in inventory are sold first as each additional sale takes place. **LIFO** assumes that sales come from the newest items in inventory. **Weighted Average Cost** gives each item in inventory a cost equal to the average value of all inventories.

You will be asked to both calculate inventory and cost of goods sold values, given sales and balance sheet items, and also to explain the impact that using one valuation method over another would have on things like Profit and Tax Expense. Assuming an environment of rising prices over time, inventory values will be sorted as FIFO > Weighted Average > LIFO. Net profit is affected the same way. Your formula sheet contains the necessary equations to calculate these figures.

For companies under GAAP that use the LIFO method, there is a requirement to disclose what is known as the **LIFO Reserve.** This refers to the difference between the reported LIFO inventory and what the inventory amount would have been under FIFO. The equation is:

$$LIFO\ Reserve\ =\ FIFO\ Inventory\ Value - LIFO\ Inventory\ Value$$

This value is important because being able to convert between financial statements using the different systems will definitely appear on the exam. Related to this is the phenomenon known as a LIFO liquidation. This occurs when more inventory is sold than is added back, so some of the older (typically less costly) items in inventory are sold. This will cause a rise in profit margins that is not sustainable into future periods.

IFRS and GAAP differ on how inventory balances are to be calculated, with IFRS requiring the lesser of Cost and Net Realizable Value and GAAP requiring the lesser of Cost and Market Value. The Net Realizable Value calculated for IFRS must be reassessed each reporting period, and any declines below the previous carrying amount must be reflected as a writing down of the inventory value and recognized as an expense on the income statement. Future increases (reversals) of the Net Realizable Value must also be recorded, though they can only go back to the original value before the write down was made. Under GAAP, value write-downs cannot be reversed.

# Reading 23 – Long-Lived Assets

Long-lived assets are resources that are expected to provide benefits to the company over multiple reporting periods. The costs of these assets are usually capitalized, rather than expensed at once. This means that their costs are allocated across multiple reporting periods. This applies to the costs for purchasing or building property or other resources, including the interest costs incurred if the company takes out loans.

**Intangible Assets** are another type of long-lived asset. The accounting treatment for these will vary depending on how the intangible asset was acquired. When they are acquired in business situations that are not business combinations, the assets are recorded at fair value at the time of acquisition. If they are developed internally, the costs of development are expensed as incurred under both GAAP and IFRS. GAAP differs slightly in that it requires the capitalization of certain costs related to software development. For intangibles acquired through a business combination, the purchase price is allocated to each asset acquired based on its fair value at the time, with any excess price recorded as the intangible asset Goodwill.

It’s important to understand the impact of expensing compared to capitalizing expenditures on a company’s financial statements. Expensing a cost rather than capitalizing it will reduce profits in the first year, but then increase it in the following years. Capitalizing will result in higher shareholders’ equity in the early years because the higher first-year profits will increase retained earnings. Capitalizing expenses also increases the reported cash flow from operations.

Long-lived (tangible) assets are depreciated over time to reflect the allocating of its cost over its entire useful life. There are three primary methods of depreciation used in financial reporting. The **Straight-Line Method** allocates the cost by an equal portion in each year of the asset's useful life. The **Accelerated Method** uses a greater cost in the early years that declines as the asset reaches the end of its life. In the **Units of Production** method, the cost is allocated based on the actual of the asset. It's important to note that, in some countries, the methods used in the financial statements will differ from what is used for tax purposes. This will cause the tax expense computed on the income statement to be different than actual tax expenses in the period.

Similarly, to the depreciation of tangible assets, intangible assets are amortized if they have a calculable, finite useful life. This would include items like a patent or copyright that has a specific expiration date. The calculation of amortization is similar to that of depreciation, including having the same three methods for calculating it. While the total accumulated amortization expense will be the same no matter what method is used, it can have a significant impact on the financial statements, especially in the early years.

While not allowed under GAAP, IFRS permits the use of the **Revaluation Model** as an alternative for calculating the carrying value of an asset. This model uses the current fair value (less accumulated depreciation or amortization) as the carrying value of the asset. Under this approach, it's possible for the value of a long-term asset to exceed its historical cost. Decreases in carrying value and future increases (up to the original value) are treated as profit or loss on the income statement. An increase in value above the original asset value is recorded as equity in a Revaluation Surplus Account.

**Impairments** are recorded if the carrying amount of an asset exceeds the recoverable costs. Intangible assets with indefinite lives are tested at least annually for impairment. If an asset is no longer used for operations and is listed as Held for Sale, it must be tested for impairment. IFRS allows reversals of impairments if the recoverable amount exceeds the previous carrying amount, but GAAP does not allow for a reversal of an impairment once it has been recorded.

Investment properties, which are used to generate non-operating returns, can be valued using either a cost model or fair value model under IFRS. The company must make additional disclosures about its fair value determinations if it uses that approach. The company must also use the same model for all of its investment property. If the company uses the fair value model and transfers a property from inventory to investment property, then the change between the inventory carrying amount and the fair value is recognized as a profit or loss.

## Reading 24 – Income Taxes

**Income Before Taxes** is reported on a company's income statement and is calculated based on the applicable reporting standards, while **Taxable Income** is the portion of total income subject to income taxes by the taxing authority. These two may differ depending on how tax policy overlaps with reporting standards. These differences can include things such as revenues and expenses being recognized in different reporting periods for tax and accounting purposes, different carrying amounts for assets and liabilities, or tax losses from prior periods being brought forward and reducing taxable income in later periods.

**Deferred Tax Assets** arise when a company's taxable income is greater than its accounting profit. This creates an excess tax payment by the company over its expected tax expense that the company can recover in future periods. When the opposite occurs, and a tax deficit is created, this is a **Deferred Tax Liability**.

Any asset or liability that will create taxable economic benefits (positive or negative) to the company will have a **Tax Base**. This is the amount that will be deductible for tax purposes once the benefits are realized in the future.

When income tax rates change, all deferred tax assets and liabilities are adjusted to reflect the new tax rate. As a result, the balance sheet of a company is affected by changes in tax rates, just like the earnings and cash flow statements. If it is likely that a deferred tax asset may not be fully realized in the future, GAAP requires the creation of a **Valuation Allowance** to account for the loss of value. This new item reduces the value of the deferred tax asset down to its new anticipated recovery value.

**Temporary Differences** occur whenever there is a difference between the tax base and the carrying amount of assets and liabilities on the balance sheet.

**Permanent Differences** are differences between the tax and financial reporting of revenue or expense items that will not be reversed in future.

When calculating the value of certain balance sheet items related to taxation, there are differences in how tax rates are applied. Current taxes payable are determined using the tax rates as of the balance sheet reporting date, while deferred taxes are measured at the rates expected to apply when the assets are realized.

Companies must include disclosure about many of their tax-related financial statement items. These include the specific tax provisions that led to their calculations and treatment of specific items, detailed information about how tax assets and liabilities were determined, and whether any valuation allowances are applied.

There are several important differences in the impact of tax-related items, depending on whether GAAP or IFRS standards apply. For example, IFRS allows revaluations to be calculated as deferred tax assets, whereas GAAP does not allow revaluations at all. Also, IFRS allows deferred tax items to use projected tax rates that have been "Substantively Enacted" even if they are not fully in effect, but GAAP requires the use of rates that are fully enacted in law.

# Reading 25 – Non-Current (Long-Term) Liabilities

**Bonds** are contractual promises made by a company to pay bondholders in the future in exchange for receiving cash in the present. There are two cash flows associated with bond repayment: periodic interest payments and the final face value of the bond. The market value of the company's bonds is the present value of the future payments discounted to a present value. The cash proceeds received from issuing bonds are reported as cash flows from financing. Bonds are initially reported as a liability on the balance sheet at the amount of cash proceeds net of issuance costs under IFRS, whereas GAAP requires bonds to be reported at the amount of proceeds with no concern given to costs of issuance.

The discount or premium for bonds issued at a price other than par is amortized over the life of the bond using one of two methods. The **Straight-Line Method** amortizes equal portions of the premium or discount over each reporting period. The **Effective Interest Rate Method** (which is required by IFRS and preferred under GAAP) applies the market rate in effect when a bond is issued to the current amortized cost to obtain the interest expense for that period.

When bonds are redeemed (repaid), the difference between the cash required to redeem the bonds and the carrying value of the bonds on the balance sheet is booked as a gain or loss on the extinguishment of the debt. As stated before, IFRS includes debt issuance costs as part of the bond's carrying amount. Under GAAP, these issuance costs are accounted for as a separate item and are amortized over the life of the bonds. The gains or losses resulting from the extinguishment of debt are disclosed as a separate line item on the income statement if the amount is material.

There is typically a line item under non-current liabilities on the balance sheet for the total amount of a company's long-term debt that is due more than one year in the future. The portion of long-term debt due in the next 12 months is included as a current liability. The notes for the financial statements will include more detailed disclosures of a company's outstanding debt. These disclosures can also include details about more complex debt contracts, such as convertible debt or bonds issued with warrants.

Companies that wish to obtain the use of an asset can choose to purchase or lease them. In a lease agreement, a lessor grants the right to use an asset to a lessee for a specified period of time in exchange for periodic payments. There are a number of advantages to leasing an asset compared to purchasing it. Leases typically provide less costly financing compared to a purchase. Lessors may be in a better position to take advantage of tax benefits of ownership, be better able to bear risks associated with ownership, or may enjoy economies of scale for servicing assets.

A lessee will prefer an operating lease to a financial lease. This is because the reported assets, debts, and expenses are higher in the early years under a finance lease.

There are two types of retirement plans which a company can provide for its employees. In a **Defined-Contribution Plan**, the company contributes a defined amount to the plan. In a **Defined-Benefit Plan**, the company is responsible for providing a specified future benefit to its employees in retirement.

The financial reporting requirements for defined-benefit plans are more involved than for defined-contribution plans. A company balance sheet will reflect a net pension asset (liability) if the fair value of the pension fund's assets is higher (lower) than the present value of the estimated pension obligation. In each reporting period, the change in net pension asset or liability is recognized as profit or loss under Other Comprehensive Income.

There are several types of solvency ratios that you can use to analyze a company's ability to meet its long-term debt obligations. The two types of these formulas are leverage and coverage ratios. **Leverage Ratios** are based on balance sheet items and measure how equity capital is used to finance a company's assets compared to debt capital. **Coverage Ratios** use income statement information to measure a company's ability to cover its debt-related payments. You can find all of these ratios on AnalystPrep's ratio sheet.

# Reading 26 – Financial Reporting Quality

The goal of financial reporting statements is to provide a complete picture of the financial performance and position of a company. Two components of how well this is done are the quality of financial reporting and the quality of reported results. The **Quality of Financial Reporting** refers to how relevant, complete, neutral, and error-free the actual financial statements are. Low-quality reporting makes it difficult to compare a company to its peers or to get an accurate understanding of how well the company is doing. Poor **Earnings Quality** means the actual results of the company are misleading or unsustainable.

High-quality earnings indicate strong returns from the company's assets and sustainable sources of revenue. These earnings are a good indicator of the company's ability to continue operating well into future reporting periods. High-quality reporting allows an analyst to make accurate assessments of a company's performance. The quality of earnings and the quality of reporting are interrelated but exist as separate concepts to be judged in exam questions.

Companies have to use judgment within the guidelines of the reporting standards in calculating their financial performance. More aggressive accounting techniques use creativity to overstate financial performance. This approach impacts the sustainability of the company's earnings because decisions that overstate current period performance will cause an understatement of future reporting periods. Conservative accounting tends to understate current period results but does not cause sustainability concerns. Using conservative methods has several benefits, including reducing the possibility of litigation and protecting the interests of regulators.

Company management can have incentives that encourage the production of low-quality earnings reports. Concerns about missing market expectations for earnings can lead to more aggressive accounting techniques in order to artificially meet public targets. Certain market conditions make it easier for low-quality reporting to occur. Poor internal controls or lax regulatory oversight can create opportunities for companies to overstate performance. Poorly designed managerial incentives create motivations for questionable decisions.

Market authorities create regulatory regimes that enforce the quality of reporting for public companies. Requirements for disclosure of specific details in financial statements, managerial commentary about judgments and calculations, and review of reported performance by market regulators help to maintain the quality of reporting by market participants.

There are a number of accounting choices that company managers make when preparing financial statements. Assumptions regarding inventory cost (FIFO vs LIFO), accrual vs. cash accounting methods, and depreciation methods can have a significant impact on reported results without changing the underlying strength or weakness of the company's performance.

## Reading 27 – Applications of Financial Statement Analysis

Companies can follow several strategies to achieve successful performance that you can identify through their past financial statements. Companies that focus on product differentiation and minimizing input costs lead to higher gross margins. Strategies that manage liquidity through maintaining adequate coverage of cash flow obligations and avoiding excessive debt constraints will have stronger balance sheet ratios but may show lower resource utilization ratios.

Reported financial results can be used to forecast future performance. Forecasting company sales is typically based on a top-down approach that uses industry and macroeconomic trends to determine future period growth. Individual company past sales are also used to determine potential results for specific business units. Performance ratios from prior periods can then be used to calculate forecasts for profitability and cash flow measures. Valuation models use this approach over several periods to construct estimated values of a company or its equity.

Financial statements are also used to determine the credit quality of companies. The quality of a company's financial performance is a good indicator of its ability to continue meeting its financial obligations in the future.

Analysts routinely use information from financial reporting to influence equity investment decisions. The standardization of financial statements is intended to make it easier to compare relative performance and positions of different companies. Methods such as stock screeners use various valuation, solvency, and performance ratios to filter out fewer desirable companies for potential equity investments.

It is often necessary to adjust financial statement information to facilitate comparisons between companies. Items like inventory cost methods that differ between otherwise related companies need to be accounted for in making sure that comparisons are as valid as possible. Other common areas requiring adjustments include depreciation methods, accounting values of intangibles such as goodwill, and off-balance sheet finance arrangements such as operating leases. It is important to understand how these various approaches differ in their impact on financial results (covered earlier) in order to make these adjustments and improve the accuracy of relative performance comparisons.

# Corporate Issuers

## Reading 28 – Corporate Structures and Ownership

A business is initiated by a founder(s) who possesses significant know-how or expertise. However, the founder may, nevertheless, lack the skills to manage the business as it grows. Moreover, the founders may lack the capital to grow their business and thus, outsource capital from private channels.

There are *four factors* that defines a business including **Legal Identity** which defines the legal relationship between the owner(s) and the business, **Business Liability** which defines the level of liability an individual assumes dues to a business' actions or debts, **Owner-operator Relationship** which describes the relationship between the owner(s) of a business and the management, and taxation which defines the tax regime applicable to a business structure.

There are different types of business structure each defined by their unique characteristics. A **Sole Proprietorship** or **Sole Trade** are where the owner raises the business capital and fully controls business operations. He also fully benefits fully from financial returns and assumes all the business risks. A **General Partnership** business structure on the other hand there are at least two owners (partners) whose roles and responsibilities are stated in the partnership agreement. It is characterized by multiple sources of funds, and the returns and risks are shared. In **Limited Partnership** business structure, the partners are divided into two: at least one **General Partner (GP)** (has unlimited liability) and **Limited Partners (LPs)**. The general partner is responsible for the management of the business while a limited partner do not participate in management and their losses are capped at their investment amounts in the limited partnership. A **Corporation** (or Limited Liability Company) is an advanced type of limited partnership in which the business owners have limited liability. A corporation has higher access to capital and expertise. This facilitates its quick growth. It is commonly a preferred business structure for larger companies. A corporation can either be **Nonprofit**, **Public For-Profit**, and **Private For-Profit**.

The factors that determines whether corporation's classification as either private or public. First, it can be classified as private, or public based on its **Issuance of Shares**. For example, public companies issue additional shares in the capital markets to raise huge amounts of capital from investors on the other hand, a private company invite investors to purchase their shares through a private placement memorandum (PPM) (also called offering memorandum). This is a document that describes a business, the terms of the offering, and the risks involved in investing in the company. Private securities are typically unregulated. As such, only accredited investors may be invited to purchase the shares.

Secondly, it can be classified as public companies or private based on its **Exchange Listing** and **Share Ownership Transfer.** For instance, the shares of a public company are listed and traded on an exchange. This allows the owners of a company to be easily transferred since buyers and sellers transact directly with one another in the secondary market. On the other hand, in private companies, shares are not listed on an exchange. For this reason, there is no noticeable valuation or price transparency, making it difficult to buy and sell shares. However, if the owner of the private company wants to sell shares, they will have to find a willing buyer and then agree on the price.

Thirdly, companies can be listed as public or private based on its **Registration** and **Disclosure**. For instance, public companies are obligated to register with a regulatory authority, and they must disclose certain information, such as a stock transaction made by directors. However, private companies are not subject to the same level of regulatory authority as public companies and have no obligation to disclose particular information to the public. Despite that, some pertinent rules such as filing tax returns and prohibitions against fraud are still applicable.

A private company can go public in three ways: **Initial Public Offering (IPO), Direct Listing**, and **Acquisition**. On the other hand, a company can go from public to private when investors (or groups of investors) purchase all the shares of the company and then delist it from the exchange. This can happen through two methods: **Leveraged Buyout (LBO)** or **Managed Buyout (MBO).**

Every business goes through four phases from **Start-up** to **Growth,** to **Maturity,** then to decline. At the **Start-Up Stage**, the company is simply an idea and with a high business risk hence sourcing finances/ capital is difficult. At this stage, friends and families may provide funds through purchasing a piece of the business.

At the **Growth Stage**, more capital is required to run the business as cash flows and revenues may be increasing with moderate business risk. The company cannot count on internally generated income as the company is not yet profitable. To source for more capital, the owners may choose to give up ownership by going public via an IPO if it meets all the requirements or source funds through series B and C capital providers.

At the **Maturity Stage**, the company is profitable with more predictable revenue and thus has the potential to borrow money from private and public market. It also enjoys reduced business risk and minimized need for external funding.

At the **Declining Stage**, business risk is increasing since the company is trying to reinvent itself by developing new lines of business or acquiring companies that are growing significantly. The cost for reinvention might be high because at this point, the company is experiencing diminishing cash flows causing an increase in difficulty of acquiring financing.

# Reading 29 – Introduction to Corporate Governance and ESG Considerations

**Corporate Governance** is the managerial system by which a company is controlled. There are two primary theories that drive corporate governance structures, though they have been converging more closely together in their influence more recently. The **Shareholder Theory** operates under the assumption that the most important responsibility of company management is to maximize returns to shareholders. The **Stakeholder Theory** emphasizes the needs of all relevant stakeholders in making decisions about the company's operations. These additional stakeholders include customers, employees, and even suppliers and creditors.

There are a variety of stakeholder groups connected to a company that will have different interests in the company's results. **Creditors** provide debt financing and are primarily concerned with the company being able to keep up with interest and principal payments. They are less concerned about the growth upsides because they do not benefit beyond debt repayments. **Equity Shareholders** are the owners of the company and stand to benefit from all future growth the company experiences. Shareholders typically have voting rights regarding major company decisions, including membership in the Board of Directors**. Managers** and **Employees** are impacted by both the upside and downside risks of a company. **Customers** are less concerned with the operations of the company itself and more about the value they receive when they purchase goods and services.

Companies must work to manage their relationships with stakeholders. Annual meetings with shareholders are typically held shortly after financial statements are published so that results and concerns can be addressed by company management. Extraordinary shareholder meetings can also be held when there are major resolutions that require a certain threshold of shareholder approval.

The shareholders' interests are represented by the **Board of Directors**, who oversee the company management and are elected by the shareholders. Companies are also subject to audit controls, both by independent internal audit groups and external auditors.

Factors that can affect stakeholder relationships fall into two categories: Market and Non-Market factors. **Market Factors** relate to capital markets and include shareholder engagement through annual meetings, shareholder activism in which large holders try to force changes in the company's strategy to increase shareholder value, and takeovers in which shareholders try to change the company's structure to take advantage of opportunities with other companies. **Non-Market Forces** that can have a major impact include the legal environment in which the company does business, media and the quick spread of news and opinions (especially social media), and the growth of the corporate governance industry itself.

Poor corporate governance can pose significant risks to a company's success. Managers with too little oversight can make short-sighted decisions that benefit themselves at the expense of shareholders. One stakeholder group could benefit unfairly at the expense of another due to a lack of proper controls. The company could be exposed to legal, regulatory, or reputational risks that could include financial penalties and loss of business.

Several factors that have been gaining in importance for companies throughout the world are what is known as ESG, which stands for Environmental, Social, and Governance. ESG analysis includes looking at factors that may not have historically been included in research done about companies but represent items that are important to a broader set of stakeholders of all companies. This analysis is often done by using screeners on these criteria to exclude companies with weak policies or histories on these factors from potential investment decisions.

# Reading 30 – Business Models

There are different features that define a business model. First, a business model should define its target customers. This mainly focuses on the geographies, market segments, and customer segment that a company will serve. Secondly, it should define the product/service offered. Based on its target customers' needs, the company should be able to outline how its products and service is different from its competitors. Thirdly, the business model should sufficiently explain the company's pricing details. The business model should explain the pricing details so that the business logic of a company is understandable.

A company's pricing system should also justify the pricing structure and be relative to its competitors. It's important to note that there are two types of pricing models. **Value-based Pricing Models** are based on the value acquired by the customer. For instance, a car manufacturing company may price cars at a premium because their cars have low operating costs. **Cost-based Pricing Models** are set based on the costs incurred by the firm. For instance, a firm that offers its service on a per-hour basis.

Lastly, a business model should define the channels of selling its offerings such as the jurisdiction or geographical location within which a firm offers its products and how it reaches its customers. It may be direct, through intermediaries, or digitally. There are three different types of channels strategies: traditional, direct and omnichannel strategy.

Price discrimination is a strategy used to maximize revenues based on the customer's will to buy. This occurs when firm charge different prices to different customers. **Dynamic, Tiered** and **Auction** and **Reverse Auction Pricing** are the three main price discrimination. In dynamic pricing, prices are determined based on different buyers, e.g., off-peak pricing in airline tickets, while in a tiered pricing gives different prices to different buyers depending on the volume of goods purchased. In auction and reverse auction models, auction establishes prices either through bidding or sellers during reverse actions.

There pricing models used to price multiple products. includes; (i) **Bundling** where customers are incentivized to combine multiple products (usually complementary goods with high incremental margins and marketing costs), (ii) **Optional Product Pricing** that is applicable when a customer purchases extra services or products on the purchase date (for example, an order in a restaurant) or afterward (for example, a change in subscription package), and (iii) **Razors-And-Blades Pricing** which occurs when a low price is charged on equipment and a high price on a dependent accessory.

The pricing models used to price increasing growth include (i) **Freemium** pricing allows customers to access a certain level of usage or functionality for free, commonly used in digital content and services, (ii) **Penetration** pricing is a form of discount pricing where a firm sacrifices margins with the intention of building scale and market share and (iii) **Hidden Revenue Business** models occur when a firm provides services for free and generates revenue in a different way.

Further, business models can apply techniques such as fractionalization, leasing, recurring revenue/subscription pricing, licensing, and franchising as an alternative form of owning an asset or a product instead of purchasing.

Another crucial characteristic of a business model is that it should outline the **Value Proposition** of a firm. The value proposition of a company describes the effective customer services offered, the sale process, features of the product, and pricing relative to competitors.

A successful business model should also outline a company's business organization and capabilities which is then used to define how the company is structured to deliver the value proposition. It should define its assets and capabilities, which are skilled labor and modern technology required in the firm for implementation of its objectives. The company's value chain should also be well-outlined based on its competitive advantage opportunities, key activities such as marketing, sales, inbound and outbound, etc., conducted by a firm.

A business model should highlight how a firm intends to generate profit. Profit expectations can be analyzed by examining margins, break-even points, and unit economics. Unit economics involves expressing revenues and costs on a per-unit basis.

There are different types of business models. For instance, in the goods-producing industry, firms can be classified according to how they fit into the supply chain such as manufacturers, wholesalers, retailers, and suppliers of raw materials, components, equipment, and services. For the service industry, business models vary; some are **Business-to-Consumers** (B2C), while others are **Business-to-Business** (B2B). In some instances, business models are specific to a certain sector. For example**,** the financial service sector includes business models such as insurance, payment processing, brokerage, merchant trading, asset management, lending, and deposit taking, or a combination of more than one business model.

New business models can be introduced jointly with a new business or introduced to an existing business. For instance, technological innovation has led to the emergence of new business models such as content development, digital advertising, data and related services, and software development. On the other hand, there are **E-Commerce** business models such as marketplace business where a company develops a network of buyers and sellers without owning the products**, Affiliate Marketing** where one earns commission revenues for sales made on a different website of a marketer or agency, and aggregators which has similar features to marketplace however an aggregator must remarket the product and services using its brand.

The **Platform Business Model** which is a business based on a network where value is created from a network outside the firm. On the other hand, in **Crowdsourcing Business Models**, users can contribute directly to the value of a product. Crowdsourcing involves user communities that allow voluntary collaboration among users of a product or users with either low or no regulation. A **Hybrid Business Model** combines both linear and platform business. Other forms of business models include licensing arrangement, franchise models, and private label (contract manufacturers).

A firm's financing needs, and risk profile depend on its business models and other factors. These factors can be classified into **External** and **Firm-Specific Factors**.

External factors include economic conditions, demographic trends, industry cost characteristics, sector demand, socio-political trends, and political, legal, and regulatory environments.

Firm-specific factors include stage of development of the business (firm maturity), competitive position, and type of business model.

Naturally, businesses face various types of business and financial risk. For example, **Macro Risk** originates from political, economic, legal, and other institutional factors in an economy, country, or region. As such, some of the factors that catalyze macro-risk include exchange rates, political instability, and gaps in legal or financial structure.

**Business Risks** refer to the risk that the operating outcomes deviate from expectations independently of the financing method. Business risk encompasses both **Industry Risk** (risks that affects all competitors in an industry) and **Company-Specific Risk**.

Industry specific risks factors include cyclicality, industry structure, competitive intensity, competitive dynamics within the value chain, long-term growth and demand outlook, and impact of other industry risks.

Company-specific risks are founded on the nature, scale, and maturity of a business and they include competitive risk, product market risk, execution risk, capital investment risk, ESG risk, and operating leverage.

**Financial Risk** arises from a firm's capital structure. Primarily, it arises from debt and other liabilities (such as leases) that need fixed contractual payments. Fixed contractual payments cause greater upside and downside variations in net profit and cash flow than operating profit (**Financial Leverage**). Moreover, fixed contractual payments make firms vulnerable to a lack of financing and default risk. Financial risks can be quantified using **Total Leverage**.

Leverage (Total leverage) consists of operating leverage and financial leverage connected as follows:

$$Total\ Leverage = Operating\ leverage \times Financial\ leverage$$

Operating the sensitivity of operating profit (EBIT) to change in revenues:

$$Operating\ leverage = \frac{Contribution}{EBIT}$$

The financial leverage sensitivity of the net profit to a change in operating profit:

$$Financial\ leverage = \frac{EBIT}{EBT}$$

# Reading 31 – Capital Investments

Capital investments are undertaken to either maintain the existing business or/and grow it. There are four main types of capital investments:

- **Going Concern** (or maintenance) projects are projects required to sustain present operations, keep a firm at its current size, or boost business efficiency. Infrastructure improvement is an example of a going concern project.
- **Regulatory** or **Compliance Projects** are projects usually undertaken due to a requirement by a governmental agency, insurance company, or some other external party. They often do not generate revenue for a company. Instead, however, they generate regulatory or compliance expenditures. This can be an obstacle to entry into a market, and an example is factory pollution control installation.
- **Expansion Projects** are projects that increase a firm's size generally come with higher levels of risk and uncertainty than going-concern projects. An example would be a merger and acquisition.
- **Other Projects** are projects beyond a company's traditional business areas include high-risk investments and new growth efforts, e.g., innovation projects.

**Capital Allocation** describes the process companies use to make decisions on capital projects, i.e., projects with a lifespan of one year or more. It is a cost-benefit exercise that seeks to produce results and benefits which are greater than the costs of the capital allocation efforts. Typically, steps involved in capital allocation include (i) generating a good idea, (ii) gathering information, which helps forecast cash flows for each project and its profitability, (iii) capital allocation planning**,** which involves looking at project timing, scheduling, prioritizing, and coordinating and (iv) monitoring and post-audit where project performs is assessed, and actual results (revenues, expenses, cash flows, etc.) are compared with planned or projected results.

There are capital allocation principles which include:

- Capital allocation decisions are based on cash flows rather than accounting earnings.
- The timing of these cash flows is critical.
- Cash flows are always analyzed on an after-tax basis.
- Cash flows are based on opportunity costs.
- Financing costs are ignored because they are reflected in the required rate of return.
- The capital allocation cash flows are not the same as accounting net income or operating income.

Some of the important capital allocation concepts include (i) **Sunk Costs** (costs that have already been incurred), (ii) **Incremental Cash Flow** (cash flow that is realized because of a decision made), (iii) **Externality** (the effect of an investment on other things besides itself), and (iv) **Conventional Cash Flows** (has an initial cash outflow followed by a series of cash inflows) versus **Non-Conventional Cash Flows** (the initial cash outflow is not followed by cash inflows only).

Capital budgeting decisions are impacted by the nature of the projects under analysis. Often, projects may be **Mutually Exclusive,** and you will need to choose only one from a set of choices due to limitations on resources. **Project Sequencing** is another important concept, in that some projects may create opportunities for other projects in future that might not otherwise be available**. Capital Rationing** occurs when there are fixed or limited resources available for investment in potential projects. Many exam questions in this area will ask you to choose the best investment opportunity, given a limited capital budget.

The **Net Present Value (NPV)** and the **Internal Rate of Return (IRR)** are the most important decision criteria are used to evaluate capital investments. The net present value (NPV) of a project is the potential change in wealth resulting from the project after accounting for the time value of money. It is calculated using the formula:

$$NPV = \sum_{t=1}^{n} \frac{CF_t}{(1+r)^t} - Outlay$$

Where:

$CF_t$ = *After-tax cash flow at time t*

$r$ = *Required rate of return for the investment*

$Outlay$ = *Investment cash flow at time zero*

Note that,

- Invest in a project if NPV > 0.
- Do not invest in a project if NPV < 0; and
- Stay indifferent if NPV = 0.

The internal rate of return (IRR) is the discount rate that makes the net present value (NPV) of all cash flows from a particular project equal to zero. For a project with one initial outlay, the IRR is the discount rate that makes the present value of the future after-tax cash flows equal to the investment outlay.

The IRR solves the equation:

$$\sum_{t=1}^{n} \frac{CF_t}{(1+IRR)^t} - Outlay = 0$$

The decision rule for the IRR is:

- To invest in the project if the IRR exceeds the required rate of return for the project, i.e., invest if *IRR* > *r*; and
- Not to invest if *IRR* < *r*.

There are some common capital allocation mistakes and they include;

i. **Inertia**: By comparing the current capital investment to the amount from the previous year and the return on investment, analysts can determine the presence of inertia.

ii. **Sources of Capital Bias**: The capital allocation process should be used for all capital investments, whether made using internally produced cash, debt, or equity, however, some management teams may plan for internally generated cash differently and as if it is "free" from externally raised capital such as equity or debt.

iii. Failing to consider **Investment Alternatives/Alternative States**: the most basic phase in the capital allocation process is the generation of solid investment ideas. In some organizations, however, many good alternatives are never even explored.

iv. Pushing **Pet Projects**: These are projects that influential managers want the company to invest in even though they may not be profitable.

v. Basing investment decisions on **EPS**, **Net Income**, or **ROE**: since managers often have short-term incentives, they may choose projects that do not align to a company's long-term interests.

vi. **Internal Forecasting** errors: companies may make internal forecasting errors that are hard, if not impossible, for external analysts to spot. Consequently, this might lead to unsuccessful investment results.

# Reading 32 – Working Capital & Liquidity

A company's short-term assets and obligations are managed through **Working Capital Management**. Its objective is to prevent excess reserves that might be expensive and, consequently, have a detrimental impact on shareholder returns. At the same time, working capital management ensures that a firm has easy access to the money it requires for day-to-day operations.

$$Working\ capital\ =\ Current\ assets\ -\ Current\ liabilities$$

**Working Capital** is represented by cash, accounts receivable, inventories, and marketable securities (short-term investments). In contrast, **Short-term Financing** is provided by accounts payable, credit lines, short-term loans, and short-term instruments, such as commercial paper, that a firm issues.

Companies can generate internal financing and liquidity from shorter-term operating activities in several ways, including:

- Generating larger after-tax operating cash flow.
- Increasing working capital efficiency, for example, by shortening the asset conversion cycle.
- Converting liquid assets into cash.
- Other forms of internal financing include accounts payable, accounts receivable, inventory, and marketable securities.

Companies can generate external financing through bank and non-bank lenders. Short-term bank financing includes.

- **Uncommitted Bank Lines of Credit**: A bank can provide an uncommitted line of credit for a lengthy period, but it retains the authority to refuse any request to use it. They are the least reliable form of external financing.

- **Committed Bank Lines of Credit**: These are reliable because of a bank's formal commitment. They are in effect for less than a year, ensuring that they are classified as short-term liabilities on financial statements. In addition, they are unsecured, and
- **Revolving Credit Agreements**: They are the most reliable form of short-term bank borrowing. Borrowers can draw down and pay back amounts periodically.

In reference to loans and factoring is another source of external financing. It includes **Asset-Based Loans**; require companies to provide collateral against a loan. Through the assignment of accounts receivables, receivables can be used as collateral for loans, enabling a business to generate cash flows from its accounts receivables. In a factoring deal**,** a business can also sell its accounts receivable to a lender, also known as the **Factor**, for a discount. The factor subsequently assumes the task of collecting the receivables.

Another form of external financing is the capital markets which includes.

- **Short-Term Commercial Paper** is a short-term unsecured loan instrument usually issued by a large company with a good credit rating. They typically range from a few days to 270 days.
- **Long-term Debt** is a debt with a maturity of over one year.
- **Common Equity** represents ownership in a company and is considered a long-term source of funds.

The financial impact of a company's working capital strategy is assessed using the DuPont equation, where:

$$Return\ on\ Equity\ (ROE) = \frac{Net\ income}{Average\ shareholder's\ equity}$$

$$= Net\ income\ margin\ \times\ Total\ asset\ turnover\ \times\ Leverage$$

$$= \frac{Net\ incomes}{Revenues} \times \frac{Revenue}{Average\ total\ assets} \times \frac{Average\ total\ assets}{Average\ shareholder\ equity}$$

**Primary Sources** of liquidity refer to funds that are readily accessible to a company at a relatively low cost. These include:

- Cash available in bank accounts.
- Short-term funds, such as lines of credit and trade credit; and
- Cash flow management.

**Secondary Sources** of liquidity include:

- Negotiating debt contracts to reduce the burdens of high interest, payments, or principal repayment.
- Liquidating assets; and
- Filing for bankruptcy protection and reorganization.

A company's liquidity is measured by the extent to which its current assets, two of the most popular liquidity ratios are the current ratio and the quick ratio:

$$Current\ ratio = \frac{Current\ assets}{Current\ liabilities}$$

$$Quick\ ratio = \frac{\begin{array}{c}Cash\ \ cash\ equivalents + Short\ term\ investments + \\ Account\ receivable\end{array}}{Current\ liabilities}$$

Liquidity ratios assist in measuring the ability of a company to satisfy short-term obligations when they fall due. Comparing a company's liquidity ratios with those of peer companies in the same industry can determine the relative creditworthiness of the company.

**Note:** The greater the value of a company's current ratio, quick ratio, or cash ratio, the greater its level of liquidity.

Common activity rations include:

I. The accounts receivable turnover ratio measures the average number of times accounts receivables are created from credit sales and collected during a year.

$$Account\ Receivables\ Turnover = \frac{Credit\ Sales}{Average\ receivables}$$

II. The inventory turnover ratio measures the average number of times new inventory is obtained and sold during a year.

$$Inventory\ Turnover = \frac{Cost\ of\ goods\ sold}{Average\ inventory}$$

III. The number of days of receivables ratio indicates how well a company manages the extension and collection of credit it gives its customers.

$$\begin{aligned} &Number\ of\ days\ of \\ &receivables\ (Days\ sales\ outstanding) = \frac{Average\ accounts\ receivable}{Average\ day's\ sales\ on\ credit} \\ &\qquad = \frac{Average\ accounts\ receivable}{Sales\ on\ credit\ /\ 365} \end{aligned}$$

IV. The number of days of inventory ratio indicates the average length of time that inventory remains with a company during its financial year.

$$\begin{aligned} &\text{Number of days of inventory} \\ &(\text{Days of inventory outstanding}) = \frac{\text{Average inventory}}{\text{Average day's cost of goods sold}} \\ &= \frac{\text{Average inventory}}{\text{Cost of goods sold} / 365} \end{aligned}$$

V. The number of days of payables is another ratio that tells us how long it takes a company to pay its suppliers.

$$\begin{aligned} &\text{Number of days of payable} \\ &(\text{Days of payable outstanding}) = \frac{\text{Average accounts Payable}}{\text{Average day's Purchases}} \\ &= \frac{\text{Average accounts Payable}}{\text{Purchases} / 365} \end{aligned}$$

$$\text{Cash conversion cycle} = \text{Days of inventory} + \text{Days of receivables} - \text{Days of payables}$$

The objectives of a short-term borrowing strategy include the following:

- Ensuring that there is the capacity to handle sudden cash needs.
- Maintaining sufficient credit sources to fund cash needs.
- Ensuring that the interest rates obtained are competitive.
- Ensuring that a company's effective cost of borrowing takes both hidden and explicit funding charges into account.

There are several factors that will likely influence a company's short-term borrowing strategies including:

- A company's size and creditworthiness
- Legal and regulatory considerations on the amount a company can borrow.
- Asset's nature, in that some assets are considered as attractive as collateral for secured loans.
- The flexibility of borrowing options.

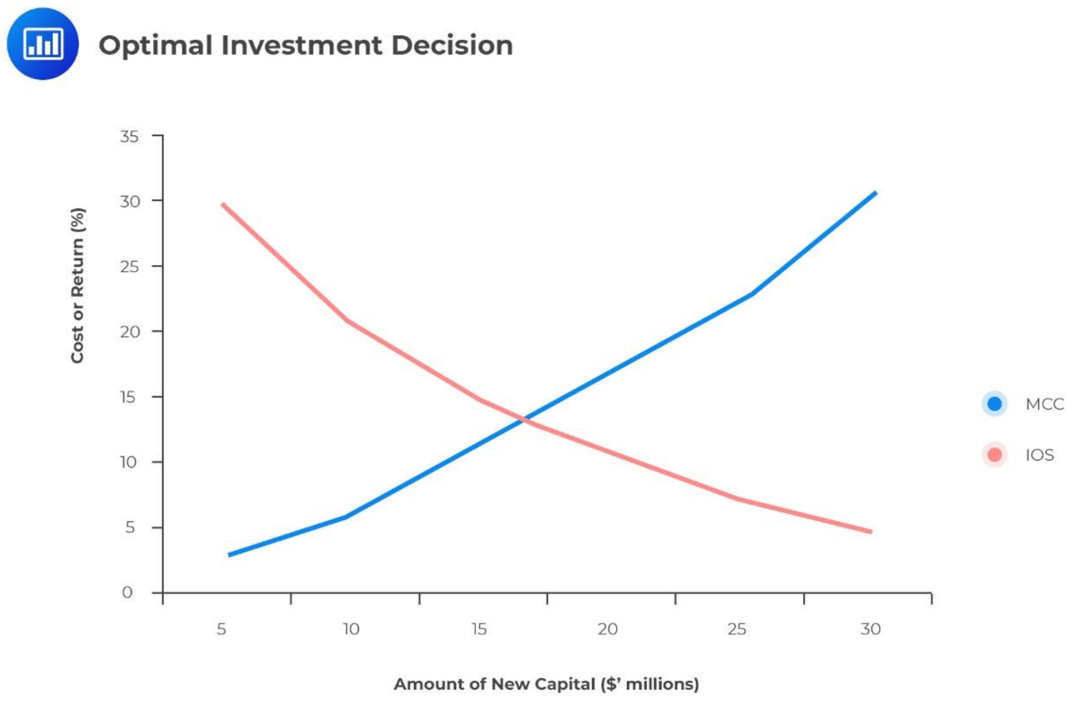
# Reading 33 – Cost of Capital – Foundational Topics

The cost of capital is the price to a company for investment resources. For a company to remain in business and grow over time, it must be able to earn more from its projects and operations than it spends to get that capital. The cost of capital to the firm is not observable but must be estimated using assumptions about the marginal cost of additional funding. Since a company can have many sources of capital funding, the common metric used is the **Weighted Average Cost of Capital**, which incorporates all possible sources of new marginal capital.

$$WACC = w_d r_d (1 - t) + w_p r_p + w_e r_e$$

The three components of WACC are **Debt**, **Preferred Stock**, and **Equity**. The formula combines the proportion and rate of each component to create a summary figure that applies to the firm as a whole. The rate applied to debt is multiplied by 1 minus the marginal tax rate because interest payments on debt are tax-deductible. The proportions of each component are the market value of that component divided by the sum of the market value of all three capital components. They should always sum to 100%.

There is a relationship that exists in determining the optimal amount of capital a company should take on to pursue new investment opportunities. As a company accumulates more capital, it will face increasing costs on each additional capital amount while it will also face decreasing returns on new projects. This relationship is highlighted in the graph below:




Companies can also adjust their cost of capital assumptions to account for higher or lower levels of risk in potential projects.

When trying to calculate the cost of debt financing, there are two primary approaches. The **Yield-to-Maturity Approach** is the yield that equates the present value of the bond's future payments to its market price.

$$P_0 = \left(\sum \frac{PMT_t}{\left(1 + \frac{r_d}{2}\right)^t}\right) + \frac{FV}{\left(1 + \frac{r_d}{2}\right)^n}$$

$P_0 = Current\ market\ price\ of\ bond$

$PMT_t = Interest\ payment\ in\ period\ t$

$r_d = Yield\ to\ maturity$

$n = Number\ of\ periods\ remaining\ to\ maturity$

$FV = Maturity\ value\ of\ the\ bond$

The **Debt-Rating Approach** uses the yield on similar debt securities when accurate market price data for a company's debt is not available. The applicable yield is equal to the stated yield of the debt instrument multiplied by 1 minus the marginal tax rate. It's important to remember that the rate on debt capital is always net of tax.

**Preferred Stock** is the second component that a company could use to raise capital. These shares represent different claims of ownership on the company than common equity. They typically offer higher (and often fixed) dividend rates and usually have no voting rights. The cost of these to the company is simply the dividend yield on the shares:

$$r_p = \frac{D_p}{P_p}$$

Where:

$r_p = Rate\ of\ preferred\ stock$

$D_p = Preferred\ stock\ dividends\ per\ share$

$P_p = Preferred\ stock\ price\ per\ share$

**Common Equity** is the third component of a company's capital base. A company can increase its common equity by reinvesting earnings or issuing new stock. The cost of equity capital is the required rate of return to the shareholders. There are three methods of calculating the cost of equity. The first is the **Capital Asset Pricing Model**:

$$E(R_i) = R_f + \beta\left[E(R_m) - R_f\right]$$

$E(R_i) = Cost\ of\ equity$

$R_f = Risk\ free\ interest\ rate$

$\beta_i = Equity\ beta$

$E(R_m) = Expected\ market\ return$

The second method is the **Bond Yield plus Risk Premium Approach**. This involves adding an equity risk premium to the company's bond yield to estimate the cost of equity. It is calculated as:

$$r_e = r_d + Risk\ premium$$

Where:

$r_e$ = *Cost of equity*

$r_d$ = *Bond yield*

*Risk premium = Compensation which shareholders require for the additional risk of equity compared with debt*

For public companies, beta is estimated through ordinary least squares (OLS) regression of the returns on a stock on the returns on the market. The value of beta obtained through OLS regression is known as raw or unadjusted historical beta which tend to revert to 1.0 overtime. It is adjusted as:

$$Adjusted\ beta = \left(\frac{2}{3}\right)(Unadjusted\ beta) + \left(\frac{1}{3}\right)(1.0)$$

Analysts can indirectly estimate the beta of nonpublic companies using the beta of publicly traded comparable companies through:

**Step 1**: The comparable company's beta is unlevered to reflect the systematic risk. This is known as asset beta, ($\beta_U$).

$$\beta_U = \beta_E \left[\frac{1}{1 + \frac{D}{E}}\right]$$

**Step 2**: The unlevered beta (asset beta) is then re-levered to reflect the financial leverage of the comparable company.

$$\beta_E = \beta_U \left[1 + \frac{D}{E}\right]$$

**Flotation Costs** are expenses that a company incurs during the process of raising additional capital. The value of these flotation costs is related to the amount and type of capital being raised.

Two approaches may be used in response to the question of whether or not a company should incorporate flotation costs into the cost of capital. The first approach incorporates the flotation costs into the cost of capital.

When flotation costs are specified on a per share basis, $F$, the equation below represents the cost of external equity:

$$r_e = \left(\frac{D_1}{P_0 - F}\right) + g$$

When a company specifies flotation costs as a percentage applied against the price per share, the equation below represents the cost of external equity:

$$r_e = \left(\frac{D_1}{P_0(1 - f)}\right) + g$$

where $f$ is the flotation cost as a percentage of the issue price.

The second approach is the recommended approach. This approach adjusts cash flows in the valuation computation. In other words, instead of including flotation costs in the cost of capital, a company should incorporate them into any valuation analysis as additional project costs.

# Reading 34 – Capital Structure

A firm's capital structure is the mix of debt and equity the company uses to finance its investments.

There is a link between a company's life-cycle stage, cash flow characteristics, and its ability to support debt. A company's life-cycle stages include start-up, growth, and maturity.

| **Stage life cycle** | **Start-Up** | **Growth** | **Mature** |
|---|---|---|---|
| Revenue growth | Beginning | Rising | Slowing |
| Cash flow | Negative | Improving | Positive |
| Business risk | High | Medium | Low |
| Debt availability | Very limited | Limited | High |
| Cost of debt | High | Medium | Low |
| Typical debt cases | N/A | Secured | Unsecured |
| Typical % of capital structure | Close to 0% | 0% - 20% | 20% + |

There are unique situations that do not follow the above characteristics. Such as:

- Capital intensive businesses with marketable assets such as real estate business.
- Cyclical industries such as mining industries
- ‘Capital-light’ businesses such as software-based technology enterprises.

Internal factors that influence a corporation’s capital structure include:

- Business model characteristics
- Existing leverage
- Corporate tax rate
- Capital structure policies, guidelines.
- Company life cycle stage

Internal factors that influence a corporation’s capital structure include:

- Market conditions
- Regulatory constraints
- Industry/peer firm leverage

The **Modigliani-Miller Propositions** are a set of theoretical propositions that state that under certain assumptions, the value of a firm is independent of its capital structure. In other words, the propositions suggest that the way a company finances its operations (through debt or equity) does not affect its overall value, given certain conditions.

**Proposition I Without Taxes: Capital Structure Irrelevance**

“The market value of a company is not affected by the capital structure of the firm.”

$$V_L = V_U = \frac{EBIT}{r_{WACC}}$$

Where:

$V_L$ = *Value of a levered firm*

$V_U$ = *Value of unlevered firm*

$r_{WACC}$ = *Overall marginal cost of capital*

In absence of taxes, WACC is unaffected by capital structure:

$$WACC = w_d r_d + w_p r_p + w_e r_e$$

**Proposition II Without Taxes: Higher Financial Leverage Raises the Cost of Equity**

"The cost of equity is a linear function of the company's debt/equity ratio."

The **Cost of Equity** increases linearly as the company increases its use of debt financing.

$$r_e = r_0 + (r_0 - r_d)\left(\frac{D}{E}\right)$$

Where:

$r_e$ = *Marginal cost of equity*

$r_d$ = *Before-tax marginal cost of debt*

$r_0$ = *Cost of capital if there is no debt financing*

**Proposition I with Taxes: Firm Value**

"The market value of a levered company is equal to the value of unlevered company plus the value of the debt tax shield."

Ignoring the practical realities of bankruptcy and financial distress costs, the value of a company increases with increased debt levels. The interest expense on debt provides a tax shield that results in savings that enhance the value of a company**.**

$$V_L = V_U + tD$$
$$\Rightarrow V_L > V_U$$

Where:

$t$ = *Marginal tax rate*

$D$ = *Debt*

$tD$ = *Debt-tax shield*

The implications of MM I with taxes include:

- In presence of taxes, a profitable company can increase its value by employing debt.
- The higher the tax rate, the greater the profit of utilizing debt in the capital structure.

**Proposition II with Taxes: Cost of Capital**

"The cost of equity is a linear function of the company's debt/equity ratio with an adjustment for the tax rate".

Debt financing is highly beneficial when considering taxes and ignoring financial distress and bankruptcy costs. As such, the firm's optimal capital structure is still 100% debt.

$$r_{WACC} = \left(\frac{D}{V}\right) r_d(1-t) + \left(\frac{E}{V}\right) r_e$$

Cost of equity is given by:

$$r_e = r_0 + (r_0 - r_d)(1-t)\frac{D}{E}$$

The implications of MM II with taxes include:

- In presence of taxes, the cost of equity increases with increase in debt but at slower rate than with no tax.
- As company's debt increases, its WACC decreases and its value increases.
- In presence of taxes, with no financial distress or bankruptcy costs, increase in debt increases the company value, such that 100% debt capital structure is optimal.

The expected cost of financial distress is composed of:

i. The costs of financial distress and bankruptcy, in the event they happen.

ii. The probability that financial distress and bankruptcy happen.

**Agency Costs** are the incremental costs arising from conflicts of interest between a principal and an agent. Agency cost of equity refers to the conflict of interest that arises between management and shareholders. These include monitoring costs, bonding costs, and residual loss.

The **Pecking Order Theory** suggests that manager prefer internal funds, then debt and finally equity when choosing a method of financing.

The following factors affect the capital structure and the use of leverage by management:

- Capital structure policies and targets.
- Capital investment financing.
- Market conditions
- Asymmetric information

**Shareholder Theory** posits that the most important responsibility of a company's managers is to maximize shareholder returns. Stakeholder theory emphasizes the need for a company to consider the needs of all its stakeholders and not just its shareholders.

## Reading 35 – Measures of Leverage

Companies face uncertainty around the operational results and the investments that are made in the hopes of leading to growth in the future. **Leverage** is the use of increased fixed costs that are intended to help increase company growth. Companies with high amounts of leverage will experience greater variation in net income and have more volatile earnings and cash flows. There are several types of risk associated with leverage. **Financial Risk** is associated with how the company acquires capital (i.e., debt vs equity) to fund its operations. **Business Risk** is the risk inherent in a company's core operations. **Sales Risk** is the uncertainty associated with the price and quantity of goods and services a company offers for sale. **Operating Risk** is attributed to the part of a company's operating structure related to its use of fixed costs.

There are three important ratios to help quantify types of risk associated with leverage. The **Degree of Operating Leverage (DOL)** measures how sensitive a company's operating income is to changes in sales. The formula highlights the amounts related specifically to fixed costs:

$$DOL = \frac{Q(P-V)}{Q(P-V)-F}$$

Where:

$Q = Units\ sold$

$P = Unit\ price$

$V = Variable\ cost\ per\ unit$

$F = Fixed\ cost\ per\ unit$

The **Degree of Financial Leverage (DFL)** refers to the sensitivity of the operating income of a company to changes in earnings.

$$DFL = \frac{Q(P - V) - F}{Q(P - V) - F - C}$$

The **Degree of Total Leverage (DTL)** combines both previous formulas to provide the sensitivity of net income to changes in units sold.

$$DTL = \frac{Q(P - V)}{Q(P - V) - F - C}$$

Higher degrees of financial leverage can cause a drag on earnings due to higher amounts of financial interest payments. When used properly, the leverage leads to increases in earnings and cash flows when the company is doing well. A company with more leverage is, however, always at an elevated risk of potential default compared to a similar company without financial leverage.

You'll need to be able to calculate the breakeven quantity of sales for a company, given its variable and fixed operating cost structure.

$$PQ = VQ + F + C$$

Where:

$P = Price\ per\ unit$

$Q = Units\ sold$

$V = Variable\ cost\ per\ unit$

$F = Fixed\ cost$

$C = Financing\ costs$

Given this relationship, the breakeven quantity is determined as:

$$Q_{BE} = \frac{F + C}{P - V}$$

$Q_{BE} = Breakeven\ quantity$

The breakeven quantity of sales refers to the point at which company net income is zero, but there is a similar breakeven point for when operating profit is at zero. That formula is:

$$Q_{BE} = \frac{F}{P - V}$$

# Equity Investment

## Reading 36 – Market Organization and Structure

Financial markets serve a wide variety of purposes, but there are six main categories of activity. They are used for **Saving** by individuals and organizations in order to grow their assets over time. **Borrowing** is also done when money is needed now that will be repaid in the future. Companies also **Raise Equity Capital** by selling ownership shares in exchange for cash today. Derivative securities like futures and forwards contracts can be used to **Manage Risk**. **Spot Market Trading** allows for the immediate exchange of one currency or other good for another. **Information Motivated-Trading** is done by investors who intend to use informational advantages to buy undervalued securities.

Assets sold in financial markets fall into several categories. **Debt and Equity Securities** are the bonds and stocks that usually get the most attention from the public. **Currencies** are the different monies issued by governments.

**Contracts** are agreements to trade other assets in the future. **Commodities** are items with non-trading uses, such as metals and agricultural products. There are also **Real Assets** such as real estate, industrial machinery, and other assets with long useful lives.

There are a number of types of markets in which assets are bought and sold. **Derivatives Markets** are where contracts are traded that require delivery of an underlying asset in the future. **Spot Markets** require immediate delivery of goods traded. **Primary Markets** are where new financial assets are created and sold by the issuers. In **Secondary Markets**, financial assets are exchanged between various traders. **Money Markets** involve debt instruments maturing in less than one year, while **Capital Markets** involve assets with greater than one year to maturity.

**Fixed Income** securities are based on the expected repayment of borrowed money along with interest. They are typically classified as short term (less than two years to maturity), intermediate term (2-5 years) and long term (greater than five years). Fixed income investments include notes (maturity under 10 years), bonds (greater than 10 years), convertibles (can be converted into stock in certain conditions), and others.

**Equity** securities represent ownership claims in companies. These can include **Common Stocks**, which typically include a right to dividends and voting on company matters, and **Preferred Stocks**, which usually do not include voting rights but often have specified dividends and have priority over common stock in a liquidation event.

Other types of investment securities include **Pooled Investments**. These involve indirect ownership of assets through shares or units. Investment pools can be either open-ended or close-ended, depending on whether they create and redeem shares for investor cash flows (open) or if there are a fixed number of shares and new investors must purchase them from existing shareholders (closed). Pooled investments include Mutual Funds, Exchange-Traded Funds (shares are traded on secondary markets), and Asset-Backed Securities.

**Contracts** include a variety of derivatives, which are so named because they derive their value from an underlying asset. **Forward Contracts** represent an agreement to trade the underlying asset at some future point at a price agreed upon today. **Futures Contracts** are also an agreement to trade an asset in the future but are more standardized and are traded through clearinghouses that remove most of the counterparty risk. **Swap Contracts** involve the exchange of periodic cash flows that are based on the prices of specified assets or interest rates. **Options Contracts** allow the buyer to buy (call option) or sell (put option) the underlying asset at a specified price by a certain date.

There are also non-financial assets that are traded in investment markets. **Commodities** include items like precious and industrial metals, agricultural products, and even energy production. These are commonly traded in forward and futures markets as a way of managing risks for producers whose businesses are exposed to the risk of price changes in these materials. **Real Assets** are tangible properties usually held by operating companies. They typically have larger holding and transactions costs than other asset types but offer income and tax benefits and usually offer diversification benefits due to their low correlation to other asset types.

Financial Intermediaries are firms that provide products and services to participants in investment markets. **Brokers** fill orders for their clients by matching them with someone willing to take the other side of their trades. **Investment Banks** help corporate clients issue new investment securities. **Exchanges** provide places where traders can arrange their trades. **Dealers** fill client orders by buying and selling directly to them. **Depository Institutions** such as banks and credit unions raise money from depositors to lend to borrowers. **Insurance Companies** sell protection to people and institutions to hedge risks.

Related to investment assets, an investor can take several different positions. Long positions mean the investor has possession (cash securities) or will take delivery of the underlying asset (derivative contracts). A short position means the investor must make delivery or is exposed to the inverse of the price change of the security. For example, if an asset moves from $100 to $150, the long party will earn money while the short party stands to lose from his investment. Investors can also take leveraged positions by borrowing money to fund a long or short purchase. This allows an investor to magnify their potential gains (or losses) by taking much larger positions then they could otherwise.

The overall exposures of a leveraged position can be calculated using the **Leverage Ratio** and **Maximum Leverage Ratio**.

$$Leverage\ ratio = \frac{Total\ value\ of\ position}{Equity\ value\ of\ position}$$

$$Maximum\ leverage\ ratio = \frac{1}{Initial\ Margin\ Requirement}$$

Execution is the manner in which an order is filled. There are a large number of ways that orders can be executed. A common method is the Market Order, where the best price is immediately used. Another is the Limit Order, which is like a market order except a price is specified that the trade must be executed above or below. Brokers sometimes perform Market Taking orders, wherein they take the other side of a market trade by one of their clients.

The two primary categories of markets are primary and secondary markets. In primary markets, new securities are offered to investors for the first time. This can include shares offered to the public for publicly listed companies or smaller, private listings to qualified investors. The secondary markets are where existing securities are bought and sold between investors.

In Quote-Driven markets, customers trade at prices quoted to them by brokers that facilitate those markets. In Order-Driven markets, trades are arranged by rules to match buy orders with sell orders.

A well-functioning financial system has effective financial intermediaries and financial instruments that allow investors to move money between the future and present at a fair rate of return. They allow borrowers to easily obtain capital, hedgers to offset risks, and traders to easily exchange currencies and commodities. Markets typically require proper regulation. The objectives of market regulation are to control fraud, promote fairness, and prevent undercapitalized financial firms from making excessively risky investments that threaten the stability of the market.

# Reading 37 – Security Market Indices

A security market index consists of a given security market segment or asset class. They allow investors to track the performance and risk of these given asset types. They are often used as benchmarks for investment funds or tracked directly by index funds. The value of the **Price Return Index** is calculated by the following formula:

$$V = \frac{\sum nP}{D}$$

$V = Value\ of\ index$

$n = Units\ of\ security\ in\ index$

$P = Price\ of\ security\ in\ index$

$D = Index\ divisor$

There are several steps to constructing a proper index. The first is defining how broad or narrow the target market is. The next is selecting the appropriate securities from within that market. Then the weightings of each security within the index must be determined. The final step is defining how frequently the index will be rebalanced to maintain the correct exposure.

Each method of index weighting has strengths and weaknesses. **Price Weightings** are the easiest to calculate but have arbitrary weightings. **Equal Weightings** provide diversified exposure but require frequent rebalancing to maintain. **Market Capitalization** indices are used commonly by index funds due to less frequent turnover but will cause investors to buy and sell in response to market changes on a lag. Market capitalization weightings also typically have to adjust for the market float to make sure they remain properly investable.

Indices require periodic adjustments to keep their exposures consistent. **Rebalancing** is when the weights of securities in the index are adjusted, while **Reconstitution** occurs when the securities in the index are changed. Reconstitution is necessary when certain securities no longer meet the criteria of the index or when new securities do meet them. A common example is the S&P 500 changing its constituent securities when companies at the margin grow or shrink enough to move into or out of the 500 largest in the US.

Indices serve a number of functions for financial markets. They act as gauges of public sentiment as the public tracks their rise and fall over time. Their performance is a useful input in models that use market return and risk, such as the capital asset pricing model or Beta models. Indices act as proxies for the performance of specific asset types or asset classes. They make useful benchmarks for investment managers. They are also useful as model portfolios for passive funds.

Common types of equity index include broad market, multi-market, sector, and style. **Broad Market** means the index includes more than 90% of a selected market. **Multi-market** comprises indices from different countries.

**Sector** represents different economic sectors and **Style** represents groups of securities classified by characteristics like size, value, or growth.

Fixed income indices also come in various types. They can be classified by the issuer's economic sector, geographic region, currency of payments, or the maturity, credit quality, or other characteristics of the issuance. Fixed income indices can also include specialized categories like high-yield, inflation-linked, and emerging markets.

In addition, indices exist for alternative investments. Commodity indices consist of futures contracts for one or more commodities. These can vary widely since there is no clear best way to weight an index of futures. There are also real estate indices that can be categorized as appraisal, repeat sales, or Real Estate Investment Trusts (REITs).

## Reading 38 – Market Efficiency

Market efficiency is the measure of how new information becomes reflected in market prices. In a completely efficient market, security prices will immediately incorporate all new information and it is nearly impossible to create sustained outperformance. The less efficient a market is, the easier it is for active managers to outperform the market. One indicator of market efficiency is how well market prices match up to their intrinsic values. The market price of a security is the price at which it can be bought or sold, while the intrinsic value is the value that would be placed on that security by investors if they had a complete understanding of that asset's investment characteristics.

Several characteristics have a big impact on the level of efficiency in a market. The number and sophistication of market participants is one. As these go up, the market becomes more efficient. The availability of information is also a big determinant. When information is limited, there will be more inefficiencies in a given market. Limits on trading activity also decrease market efficiency. Activities like short selling make it easier for investors to act on relevant information, which is how that information becomes reflected in market prices.

According to the framework laid out by Eugene Fama, there are three forms of market efficiency. In the **Weak Form**, the market prices of securities reflect all historical information. Therefore, technical analysis cannot earn abnormal returns. In the **Semi-Strong Form**, security prices reflect all publicly known information, so an investor cannot earn abnormal returns by analyzing public financial disclosures. The **Strong Form** posits that security prices reflect all public and private information. Research has shown that markets are generally not strong-form efficient, as investors have been able to earn abnormal returns through the application of private information. The table below summarizes what methods can be expected to earn abnormal returns in the different forms of market efficiency.

**Market Prices Reflect**

| **Forms of Market Efficiency** | **Past Market Data** | **Public Information** | **Private Information** |
|---|---|---|---|
| Weak form | ✓ | | |
| Semi-strong form | ✓ | ✓ | |
| Strong form | ✓ | ✓ | ✓ |

Market anomalies are exceptions to the idea of market efficiency. These are persistent occurrences that cannot be linked to relevant information in the markets. They exist in forms such as time series anomalies like the January effect, in which stocks tend to outperform in the month of January. There are also cross-section anomalies such as the size effect, where smaller companies outperform larger companies. Other anomalies include the excess returns that are routinely earned by investors that are able to purchase a stock at its IPO price.

There are behavioral finance concepts that influence our understanding of market efficiency. Traditional finance presumes rationality of market participants, but, generally, markets can behave rationally in aggregate even if the participants do not always do so. Investors tend to exhibit behavioral characteristics like **Loss Aversion Bias**, wherein they dislike losses more than comparable gains. The **Herding Bias** is the tendency of investors to trade along with others, which can explain some under- or overreaction behaviors to the news. The **Overconfidence Bias** leads investors to overestimate their own ability to accurately determine intrinsic values. These and other biases are used to explain behaviors by market participants that would otherwise appear to be irrational.

# Reading 39 – Overview of Equity Securities

Equity securities provide investors with a claim on a company's net assets and ownership rights, such as voting on company matters. Common shares represent an ownership interest in the company. Common shareowners are eligible to vote on Board membership and other major company issues. Preferred shareowners receive preference in dividend distributions and in claims on company assets, but typically do not also have voting rights. Companies often issue multiple classes of share within these categories that have different levels of voting and ownership rights.

Shares of companies can be either public or private, depending on whether they are available to the general public. The public stock market is much larger than private companies, but the number of private equity companies has been increasing in recent years. In exchange for being more easily available for sale, public companies are subject to more stringent reporting requirements. While public company stock securities are more liquid and easier to purchase in the secondary market, private equities are typically purchased in negotiated sales with existing investors.

Technological and communications advancements have made it easier for investors to purchase equities in markets around the world. There are several options for investors that want to invest in equity markets outside of their own country. One option is **Direct Investment**, in which the investors directly purchase securities in the foreign markets. They can also purchase **Depository Receipts**, which are securities that trade in a local exchange but represent ownership in a foreign company. There are also **Global Registered Shares** that are common shares that are traded on different stock exchanges around the world in different currencies.

There are two sources of returns to equity securities: capital appreciation and dividend income. Generally, preferred share investors will expect more of their total return to come from the dividend income on their shares over time.

$$R = \frac{P_t - P_{t-1} + D_t}{P_{t-1}}$$

$R_t = Total\ return$

$P = Sales\ price$

$P_{t-1} = Purchase\ price$

$D_t = Dividend\ income$

The goal of issuing equity securities is to raise new capital for investment in company operations. Companies use this capital to maximize long-term shareholder wealth by growing the total company's value and returning wealth through dividend distributions.

The equity value of a company can be measured by either the book value or market value. The **Book Value of Equity** reflects the historical operating and financing decisions of its management, while the **Market Value** reflects investors' collective assessment and expectations about the company's future cash flow from operations. If a company has a high price-to-book ratio (meaning its market price is high relative to its book value), then the market has high growth expectations for the company.

A company's cost of equity is the minimum expected rate of return that it must offer its investors to purchase shares in the primary market and hold them in the secondary market. The return on equity is the measure by which investors determine whether management is effectively using their equity capital to generate profits. Return on equity is calculated by dividing the company's net income by its average book value over the time period being analyzed.

$$ROE = \frac{NI}{\frac{(BVE_t + BVE_{t-1})}{2}}$$

# Reading 40 – Introduction to Industry and Company Analysis

Industry analysis is used to understand a company's business and its business environment. Analysts use it to identify potential active equity investments and for measuring how sector selection impacted the performance of an active strategy. Industries are a classification that groups companies by common characteristics. Industries can be categorized by the nature of the products or services they provide, by how cyclical their operations are, or by statistical similarities of past returns.

There are several standard industry classification systems. The **Global Industry Classification Standard** was developed by Standard & Poor's and MSCI Barra and assigns companies to one of 154 sub-industries based on principle business activity. The **Russel Global Sectors** puts companies into one of 141 industries based on the products or services a company provides. The **Industry Classification Benchmark** developed by Dow Jones and FTSE places companies into one of 114 subsectors based on their primary source of revenue. Each classification system sorts their sectors and subsectors into broader sectors and industries that combine for higher-level attributes.

Companies can be sorted by how sensitive their operations are to economic and business cycles. Cyclical companies will experience greater volatility in earnings and growth over time. A limitation of cyclicality as a classification is that the sensitivity to cycles is a spectrum rather than a discrete difference. Groupings by statistical similarities can also have the same problem of drawing arbitrary distinctions between companies that might otherwise be very similar. It is up to the analyst to understand the important differences between companies in different industries and how that will impact their inclusion in an investment portfolio.

A thorough industry analysis must include identification of an industry's life cycle, specific important segments within that industry, and the range of economic forces that have the biggest impacts on operations for companies in that industry.

An important framework for strategic analysis of companies and industries is **Michael Porter's Five Forces:**

1. Threat of Entry
2. Power of Suppliers
3. Power of Buyers

4. Threat of Substitutes
5. Rivalry Among Existing Competitors

Factors that influence industry profitability include barriers to entry. High barriers that prevent new entrants from the market keep profit margins higher for existing firms. Lower barriers to entry lead to more competition and make it difficult for firms to maintain market share. Another important factor is the level of concentration in the industry. Highly concentrated industries tend to have more pricing power and can sometimes coordinate with their competition to keep out entrants and maintain higher prices.

There are five stages to the life cycle of an industry. The **Embryonic** stage is characterized by slow growth, high prices, and low sales volumes.

The **Growth** stage is characterized by increasing demand, improving profitability, and relatively low competition. The **Shakeout** stage features slowing growth, increasing competition, and declining profitability. **Mature** industries have little growth, consolidation among firms, and high barriers to entry. **Decline** involves negative growth, excess production capacity, and intense competition.

External factors can have significant impacts on industry growth. Macroeconomic trends, both cyclical and structural, can speed or slow industry growth and profitability by impacting costs of money and consumer behaviors. Technology can also influence growth trends by radically changing industries or introducing entirely new business models. Demographics also change how industries perform through changing spending habits.

A thorough company analysis must include an overview of the company and its operations, identification of the characteristics and trends of its industry, analysis of demand and supply for its products or services, explanation of the pricing environment the company faces, and interpretation of all relevant financial ratios.

# Reading 41 – Equity Valuation: Concepts and Basic Tools

An equity security is considered to be fairly valued if its current market price is approximately equal to its value estimate. While there are many uncertainties associated with calculating an estimated valuation, they can provide investment insight when compared to how a company is valued by the market.

Companies are considered over- or under-valued when their market value is higher or lower than their estimated value.

There are three categories of models used to calculate equity security values: Discounted Cash Flow Models, Market Multiple Models, and Asset-Based Valuation Models.

**Discounted Cash Flow Models** are based on finding the present value of the expected future cash flows to the investor. One example is the Dividend Discount Model, which finds the present value of futures dividends and the expected eventual selling price of the stock. This model is especially useful for mature companies but is less applicable for complex companies in uncertain economic environments.

$$V = \sum \frac{D_t}{(1+r)^t} + \frac{P_n}{(1+r)^n}$$

$V = Present\ value\ of\ stock\ today$

$D_t = Dividend\ at\ time\ t$

$P_n = Expected\ selling\ price\ at\ time\ n$

$r = Required\ rate\ of\ return$

Another discounted cash flow model commonly used is the **Free Cash Flow to Equity** model. This finds the present value of future cash flows that will be available for distribution as dividends. It can be helpful in situations where dividends are not proscribed or especially volatile. It is useful because it does not rely only on the dividend policy of the firm, but more assumptions must be made compared to some other models. It is calculated the same as any other present value, but you need to first find the Free Cash Flow to Equity (FCFE):

$$CFO = NI + NCE - WC$$

$$FCFE = CFO - FCInv + NB$$

Where:

$NI = Net\ income$

$NCE = Non\ cash\ expenses$

$WC = Working\ capital$

$FCInv = Fixed\ capital\ investment$

$NB = Net\ borrowing$

Preferred stock can be valued using a similar method since their dividends are often specified and more consistent over time. You can use the same **Dividend Discount Model** as with common shares, but you use the par value (F) of the shares in place of the expected future price ($P_t$):

$$V = \sum \frac{D_t}{(1+r)^t} + \frac{F}{(1+r)^n}$$

When using Dividend Discount Models, it is important to be able to forecast the expected future growth of dividends. One approach is the **Gordon Constant Growth** model, which assumes a constant and consistent growth rate for dividends into perpetuity:

$$V = \sum \frac{D_1}{r-g}$$

Where:

$D_1 = Expected\ dividends\ in\ year\ 1$

$r = Required\ rate\ of\ return$

$g = Growth\ rate g = b \times ROE$

$b = Earnings\ retention\ rate (1 - Dividend\ payour\ ratio)$

$ROE = Return\ on\ equity$

There are also models that incorporate multiple growth rates, which is helpful for less mature companies that will grow faster in the short term before settling into a long-term sustainable growth rate in the future. They are similar to the single rate models except that a different growth rate $(g_s)$ is used to calculate the short-term dividend growth than the one used $(g_l)$ to calculate the long-run sustainable growth.

$$V = \sum \frac{D_0(1+g_s)^1}{(1+r)^t} + \frac{V_n}{r-gl}$$

$$V_n = \frac{D_{n+1}}{r - gl}$$

$$D_{n+1} = D_0(1 + g_s)^n(1 + gl)$$

**Market Multiple Models** are another approach to determining the value of a company. They use information from how the market is valuing the company and accounting information to develop a calculated value. Since these models incorporate comparable companies into calculating their multiples, it can be especially helpful when trying to value a company with limited or negative earnings. One such model is the justified price to earnings multiple.

$$\frac{P_0}{E_1} = \frac{p}{r - g} = Justified\ forward\ \ P/E$$

Where:

$p = Payout\ ratio$

$r = Required\ rate\ of\ return$

$g = Dividend\ growth\ rate$

There are a number of different multiples models that can be used for company valuation purposes. The **Price-to-Earnings Ratio** (P/E) is the most commonly referenced and is the stock price divided by the earnings per share. Other ratios often used include the price to book ratio, the price to sales ratio, and the price to cash flow ratio.

A slightly different approach to multiples valuation that doesn't rely solely on the market price of the security is to use the **Enterprise Value** (EV). This is the value of the company as it relates to what it would cost in a takeover. It is the total market value of all outstanding stock, plus the market value of all debt, and minus the value of all cash and cash equivalent securities. One of the most often used formulas involving enterprise value is the EV/EBITDA (earnings before interest, taxes, depreciation, and amortization) ratio.

The third category of valuation model is the **Asset Based Valuation Model**. These are based on the market or fair value of a company's assets and liabilities. These models are most applicable to companies with high proportions of current (and marketable) assets and few intangibles. Intangible assets can be difficult to value, and their book or holdings values may not be relevant to the market value for an investor. These models are the simplest to calculate and do not require projection assumptions. They also don’t take into account expected cash flows, earnings, or growth rates and so often aren’t the best for valuing a going concern (unless used in conjunction with other approaches).

# Fixed Income

## Reading 42 – Fixed-Income Securities: Defining Elements

A bond is a debt instrument that entitles the buyer to future cash flows from the issuing entity. Bonds can be issued by a variety of organizations, including companies, governments, and even supranational groups like the European Union. Bonds are categorized into three primary sectors based on issuer and structure: Government, Corporate, and Structured Finance. Bonds are also categorized by the level of credit risk they have. Higher-quality bonds are known as **Investment Grade** and have lower yields, while riskier bonds (more likely to default) are known as **High-Yield** (sometimes referred to as "junk bonds").

A traditional bond is a promise to pay back a borrowed amount (the "par" value of the bond) plus interest over a specified time period. This typically involves periodic payments of the interest and then a final repayment of the original principal amount at the maturity date.

The legal document that specifies all the parameters of a particular bond is the **Indenture**. It will describe all the important information such as the issuer, paramount, coupon rate and frequency, and any covenants. The covenants are rules specific to that bond issue that the borrowers and lenders agree to at the time of the bond issuance. The indenture will also outline whether the bond is secured or unsecured and what assets are backing the bond if it is secured. Bonds can also contain credit enhancements to help decrease the credit risk, such as insurance against default, overcollateralization, or even reserve accounts that can be drawn against to make payments.

Bonds are subject to numerous legal and regulatory requirements, and one categorization that determines how they are regulated is the domicile of the issuer in relation to where the issue takes place. If an entity issues a bond in their own country, it is considered a **Domestic Bond**. If an entity issues a bond in a different country, it is considered a **Foreign Bond**.

The most common structure for a bond involves periodic fixed interest payments and a lump sum of the principal at the maturity date. These periodic interest payments are known as coupon payments. There are also bonds that are amortizing, which means that each periodic payment includes both interest and repayment of part of the principal amount. The coupon rates for bonds can be either fixed or floating, which means that the interest payments will either remain consistent over time or change in relation to a specified benchmark rate.

Bonds can include contingencies that provide additional rights to the issuer and/or the bondholders. One of these is a call option. **Callable Bonds** give the issuer the right to redeem (call) the bond by repaying the principal amount before the bond reaches maturity. This is useful for issuers when interest rates drop, and they are able to repay an older bond with higher interest with funds from a new issue at the lower rates. **Puttable Bonds** give the bondholders the right to sell (put) the bond back to the issuer. Bondholders may want to exercise this option if rates have increased, and they want to sell that bond and purchase newer bonds with higher yields. **Convertible Bonds** are a hybrid of debt and equity securities. These give the right to convert the bonds into equity shares in the company in certain situations.

## Reading 43 – Fixed-Income Markets: Issuance, Trading and Funding

As mentioned in the previous section, floating rates bonds differ from their fixed rate counterparts in that the coupon rate changes over time-based on movements of a specified reference rate. The most commonly used reference rate is the London Interbank Offered Rate (LIBOR).

Similar to equity securities, there are both primary markets (where new bonds are sold to investors for the first time) and secondary markets (where existing bonds are bought and sold between investors) for bond securities. There are a number of mechanisms by which new bonds are issued in primary markets. In an **Underwritten Offering**, an investment bank guarantees the sale of new bonds at a negotiated offering price. The investment bank takes on the risk of this price guarantee because they are responsible for buying any remaining bonds that they are unable to sell at that price. For larger bond offerings, multiple underwriters may work together in what is known as a **Syndicated Offering**. Some issuers use the **Shelf Registration** approach, wherein they get approval for a new offering that may not take place right away. They retain the ability to issue new bonds as part of that offering in the near future without going through a new registration.

Secondary trading for bonds can take place either on an exchange or over the counter (OTC). On an exchange, trade orders are submitted to the exchange and matched with other orders based on exchange criteria, similar to what one sees on an electronic stock exchange. In OTC trading, buyers and sellers contact each other directly to negotiate trades. The vast majority of bond trading is done via OTC.

The difference between the bid and ask quotes in the market for a specific bond (the bid-ask spread) is an indicator of the liquidity for that security. The larger the spread, the less liquid that bond likely is. Compared to equity securities, individual bonds trade much less frequently and tend to have commensurately larger spreads. When bonds do trade, they typically settle on either a T+1 basis (for government bonds) or a T+2 or T+3 basis (for corporate bonds).

US government bonds are denoted differently depending on their time to maturity. Issues made with less than 1 year to original maturity are known as **T-bills**, while **T-notes** and **T-bonds** have longer maturities. T-bills are issued as discount notes, meaning that they are issued at a price less than par. T-notes and T-bonds are more typical bonds that are issued at par.

In addition to traditional corporate or government bonds, there are also several categories of bonds issued in conjunction with governments but are not considered government bonds. Non-sovereign bonds are those issued to help fund public projects but are not guaranteed by national governments. They typically have low default rates but have higher yields than guaranteed government bonds. There are also quasi-government bonds that are also not guaranteed but often have an implied government backing. Examples of issuers of these bonds are the semi-government organizations Fannie Mae and Freddie Mac. There are even bonds issued by agencies that exist in multiple countries which are known as supranational bonds. Organizations like the IMF and European Union issue bonds outside of any individual nation.

Companies issue debt as a way to get funding for investments in their operations. When companies need short-term funding, they can issue **Commercial Paper**, which consists of debt with maturities generally less than 3 months and as short as overnight. The commercial paper market is extremely liquid, as companies frequently pay off loans with funding from new short-term issuances (this is known as “rolling” the debt).

Banks have additional sources of short-term funding available to them, such as **Interbank Lending** and **Central Banks**, where they can borrow or loan money with other banks for terms of less than 1 year. One source of funding for banks are **Certificates of Deposit**, wherein they offer higher rates to depositors in exchange for having a fixed period before the deposits can be withdrawn.

A special type of lending facility is the **Repurchase Agreement**, also known as a 'repo.' These consist of the sale of a security (usually financial market securities) and an agreement to buy back the security at a specified future date at an agreed-upon price. These are naturally collateralized because the asset involved in the repo agreement serves as collateral for the loan.

# Reading 44 – Introduction to Fixed-Income Valuation

Similar to equity securities, the price of a bond should be the present value of the expected future cash flows. Unlike equity securities, these cash flows tend to be finite and known in advance. Especially for plain vanilla coupon bonds, the buyer will know the exact amounts and dates of the cash flows they will receive. This means that it is relatively straightforward to calculate the present value of those cash flows, discounted using the current market interest rate.

The general formulae for calculating the present value of a bond is as shown below:

$$PV_{bond} = \frac{PMT}{(1+i)^1} + \frac{PMT}{(1+i)^2} + \cdots + \frac{PMT}{(1+i)^n}$$

$PMT \ = \ Coupon\ payment\ per\ period$

$FV \ = \ Face\ value\ of\ the\ bond\ at\ maturity$

$i = \ Market\ discount\ rate$

For example, the calculation of the price for a 5-year bond that pays a 5% coupon annually when the market rate is 6% would be as follows:

| **Time period** | **1** | **2** | **3** | **4** | **5** | |
|---|---|---|---|---|---|---|
| Calculation | $\frac{5}{(1+6\%)^1}$ | $\frac{5}{(1+6\%)^2}$ | $\frac{5}{(1+6\%)^3}$ | $\frac{5}{(1+6\%)^{41}}$ | $\frac{100+5}{(1+6\%)^5}$ | |
| Result | \$4.717 | \$4.450 | \$4.198 | \$3.960 | \$78.462 | =\$95.788 |

This bond would be trading at less than its par value of 100 (known as a discount) because its current rate is less than what's available in the market. If the bond's coupon rate were higher than the market rate, we would expect it to be trading at a premium to its par value.

Since the market rate serves as the discount rate, we expect the price of a bond to decrease as rates increase, since it means we are more heavily discounting the cash flows. Intuitively, this should also make sense because a bond whose coupon rate is below the new market rate should be worth less than it was before since the investor could now sell that bond and buy another one with a higher coupon.

You will likely be asked to calculate the **Yield to Maturity** for a bond security, which is the internal rate of return that sets the present value of the future cash flows equal to the current price. You will use the time value of money functionality on your calculator to solve for this value the same way that you solved for IRR values in the Quantitative Methods section. The only change you may see that might not have occurred in that section is to change the "P/Y" value, because most common bonds make two interest payments per year. You change this value on your calculator the way you change any other time value, by pressing 2 and then the P/Y button. Be sure to keep track of what you have this value set to because you will definitely see questions involving bonds with varying numbers of payments per year. The curriculum refers to this annual frequency of payments as the Periodicity.

It is also important to note that the price of a bond moves closer to the par value as the maturity of the bond nears. Also, the prices of longer-term bonds vary more than those of shorter-term bonds.

In addition to finding the Yield to Maturity for a given bond, you can also find its current price using spot rates. This is a similar calculation except that you will use different discount rates for each cash flow that you are discounting. Rather than using the time value of money functionality on your calculator for these problems, you'll manually calculate the present value of each cash flow using the spot rate given for that time period, as per the below formula:

$$PV_{bond} = \frac{PMT}{(1+Z_1)^1} + \frac{PMT}{(1+Z_2)^2} + \cdots + \frac{PMT + Principal}{(1+Z_n)^n}$$

Bond prices can be quoted in two different ways due to the nature of their interest accruals. The **Flat Price** of the bond is calculated the way we've covered up until now.

The **Full Price** (also called the "dirty price") includes the Flat Price and the interest that has accrued since the last interest payment, $PV^{full} = PV^{flat} + Accrued\ Interest$. If given the market discount rate per period (r), the full price can be calculated using the formulae: $PV^{full} = PV \times (1 + r)^{t/T}$, where t/T is the proportion of the coupon period that has passed since the last payment.

The **Accrued Interest (AI)** is the proportional share of the upcoming interest payment. This is included in the **Full Price** so that the investor selling the bond doesn't lose out on the amount of interest they would have received for the part of the accrual period in which they owned the bond. The accrued interest is calculated by taking the next coupon payment (PMT) and multiplying it by the percentage of the coupon period that has passed since the last payment (t/T), i.e.:

$$Accrued\ \mathrm{I}nterest = \frac{t}{T} \times PMT$$

When calculating this elapsed time, it's important to know the day-count methodology being used. There are 2 primary methods used for this: actual-actual and 30/360. In the 30/360 method, each month is assumed to have 30 days and a year is assumed to have 360 days. In the actual-actual method, the exact number of calendar days that are in each month and year are used.

As mentioned earlier, bonds do not trade as frequently as equities, so it is more often necessary to calculate the correct current price for them. One method is to estimate the market discount rate and price based on the quoted or traded prices of similar securities that have recently traded. This method is known as **Matrix Pricing**. It utilizes the market prices of securities with similar maturities, credit quality, and other characteristics to calculate a price for a given security that may not have traded. Since matrix pricing using many securities is computationally intensive, you will likely only be asked to use a few securities in a question on this topic. For example, with quantitative factors like maturity, you can find the appropriate yield to maturity for a bond by finding the average of the yields to maturity for securities with a longer and shorter maturity.

In order to compare the yields of bonds with different payment frequencies, bond yields are typically annualized. This is the same calculation principle from the Quantitative Methods section. When you use the time value of money functionality on your calculator to find the yield to maturity, that value is already annualized. You will likely be asked to compare the yields for bonds that have similar characteristics but different periodicities. The periodicity of a rate of r% compounded semi-annually is 2. To get the annual rate for a periodicity of 2, we take r%×2. The periodicity of a rate of r% compounded quarterly is 4. To get the annual rate for a periodicity of 4, we take r%×4, and so on and so forth. To convert rates between different periodicities, we use the general formula:

$$\left(1+\frac{APR_m}{m}\right)^m = \left(1+\frac{APR_n}{n}\right)^n$$

$Where$ $m$ and $n$ are the different periods within a year.

When valuing bonds, it is important to consider that bonds can either be **Fixed-Rate Bonds** (pay the same amount of interest for a specified term) or **Floating-Rate Bonds** (interests vary depending on the level of the reference interest rate).

Fixed rate bonds can be valued using **Street Convention** (assumes that payments are made on scheduled dates excluding weekends and holidays), **True-Yield** (IRR calculated using a calendar and includes weekends and holidays), **Current Yield** (calculated as the sum of the bond's coupon payments over one year divided by the flat price of the bond) or **Simple Yield** (calculated as the sum of the coupon payments and straight-line proportion of the gain or loss, divided by the flat price of the bond.)

For money markets instruments, bond yields to maturity are annualized but not compounded, and are quoted using non-standard rates (discount rates and add on rates).

When using the discount rates to price money market instruments.

$$PV = FV \times (1 - \frac{\text{Number of days between settlement and maturity}}{\text{Number of days in a year}} \times \text{Annualized discount rate})$$

To calculate the discount rate,

$$\text{Annualized discount rate} = \left(\frac{\text{Number of days in the year}}{\text{Number of days between settlement and maturity}}\right) \times \left(\frac{FV - PV}{PV}\right)$$

When using add on rates,

$$PV = \frac{FV}{\left(1 + \frac{\text{Number of days between settlement and maturity}}{\text{Number of days in the year}} \times \text{Annualizedaddonrate}\right)}$$

To calculate the Annualized add on rate,

$$AOR = \left(\frac{\text{Number of days in the year}}{\text{Number of days between settlement and maturity}}\right) \times \left(\frac{FV - PV}{PV}\right)$$

To compare the yields of money market instruments on the same basis, we have to use the bond equivalent yield. The bond equivalent yield is quoted on a 365-day add-on rate basis and is obtained by converting one rate to another.

A graph of all of the yields for a type of security along the range of maturities is known as a **Yield Curve**. Yield curves are upward sloping because longer-maturity bonds have higher yields, but the slope flattens out at longer maturities because the yield differences become much smaller for each additional time to maturity at the long end. This version of the yield curve is known as the **Spot Curve** because it shows the spot rate for each maturity period. There is also a **Par Curve** which is the same except it shows the yield-to-maturity for bonds priced at par at each point in the maturity curve. Par curves are calculated based on spot curves and represent the difference in yields due to changes in maturity but holding every other bond characteristic the same.

Bond rates can be quoted in terms of spot rates or forward rates, depending on whether the bond funding is being provided right now or at some point in the future. A spot rate applies when the bond is to be issued now and the rate will apply immediately, while the forward rate is used when the bond will be issued in the future. Implied forward rates are calculated based on spot rates. The implied forward rate is the rate that sets the current spot rate and the spot rate for the period covering both the current spot and forward periods equal. For example, the implied forward rate for a 3-year bond to be issued 2 years in the future would be calculated as follows:

$$(1 + r_2)^2 \times \left(1 + IFR_{2,3}\right)^3 = (1 + r_5)^5$$

Where:

$r_2 = \textit{2-year spot rate}$

$r_5 = \textit{5-year spot rate}$

$IFR_{2,3} = \textit{Implied forward rate}$

You can also create a curve of these forward rates extending along the maturity dates and this would be known as the **Forward Curve**.

Yield spread is the difference between any two bond issues. There are 4 types of spreads:

i. **G-Spread** is the difference between the yield on the corporate bond and the yield on a government bond.

ii. **I-Spread** is the difference between the yield on a bond and the swap rate.

iii. **Z-Spread** is the constant spread that makes the price of a security equal to the present value of its cash flows when added to the yield at each point on the spot rate Treasury curve.

iv. **Option-Adjusted Spread** (OAS) is the difference between the Z-spread and the option value.

# Reading 45 – Introduction to Asset-Backed Securities

**Asset-Backed Fixed Income Securities** (ABS) are different than the bonds already covered here because, rather than being based on an amount lent to an issuer that they pay back over time, they are based on pools of assets that are securitized and combined into one fixed income security. The pool of assets from which the security's cash flows derive is known as its collateral. These are very common for mortgages and other consumer lending products since it allows these (relatively) smaller loan amounts to be bundled together to create fixed income assets large enough to be purchased by institutional investors.

The pools of assets that underly an asset-backed security are held in legal ownership by an entity created just for that purpose rather than owned by the issuer. These entities (known as **Special Purpose Entities**, or SPEs) control the assets, receive the cash flows from the individual borrowers (for example, mortgagees), and distribute these cash flows to the bondholders. The SPEs can sometimes use another organization as the servicer, in which case that servicer is the one that receives and distributes the cash flows even though the SPE maintains the ownership of the assets.

ABS can be divided up into multiple classes of bonds based on characteristics like credit and default risk. The more subordinated (lower in rank) a class is, the higher the coupon payments but also the higher the default risk. These classes typically function in a **Waterfall Structure**, where the cash flow distributions are made in order of class, starting with the most senior, and the default losses are assessed in the reverse order. Investors can choose the level of risk and yield they wish to expose themselves to by purchasing at different class levels.

One common source of assets for ABS bonds are mortgage loans. These are categorized as either prime or sub-prime depending on how creditworthy the borrowers and how risky the loans are. An important risk to consider in mortgage ABS is that of prepayment risk. Since many mortgage borrowers are able to pay their mortgages off early or refinance when rates go down, there is an increased risk that the ABS will lose assets early and the investor will have to reinvest that money into newer, lower yielding securities. A potentially more costly risk regarding mortgages is borrowers defaulting and the properties being foreclosed upon. The ABS investor is entitled to proceeds from foreclosure sales but there is often a shortfall between those proceeds and the original value of the mortgage loan.

Mortgages are categorized by whether they are agency or non-agency. **Agency Mortgages** are those guaranteed by a federal agency or a government-sponsored entity like Fannie Mae. **Non-agency Mortgages** are those that do not have any of these guarantees and are issued by private entities, so these mortgages generally carry higher yields and represent a higher credit risk to the investor.

Since the rates of prepayment for mortgage ABS (MBS) will impact how long it takes for the security to mature (because the security matures when all underlying mortgages are paid off), there are inherent risks that don't apply to other bond securities. If rates go down, there is a risk of **Maturity Contraction** because mortgage borrowers can refinance their loans at lower rates, which pays off the original loans early. The opposite risk is that of **Extension**, where the mortgage security takes longer to fully mature because rates go up and mortgage borrowers prepay more slowly.

One measure of this is the **Single Monthly Mortality Rate** (SMM), which represents the monthly repayment of the mortgage pool. It is the prepayments for a specific month divided by the outstanding mortgage pool balance. This rate can also be annualized, which is known as the **Conditional Prepayment Rate** (CPR).

Another source of funding for ABS is from commercial property mortgages. These are known as **Commercial Mortgage-Backed Securities** (CMBS). The credit risk of these securities is quantified using the loan-to-value ratio (LTV) and the debt-service-coverage (DSC). The higher these values are, the more credit risk these loans represent. Commercial mortgage loans are more likely to have prepayment penalties, so the prepayment risk is less than what it is for MBS bonds. There are several other common sources for funding in ABS bonds. There are securities based on auto loans, auto lease receivables, and credit card receivables (somewhat riskier because these are unsecured loans).

**Collateralized Debt Obligations** (CDO) are securities backed by a diversified pool of one or more debt obligations. They can be backed by corporate bonds, ABS, or even other CDOs. These are usually structured in several tranches that represent different levels of credit risk and seniority in receiving cash flows. These involve the creation of SPEs who own the underlying assets and exist only as a vehicle for managing the pool of assets until the CDO matures. The riskiest tranche of a CDO is known as the equity tranche and is the first class to be impacted by defaults in the pool of assets.

A covered bond is a type of senior debt obligation issued by a financial institution. It is usually backed by a segregated collection of assets consisting of commercial, public sector, or residential mortgages.

Characteristics of covered bonds include:

- Covered bonds are like asset-backed securities (ABS) but with dual recourse.
- Covered bonds consist of one class of bonds per cover pool.
- The cover pool in the covered bonds is dynamic in nature.
- Covered bonds are associated with the redemption regimes.

Types of covered bonds include:

i. Hard bullet covered bonds.

ii. Soft bullet covered bonds.

iii. Conditional pass-through covered bonds.

# Reading 46 – Understanding Fixed-Income Risk and Return

The return from a fixed income security can be broken down into three sources: the coupon and principal payments received, reinvestment of the coupon payments, and capital gains (or losses) on the sale of the bond before maturity. Bonds that trade at a premium or discount to par value will amortize to par as the bond approaches maturity.

The calculation for returns on reinvested coupons involves finding the future value of the coupon payments using the yield to maturity of the bond. You can use the time value of money functions on your calculator using the coupons as the PMT amount and the yield to maturity as the I/Y value to find the total return from reinvestment. The amount of this future value in excess of the coupons themselves is known as the "interest-on-interest" gain from compounding. The internal rate of return between the total return (the sale price plus the sum of coupons and interest-on-interest gains) and the original purchase price is known as the horizon yield.

Bond **Duration** is the measurement of a bond's full price to changes in interest rates and is a very important part of the fixed income curriculum. There are two types of duration: yield duration is the sensitivity of a bond's price to changes in its own yield-to-maturity (discount rate) and curve duration measure the sensitivity to changes in a specified benchmark rate (usually government rate curves).

There are several duration calculations that are used in the curriculum. **Macaulay Duration** is the weighted average of the present values of all future cash flows for a bond. The table below outlines the calculation of the Macaulay duration for a traditional bond. In this example, the present values were found using the bond's yield to maturity of 12.18%. The weight column is each cash flow as a percentage of the total present value (the 85 at the bottom of column 3). Once the present values are multiplied by their weights (column 5), the sum of those values (4.247) represents the Macaulay duration for the bond:

**Macaulay Duration of a 5-Year 8% Annual Payment Bond**

| Period | Cash Flow | Present Value | Weight | Period ×Weight |
|---|---|---|---|---|
| 1 | 8 | 7.131 | 0.084 | 0.084 |
| 2 | 8 | 6.357 | 0.075 | 0.150 |
| 3 | 8 | 5.667 | 0.067 | 0.200 |
| 4 | 8 | 5.052 | 0.059 | 0.238 |
| 5 | 108 | 60.793 | 0.715 | 3.576 |
| | | 85.000 | 1.000 | 4.247 |

**Modified Duration** is based on the Macaulay duration. The formula for it is:

$$Modified\ Duration = \frac{Macaulay\ Duration}{1 + YTM}$$

The **Effective Duration** is the sensitivity of a bond's total price to changes in a specified benchmark rate (usually a government yield curve)

$$Effective\ Duration = \frac{PV_{-} - PV_{+}}{2 \times (\Delta Curve) \times PV_0}$$

Where:

$PV_{-} = New\ PV\ when\ curve\ is\ raised$

$PV_{+} = New\ PV\ when\ curve\ is\ lowered$

$PV_0 = Original\ PV\ of\ bond$

$\Delta Curve = Curve\ change$

Effective duration is especially useful for securities with call or other options. Since the likelihood of exercising a call or put option varies with changes in interest rates, it is important to know how sensitive they are to these rate changes. Because there is more uncertainty regarding their future cash flows, it is also hard to define a consistent yield-to-maturity for these bonds and therefore harder to calculate a useful Macaulay or Modified duration figure.

**Key Rate Durations** measure a bond's sensitivity to changes in benchmark yield curves at specific points of maturity. Whereas the effective duration is based on parallel shifts of the entire yield curve, key rate durations break down that exposure to sections of the curve.

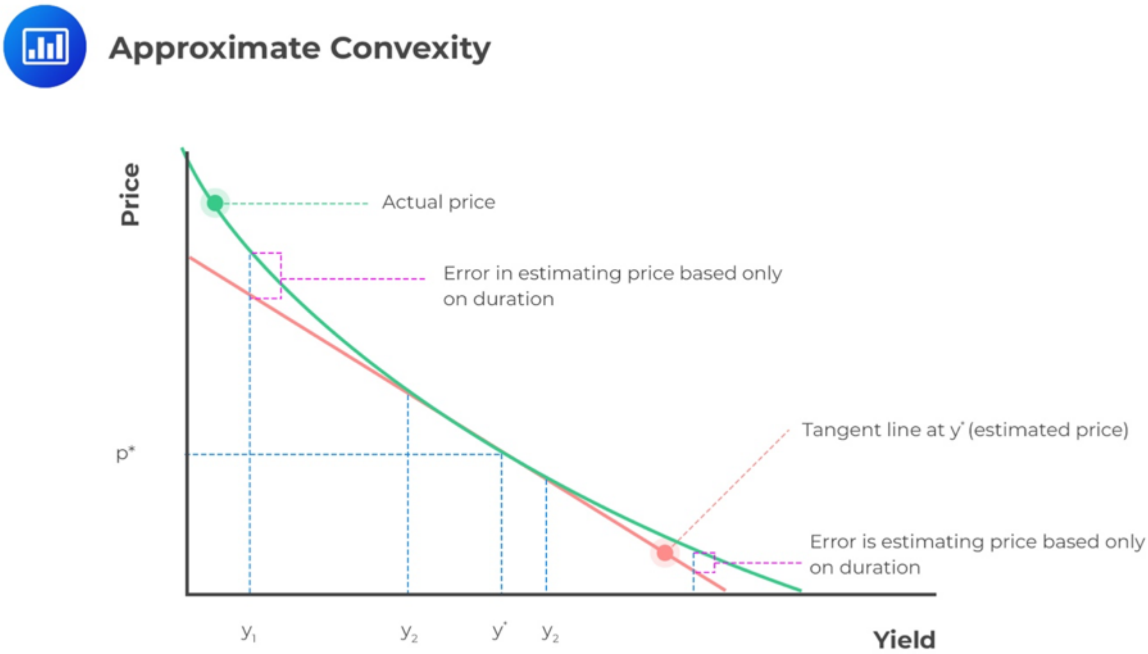
The measures of duration (which are expressed in years) will always be less than the time to maturity for a typical coupon bond. The **Maturity Effect** states that a longer-term bond is more sensitive to interest rate changes than a shorter-term bond. Longer maturity bonds will tend to also have higher duration values. Since Macaulay and Modified duration is based on the yield-to-maturity as the discount factor, higher yielding bonds will have lower duration values than ones with lower yields. Since bonds with put options can be sold when rates rise, their effective duration will be lower than bonds without similar protections for the investor.

Similarly, to the way in which duration is used for individual bond securities, the duration of a portfolio of fixed income securities is an important risk metric. The most accurate way to calculate the duration of a fixed income portfolio would be to find the weighted average of all cash flows for all bonds in the portfolio. Since this is computationally intensive, the more common approach is to find the weighted average of the durations of all bonds in the portfolio. This gives the Macaulay duration of the portfolio as a whole and the Modified duration can be found using the normal formula.

The modified duration tells you the expected percentage change of a bond's price given a change in its yield to maturity. Another way this can be expressed is to calculate the **Money Duration** (also known as dollar duration). This value is simply the annual modified duration multiplied by the Full Price of the bond. The new value will tell you the actual dollar amount of value change to be expected for a change in yield to maturity. A similar measure is the **Price Value of a Basis Point**, which is calculated by taking the difference between the prices of a bond given a 1bp increase and a 1bp decrease in yield to maturity and dividing that value by 2. This is the value, in dollars, attributed to a single basis point of yield to maturity. The calculation uses the average for both an increase and a decrease in yield because convexity means the increase and decrease could cause different changes in price. (Remember that one basis point is equivalent to 0.01% (1/100th of a percent) or 0.0001 in decimal.)

The **Convexity Effect** describes how a bond's price will change in relation to interest rate changes. It directs that the percentage price change is greater for rate decreases than rate increases.

The **Convexity Effect** shows up in graphical form when looking at how the price of a bond continues changing as rates continue to change. Rather than following a straight line, the convexity of a bond describes how the bond can become more or less sensitive to each additional change in interest rates. In the graph below, Bond A is demonstrating strong convexity and the amount by which its price change varies along the line of possible interest rates.




Convexity can either be **Approximate Convexity** or **Effective Convexity**. Approximate convexity is used to improve the estimate of the percentage price change:

$$\textit{Change in price of a bond} = \textit{Duration effect} + \textit{Convexity effect}$$

$$\%\Delta PV^{full} \approx (-AnnModDur \times \Delta Yield) + (\frac{1}{2} \times AnnConvexity \times (\Delta Yield)^2)$$

Effective convexity measures the secondary effect of a change in a benchmark yield curve:

$$Effective\ Convexity = \frac{PV_{-} + PV_{+} - (2 \times PV_0)}{(\Delta Yield)^2 \times PV_0}$$

Where:

$PV_{-}$ = New PV when yield declines

$PV_{+}$ = New PV when yield increases

$PV_0$ = Original PV of bond

$\Delta Yield$ = Yield change

The main difference between approximate and effective convexity is that approximate convexity is based on a yield to maturity change, whereas effective convexity is based on a benchmark yield change.

Interest rate changes greatly affect the prices of fixed income securities. To measure and manage the risk arising from interest rate shifts, analysts use duration to establish the sensitivity of a portfolio to changes in the level of interest rate changes. Interest rate increases with an increase in expected yield volatility.

A long-term investor is majorly concerned with total return over the investment horizon. A buy-and-hold investor has a higher total return if interest rates rise and a lower total return if rates fall. Duration measures the immediate drop in value or price of interest rates. The bond is, however, "pulled to par" as time passes. At some point in the lifetime of the bond, those two effects offset each other, and the gain on reinvested coupons is equal to the loss on the sale of the bond. That point is the **Macaulay Duration** statistic. The difference between the Macaulay duration and the investment horizon is known as the **Duration Gap**. An investor is hedged against interest rate risk when the duration gap is zero, exposed to lower interest rate risk when the duration gap is negative, and to higher interest rate risk when it is positive.

The yield to maturity of a corporate bond is dependent on a government benchmark yield and a spread over that yield. The benchmark yield changes with a change in the expected inflation rate/expected real interest rate. The spread over the yield changes with changes in **Credit Risk** of the issuer and **Liquidity** of the bond.

**Analytical Duration** refers to estimating duration and convexity using mathematical formulas. Analytical duration approximates the effect of changes in benchmark yields on bond prices by assuming that the government bond yields and spreads are independent and uncorrelated variables.

**Empirical Duration** refers to constructing statistical models using historical data to estimate duration. The statistical data includes diverse factors impacting bonds prices.

# Reading 47 – Fundamentals of Credit Analysis

**Credit Risk** is the possibility of loss to the bondholder due to the borrower failing to make the regular coupon and principal payments. The failure to keep up with these payments is known as a default. Most defaults do not result in a total loss, as there may still be a recovery of part of the original par value. The risk of this occurring is the default risk of the bond. There are also several other credit-related risks for bond investors. **Spread Risk** is the risk of the spread premium at which risky securities trade in relation to risk-free bonds (like US Treasuries) widening, which causes prices to decline. **Credit Migration Risk** (downgrade risk) is the possibility of the bond issuer's creditworthiness declining and causing an increased risk of default over the life of the bond. Market liquidity risk is the concern that illiquid securities might not trade at the fair value price of the bond.

There are two components for the measurement of credit risk: default risk and loss severity. **Default Risk** is the probability of a borrower defaulting on bond payments. **Loss Severity** is the portion of a bond's value that will be lost in the event of a default. Since there is usually recovery of partial value, the loss severity is very rarely 100%. These values are used to calculate the expected loss on a bond:

$$Expected\ Loss\ =\ Default\ Probability\ \times\ Loss\ Severity.$$

Similarly, to how companies can have different classes of equity offerings, bond offerings can have levels of seniority that dictate the order in which bondholders have a claim to the company's assets in the event of defaults. Bonds can also be either **Secured** or **Unsecured**, and only secured bonds have claims on specific assets. Bond seniority rankings include **Senior** and **Junior** levels and can have even finer categories within each.

Corporate issuers are rated according to their credit risk by rating agencies. The credit rating agencies are Standard & Poor's, Moody's, and Fitch. They play an important role in analyzing the creditworthiness of bond issuers and their ratings are used by investors as a way of gauging the risk of investing in specific issues and issuers. Higher credit rating bonds have lower default risk and issuers with higher ratings get to pay lower rates when issuing new debt. There are limits to credit ratings, however, in that credit ratings can change over time. The creditworthiness of an issuer could move up or down over the life of a bond, especially for something with a long time until maturity.

There are four primary characteristics for performing credit analysis, these are known as the "Four Cs of Credit Analysis."

1. **Capacity** is the ability of a borrower to service its debt. Determining this involves looking at the operating fundamentals of the company and its industry.
2. **Collateral** represents assets pledged against the debt issuance that can be claimed by the bondholder if the borrower defaults.

3. **Covenants** are items in the bond prospectus that outline elements that prescribe and limit the available actions of the lender.
4. **Character** is the qualitative assessment of the reputation and trustworthiness of the borrower.

Proper credit analysis includes looking at the operating results of the issuing company to determine how likely they are to remain in business and keep up with their debt obligations. The profitability and cash flow of the firm are important metrics to track, in addition to the total leverage and indebtedness of the firm, and also the interest coverage margins.

Bonds that have more credit risk also offer an additional yield in order to compensate for the additional risk from default and volatility. The total yield on a corporate bond can be broken down into component: rates and the premiums added for additional risk metrics. The yield includes elements such as the risk-free rate, expected inflation, a premium for maturity, a premium for lack of liquidity, and a credit spread to account for default risk.

Certain categories of bonds deserve extra considerations to account for different circumstances. High yield bonds ("junk bonds") are those rated below Baa3 or BBB- by the rating agencies and represent significantly more default risk due to weaker operating results, high amounts of leverage, or other factors that make the borrower more of a credit risk. Analysis for high yield bonds is more thorough than that required for investment-grade issues and will focus more on the elements that are important in a potential default scenario, such as the liquidity of company assets, the seniority of the issue at hand, and detailed financial operating projections.

**Sovereign Debt** is another category that requires special consideration. An analyst must look at the political risk and stability of the issuing country, the relevant economic structure and trends, and the flexibility of the monetary system. **Non-Sovereign Debt** can also be of extra concern because the issuing entities will not have the same level of currency and monetary control as sovereign governments. These issuers include municipalities and states that may have limited taxing authority but no monetary policy authority.

# Derivatives

## Reading 48: Derivative Instrument and Derivative Market Features

A derivative is a financial instrument that derives (obtains) its value from the performance of an underlying. The underlying may be a single asset, a group of assets, or variables such as interest rates.

A derivative is created as a contract involving two counterparties: a **Buyer** and a **Seller**. A **Derivative Contract** is a legal agreement between counterparties that defines the rights and obligations of each party. It contains specific maturity or settlement.

Derivatives can be stand-alone or embedded. **Stand-Alone Derivatives** are distinct. An example, a is a derivative on a bond. On the other hand, **Embedded Derivatives** are derivatives within an underlying, for example, callable, puttable, and convertible bonds.

Derivatives can be classified into either one of two categories:

- **Firm Commitment** (Forward Commitment): In this category, an amount is pre-determined, and the parties involved agree to exchange it at a future date. Firm commitments include forward contracts, futures contracts, and swaps.
- **Contingent Claim**: The trade settlement depends on one of the counterparties. Options are the primary contingent claims.

Derivatives are used for different purposes, including hedging, speculation, and arbitrage.

Equities, fixed income, interest rates, currencies, commodities, and credit are commonly used in derivatives. Derivatives that use equities as the underlying may reference a single stock, a group of stocks, or a stock index. Options are predominantly associated with individual stocks, while index derivatives are primarily traded as options, futures, forwards, and swaps. Fixed income instruments mostly use bonds as the underlying. Associated derivatives include futures, swaps, and options. On the other hand, interest rates are a fixed income underlying used by interest rate derivatives, such as forwards, futures, and options.

Derivatives can be used to hedge **Foreign Exchange Risk** in commercial and financial transactions. For example, exporters may use forward contracts to sell domestic currency and buy foreign currency in a way that coincides with the delivery of goods or services in a foreign country. Derivatives are commonly used on commodities to manage the price risk of an individual commodity, or a commodity index separate from physical delivery. Derivative contracts that use credit as an underlying are based on the default risk of either a single or a group of issuers.

Investors can buy and sell derivatives through an **Exchange** or **Over the Counter** (OTC). Exchange-traded derivative contracts have the advantage of being more standardized and offering more transparency. OTC contracts offer more flexibility in customizing their terms through direct negotiations between buyers and sellers.

# Reading 49: Forward Commitment and Contingent Claim Features and Instruments

A forward contract is an over-the-counter (OTC) derivative contract. In this contract, two parties agree that one party, the buyer (long), will purchase an underlying asset from the other party, the seller (short), at a later date at a fixed price (the forward price) agreed upon when the contract is initiated.

The payoff profile for a forward contract at maturity is calculated as follows:

$$Buyer\ payoff = S_t - F_0(T)$$

$$Seller\ payoff = -[S_t - F_0(T)]$$

Where:

$S_t = Price\ of\ the\ underlying\ asset\ when\ the\ contract\ at\ expiry$

$F_0(T) = Forward\ price$

Futures contracts are a standardized variation of forward contracts. The buyer of the futures contracts agrees to buy the underlying in the future at a pre-agreed price (futures price). On the other hand, the seller agrees today to sell the underlying asset in the future at a price agreed upon at the initiation of the contract. The exchange determines expiration dates, underlying assets, the size of the contracts, and other details. **Mark-To-Market (MTM)** is a distinguishing feature for future contracts and helps determine an average of the final futures trade of the day (settlement price).

Like forward contracts, no cash changes hands at the initiation of the futures contract. However, both counterparties must deposit the **Initial Margin** into a **Futures Margin Account** at the exchange. The exchange uses the futures margin account to settle daily market changes. Special financial intermediaries usually execute futures contracts on behalf of the counterparties.

When the futures margin account funds fall below the initial margin, the seller receives a margin call to top up the account back to the initial amount. The added sum is called the **Variation Margin**. If the counterparty cannot replenish the margin account, it must close the contract immediately and incur additional costs. In extreme cases where the counterparty cannot meet the obligations, the exchange covers the losses through an **Insurance Fund**.

An exchange might impose additional requirements such as increasing required margins, price limits, and **Circuit Breakers** to limit potential default-occasioned losses.

A **Swap** is an over-the-counter derivative contract in which two parties agree to exchange a series of cash flows whereby one party pays a variable (floating) series that an underlying asset or rate will determine. The other party either pays (1) a variable series determined by a different underlying asset or rate or (2) a fixed rate.

There are some common features among forwards, futures, and swaps, such as defined contract size, defined underlying, presence of one or more exchanges of cash flows or underlying on a given date or dates, and exchanges are based on pre-agreed prices.

Options are derivative instruments that give the option buyer the right, but not the obligation, to buy (call) or sell (put) an asset from (or to) the option seller at a fixed price on or before expiration. The **Payoff** of an option is positive or zero. When the option buyer decides to transact the underlying, it is called exercising the option. The pre-agreed price at which the option buyer exercises the underlying is called the **Exercise Price**. Options **Premium** is the price the option buyer pays the seller for the right to exercise the option in the future.

Options can be traded on over-the-counter (OTC) markets or exchanges based on standardized terms. Options can be **American Options** or **European Options**. European-style options can only be exercised at expiry, while American-style options are exercisable before expiry.

There are two options: **Call Options** and **Put Options**. In call options, the buyer has the right but not the obligation to buy the underlying. On the other hand, in put options, the buyer has the right but not the obligation to sell the underlying.

Let's define the following:

$c_T = Value\ of\ the\ call\ at\ expiration$

$p_T = \ Value\ of\ a\ put\ option\ at\ expiration$

$S_T = \ Price\ of\ the\ underlying\ at\ time\ T$

$X = Exercise\ price$

$c_0 = \ Call\ option\ premium$

$p_0 = \ Put\ option\ premium$

$\Pi = \ Profit\ from\ an\ option\ strategy$

For a call option, the buyer would only exercise the option if $S_T > X$. As such, the payoff to the buyer at expiration is given by:

$$C_T = \max(0, S_T - X)$$

Conversely, the payoff to the seller at expiration is:

$$-C_T = -(\max(0, S_T - X))$$

Note also that the option buyer pays the seller the call option premium $(c_0)$ at time $t = 0$ for the right to buy the underlying $S_T$ at an exercise price of $X$ at time $t = T$.

Therefore, the profit the buyer will earn is calculated as follows:

$$\Pi = \max(0, S_T - X) - C_0$$

For the call option seller's profit, it is given by:

$$\Pi = -\max(0, S_T - X) + C_0$$

For the put option, it is only exercisable if $S_T < X$.

Therefore, the payoff to the buyer is given by:

$$p_T = \max(0, X - S_T)$$

Conversely, the payoff to the put option seller is:

$$-p_T = -(\max(0, X - S_T))$$

Derivatives typically fall into two classes: forward commitments or contingent claims. **Forward Commitments** are derivative contracts between two parties that require both parties to transact in the future at a pre-specified price. The parties are obligated to transact, and a legal remedy may be enforced in the event of non-performance. A **Contingent Claim** is a type of derivative where the payoff profile is dependent on the outcome of the underlying asset or conditional on the occurrence of some events. With a contingent claim, there is the right to transact but not the obligation. Consequently, contingent claims have become synonymous with the term "option."

## Reading 50: Derivative Benefits, Risks, and Issuer and Investor Uses

Some of the benefits of derivative instruments include:

- They allow risk allocation, transfer, and management without trading an underlying.
- Derivative instrument prices provide a price discovery function outside cash or spot markets. More specifically, futures prices may give information about future cash market movement.
- They bring about operational advantages compared to cash/spot markets. Some of these advantages include lower transaction costs, high liquidity, low, upfront cash requirements, and the ability to take short positions.
- The operational efficiency of derivative markets is nature's greater market efficiency since they offer an effective way to exploit mispricing (deviation of prices from the fundamental value).

Some of the potential risks associated with using derivative instruments include:

- High potential for speculative use
- Lack of transparency.
- Likelihood for basis risk.
- Likelihood for liquidity risk.
- Likelihood for counterparty credit risk.
- Likelihood for destabilization and systemic risk.

There are different types of hedge accounting, such as:

- **Cash Flow Hedge** absorb variable cash flow of floating-rate assets or liabilities such as interest rates and foreign exchanges. They use either forward commitments or contingent claims.
- **Fair Value Hedge** occurs when a derivative is used to offset fluctuation in the fair value of an asset or liability.
- **Net Investment Hedges** arise when a foreign currency bond or a derivative such as forward is used to offset the foreign operation's exchange rate risk of equity.

Investors use derivatives for the following purposes:

- **Replicate a cash market strategy:** A feature that allows investors to replicate a chosen position using derivatives easily.
- **Perform derivative hedges:** By isolating specific underlying exposures while retaining other positions.
- **Add/Modify exposures:** Since derivative markets are flexible to take positions, an investor can use a derivative to add or modify an exposure beyond cash market alternatives.

# Reading 51: Arbitrage, Replication, and the Cost of Carry in Pricing Derivatives

**Arbitrage** refers to buying an asset in a cheaper market and selling it in a more expensive market to make a risk-free profit. Traders endeavor to exploit arbitrage opportunities when short-lived market differences exist between assets in the same or different markets. An arbitrageur will buy assets in a market with low prices and sell in another market at a higher price to make a profit.

The **Law of One Price** states that *identical things should have the same prices*. Suggesting that assets that produce identical results have only one true market price. The no-arbitrage conditions for pricing derivatives with the underlying with no additional cash flows include:

- Identical assets (assets with identical cashflows) traded at the same time must have the same price.
- Assets with known future prices must have a spot price equal to the present value of the future price discounted at risk-free interest.

**Replication** refers to a strategy in which a derivative's cash flow stream may be recreated using a combination of long or short positions in an underlying asset and borrowing or lending cash. It offsets a derivative position, given that the law of one price holds and arbitrage does not exist. It implies that a trader can take opposing positions in a derivative and the underlying, creating a default risk-free hedge portfolio and replicating the payoff of a risk-free asset.

For example, the following combinations produce the equivalent single asset:

$$Long\ asset + Short\ derivatives = Long\ risk\ free\ asset$$

The formula can be rearranged as:

$$Long\ asset + Short\ risk\ free\ asset = Long\ derivatives$$

$$Short\ derivative + Short\ risk\ free\ asset = Short\ asset$$

The **Spot Price** is the price an investor must pay immediately to acquire the asset. On the other hand, the **Future Price** refers to the projected price of an asset at a later date, say, in 6 months.

Assuming there are no costs and benefits associated with the underlying asset, spot and forward prices are related as follows under **discrete compounding**:

$$F_0(T) = S_0(1+r)^T$$

Where:

$F_0(T) = \textit{Forward price}$

$S_0 = \textit{Spot price}$

$r = \textit{Risk-free rate of return}$

$T = \textit{Time to maturity}$

Under **Continuous Compounding**:

$$F_0(T) = S_0 e^{(r)T}$$

Where:

$e =$ Euler's constant $=$ 2.71828...

An FX (foreign exchange) forward contract involves an agreement to buy a particular amount of foreign currency on a future date at a forward price $F_{0,f/d}$. The transaction is made at a pre-agreed exchange rate and is meant to protect the investor from changes in the exchange rates of that foreign currency.

The foreign exchange spot rate is denoted as $S_{0,f/d}$ where the foreign currency $f$ is taken as the **Price Currency**, while the domestic currency $d$ is considered the base currency. A forward price $F_{0,f/d}$ reflects the difference between risk-free foreign rates $(r_f)$ and the domestic risk-free rate $(r_d)$ expressed as:

$$F_{0,f/d}(T) = S_{0,f/d} e^{r_f - r_d}$$

**Cost of Carry** is the net of the costs and benefits of owning an underlying asset for a period.

**Convenience Yield** is a non-monetary benefit of holding a physical asset rather than a contract (derivative).

**Costs** include storage, transportation, insurance, and spoilage costs associated with holding the underlying asset, such as warehouse costs (rent) and insurance costs. If the asset owner incurs costs (in addition to opportunity cost), compensation is done through a higher forward price to cover the added costs.

**Benefits** (or income) refer to monetary returns (such as interest and dividends) and non-monetary returns (such as convenience yield) associated with holding the underlying asset. Benefits decrease the forward price since it accrues to the underlying.

Considering the cost of carry, the relationship between the spot price and futures price changes as follows:

$$F_0(T) = [S_0 - PV_0(I) + PV_0(C)](I + R)^T$$

Where:

$C = Costs$

$I = Benefits/income$

Under continuous compounding, the costs (c) and income (i) are expressed as rates of return so that the futures price is given by:

$$F_0(T) = S_0 e^{(r+c-i)T}$$

# Reading 52: Pricing and Valuation of Forward Contracts and for an Underlying with Varying Maturities

The value of a forward commitment changes over the life of the forward contract as the spot price and other factors change. For instance, let's consider a forward contract on an underlying that does not generate cash flows:

- **At the inception of the forward contract:** The value is zero since there is no down payment; that is, investors are willing to sign off on the contract for \$0.00.
- **During a forward contract's life:** The value of the forward contract is calculated as spot price minus the present value of the forward price discounted at the risk-free rate.
- **At maturity:** The value of the forward contract is calculated as the spot price at maturity minus the forward price.

In forward contract pricing and valuation, the following terms are commonly used:

$S_0$ = *Spot price of the underlying at the initiation*

$S_t$ = *Spot price of the underlying at time t, during the contract's life*

$r$ = *Risk free rate of return*

$F_0(T)$ = *Forward price (satisfies the no-arbitrage conditions)*

$V_0(T)$ = *Value of the forward contract at the initiation*

$V_t(T)$ = *Value of the forward contract during the life of the contract*

$V_T(T)$ = *Value of the forward contract at expiration*

At the initiation of the forward contract, no money is exchanged, and the forward contract's value is zero:

$$V_0(T) = 0$$

And the forward price at initiation is the spot price of the underlying compounded at the risk-free rate over the contract's life:

$$F_0(T) = S_0(1 + R)^T$$

The **Value** of a forward contract during the contract's life is the spot price at time $t$ of the underlying asset minus the present value of the forward price (long position).

The short forward position value is calculated as follows:

$$V_T(T) = s_t - F_0(T)(1+r)^{(-T-T)} - S_t$$

# Reading 53: Pricing and Valuation of Futures Contracts

Let's consider an underlying with no associated costs or benefits; the futures price is calculated by compounding the spot price of the underlying using the risk-free rate expressed in the equation below.

$$f_T(0) = S_0(1+r)^T$$

Where:

$f_T(0) = \textit{Futures forward price}$

$S_0 = \textit{Spot price of the underlying at time } t = 0$

$r = \textit{Risk free rate}$

$T = \textit{Time to maturity}$

Using continuous compounding, the future price is given by:

$$f_T(0) = S_0 e^{rT}$$

Under costs of carry, the price of futures whose underlying has income (I) and costs (C) is adjusted as follows:

$$f_T(0) = [S_0 - PV_0(I) + PV_0(C)](1+r)^T$$

Where:

$PV_0(I) = \textit{Present value of income or benefit associated with the underlying at time } t = 0$

$PV_0(C) = \textit{Present value of costs associated with the underlying at time } t = 0$

The MTM value of the forward contract is not settled until its expiration date, which causes **Counterparty Risk**. On the other hand, the futures price changes depending on market conditions. At the same time, the daily settlement resets the MTM value to zero and the variation margin is exchanged to cover the difference, decreasing counterparty risk.

Interest rate futures trade on a price basis, given by the following formula:

$$f_{A,B-A} = 100 - \left(100 \times MRR_{A,B-A}\right)$$

Where:

$f_{A,B-A}$ = *Futures price for the market reference rate for* $B-A$ *periods that begin in* $A$ *period* $(MRR_{A,B-A})$.

It's important to note that the long position (lender) gains as prices rise and future MRR falls. In contrast, the short position (borrower) gains as prices fall and future MRR rises.

The **Daily Settlement** of the interest futures occurs depending on the price changes, regarded as futures contract basis point value (BPV) and calculated as follows:

$$Futures\ contract\ BPV = Notional\ principlal \times 0.01\% \times Period$$

Reasons why forward and future prices differ:

- Mark-to-market (MTM), margining, settlement of gains and losses, and risks.
- Convexity bias.

What causes similarities between forward and future prices:

- Relationship between interest rates and futures prices.
- Central clearing on OTC derivatives.

# Reading 54: Pricing and Valuation of Interest Rate and Other Swaps

A **Swap** is a derivative contract between two counterparties to exchange a series of future cash flows. In contrast, a forward contract is also an agreement between two counterparties to exchange a single cash flow at a later date. Swaps and forward contracts are similar in that *both* are forward commitments with symmetric payoff profiles. Besides, in both interest swaps and forward contracts, no cash exchanges hand at initiation.

A distinguishing factor, however, is that the fixed swap rate is constant, while a series of forward contracts have different, forward rates at each expiration.

A **Forward Rate Agreement** (FRA) is a cash-settled over-the-counter (OTC) contract between two counterparties. In this contract, the buyer (long position) is borrowing a notional sum (underlying) at a fixed interest rate (the FRA rate) and for a specified period starting at an agreed-upon date.

The seller deposits interest based on the market reference rate (MRR), where the MMR is established before the settlement dates, at time $t - A$.

The FRA settlement amount is a function of the difference between the forward interest rate ($IFR_{A,B-A}$) and the market reference rate ($MRR_{B-A}$) or the MMR for $B - A$ periods.

$$Net\ payment = (MRRB - A - IFRA, B - A) \times Notional\ principal \times Period$$

In a **Swap Contract**, two parties agree to exchange a series of cash flows. In the agreement, one party pays a variable (floating) series of cash flows that will be determined by a market reference rate (MRR) that resets every period.

The **Par Swap Rate** is a fixed rate that equates the present value of all future expected floating cash flows to the present value of fixed cash flows. That is,

$$\sum_{i=1}^{N} \frac{IFR}{(1 + Z_i)^i} = \sum_{i=1}^{N} \frac{s_i}{(1 + Z_i)^i}$$

Where:

$IFR = Implied\ forward\ rates$

$s_i = Par\ swap\ rate\ for\ period\ i$

$z_i = Spot\ rates\ for\ period\ i$

# Reading 55: Pricing and Valuation of Options

Options are **Contingent Claims** that give the option buyer the right but not the obligation to transact the underlying, and the option seller is obligated to meet the obligation chosen by the buyer. The payoff of an option is either positive or zero. The profit, on the other hand, can be negative because of the option premium.

The **Exercise Value** of an option would be the value if it were immediately exercisable. On the other hand, an option's **Time Value** reflects the passage of time and variability of the underlying.

The **Moneyness** of an option describes the relationship between the underlying price and the exercise price.

The "**In-The-Money**" options refer to call options with the spot price above the exercise price ($S_t > X$) and put options with the spot price below the exercise price ($S_t < X$). An option is considered "**Deep-in-the-Money**" if the stock price is much higher than the exercise price. Conversely, "**Out-of-The-Money**" options refer to call options with the spot price below the exercise price $S_T < X$ and put options with the spot price above the exercise price $S_T > X$. An option is considered "**Deep-Out-of-the-Money**" if it is unlikely to be exercised. An "**At-the-Money**" option is a call or put option where the exercise price equals the current underlying spot price.

The moneyness of an option can be summarized in the table below:

| **Moneyness** | **Call Options** | **Put Options** |
|---|---|---|
| In-the-Money | $S_t > X$ | $S_t < X$ |
| At-the-Money | $S_T = X$ | $S_T = X$ |
| Out-of-the-Money | $S_t < X$ | $S_t > X$ |

In a call option**,** the buyer has the right but not the obligation to buy the underlying; it will only be exercised if the payoff is positive.

The buyer would only exercise (buy the underlying) the option if $S_T > X$. Therefore, the payoff (exercise value) to the buyer at expiration is given by:

$$c_T = \max(0, S_T - X)$$

While the exercise to the seller at expiration as:

$$-c_T = -\left(\max(0, S_T - X)\right)$$

In a put option**,** the buyer has the right but not an obligation to exercise the option at expiry. Exercising the option means that the buyer sells the underlying $S_T$ at the exercise price $X$ at expiration. As such, the put option is only exercisable if $S_T < X$.

Therefore, the payoff to the buyer is given by:

$$p_T = \max(0, X - S_T)$$

Conversely, the payoff to the put option seller is

$$-p_T = -\left(\max(0, X - S_T)\right)$$

Investors estimate the value of options based on the spot price and exercise price before maturity $(t < T)$.

For call options, exercise value is calculated as spot price minus the present value of exercise price:

$$c_t = \max\left(0, S_t - X(1 + r)\right)^{-(T-T)}$$

For put options, it is calculated as the present value of exercise price minus spot price:

$$p_t = \max\left(0, X(1+r)^{-(T-t)} - S_t\right)$$

If the spot price is greater than the present value of the exercise price, the exercise value of a call option is equivalent to the value of the long forward commitment at that time.

The **Time Value** of an option is the difference between the option's current price and its current payoff, or exercise value. European options can only be exercised at the expiration date but can be bought or sold before maturity at a price that reflects the option's future expected payoff.

The time value of a call option is given by;

$$Time\ value = c_t - \max\left(0, S_t - X(1+r)^{-(T-t)}\right),$$

While the time value of a put option is given by;

$$p_t = \max\left(0, X(1+r)^{-(T-t)} - S_t\right) + Time\ value$$

The current option price equals the exercise value plus the time value for both call and put options.

**Arbitrage** opportunities can arise if the **Law of One Price** does not hold. In options, no-arbitrage conditions are based on the payoff at maturity, which has an asymmetric profile. The **Option Buyer** pays a (premium $(c_0)$ for call options and $(p_0)$ for put options) to the seller, and profits at maturity are given by;

$$\Pi_{call} = \max(0, S_T - X) - c_0$$

$$\Pi_{put} = \max(0, X - S_T) - p_0$$

An option is only exercised if it is in the money, so there must be upper and lower no-arbitrage price bounds at any time t.

For a **Call Option,** the **Lower Bound** is the maximum of zero and the difference between the underlying price and the present value of the exercise price. The **Upper Bound** is the current underlying price, as a call buyer will not pay more than the underlying price for the right to buy the underlying. The no-arbitrage bounds of a call option are usually expressed as follows;

$$\max\left(0, S_t - X(1+r)^{-(T-t)} < c_t \leq S_t\right)$$

For a **Put Option,** the **Upper Bound** is the exercise price $(X)$ since a put option buyer only exercises the option if the underlying price is less than the exercise price. The **Lower Bound** is the maximum of zero and the difference between the present value of the exercise price and the underlying price. The no-arbitrage bounds of a put option are generally expressed as follows:

$$\max\left(0, X(1+r)^{-(T-t)} - S_t\right) < p_t \leq X$$

To replicate the payoff of a put option, a trader needs to sell the underlying asset short and lend the proceeds at a risk-free rate. At the expiration date, two outcomes are depending on the spot price of the underlying asset:

- If $S_T < X$, the put option is exercised, and the trader buys the underlying asset at ST using the proceeds of the risk-free loan.
- If $S_T \geq X$, the put option is not exercised, and no settlement is needed.

A similar outcome applies to call options. The **Non-Linear** nature of option payoff requires replicating transactions to be adjusted based on the likelihood of exercise. A proportion of $X(1+r)^{-T}$ is borrowed based on the moneyness of the option and the probability of exercise at time $T$.

The factors that affect the value of an option include the value of the underlying, exercise price, time to maturity, risk-free rate, volatility, and income or cost associated with the underlying.

The table below summarizes the factors that affect the value of an option.

| **Factor** | **Value of European Call Option** | **Value of European Put Option** |
|---|---|---|
| Value of the underlying | Directly proportional | Inversely proportional |
| Exercise price | Inversely proportional (as the exercise price increases, value decreases) | Directly proportional (as exercise price increases, value increases) |
| Time to maturity | Directly proportional | Inversely proportional |
| Risk-free rate | Directly proportional | Inversely proportional |
| Volatility | Directly proportional | Directly proportional |
| Benefits | Inversely proportional | Directly proportional |
| Costs | Directly proportional | Inversely proportional |

# Reading 56: Option Replication Using Put-Call Parity

**Put-Call Parity** is a no-arbitrage concept that allows the pricing and valuation of portfolios involving cash and derivative instruments without directly modeling them using non-arbitrage conditions. It involves two portfolios: **Fiduciary Call** and **Protective Put**, which have identical profiles and allow investors to benefit from the rise in underlying price without exposure to a decrease below the exercise price. Put-call parity holds for European options with similar exercise prices and expiration times. It implies that the price of the long underlying asset plus the long put must be equal to the price of the long-call option plus the risk-free asset.

The put-call parity equation is expressed as:

$$S_0 + p_0 = c_0 + X(1+r)^{-T}$$

Where:

$S_0 = \textit{Price of the underlying asset}$

$p_0 = \textit{ Put premium}$

$c_0 = \textit{Call option premium}$

$X = \textit{Exercise price}$

$r = \textit{Risk-free rate}$

$T = \textit{Time to expiration}$

The put-call parity equation relates the prices of call and put options, the underlying asset, and the risk-free asset, and implies that they must trade at the same price, ensuring a no-arbitrage relationship.

Option replication using put-call parity discusses creating synthetic relationships with options by replicating a one-part position under put-call parity. The put-call parity equation can be rearranged to solve for the put option premium or the call option premium, and these can be used to create **Synthetic Puts** or calls, respectively.

For example, we can rearrange the put-call parity equation to solve for the put option premium:

$$p_0 = c_0 + X(1+r)^{-T} - S_0$$

Additionally, the **Covered Call** position can be created by further rearranging the put-call parity equation.

The tables below show the positions and their corresponding underlying assets, risk-free bonds, call options, and put options.

| **Position** | **Underlying ($S_0$)** | **Risk-Free Bond $X(1+r)^{-T}$** | **Call Option ($c_0$)** | **Put Option ($p_0$)** |
|---|---|---|---|---|
| Underlying ($S_0$) | - | Long | Long | Short |
| Risk-free bond ($(1+r)^{-T}$) | Long | - | Short | Long |
| Call option ($c_0$) | Long | Short | - | Long |
| Put option ($p_0$) | Short | Long | Long | - |

If put-call parity does not hold, an arbitrage opportunity exists that can be exploited by selling the most expensive portfolio and purchasing the cheaper one.

The **Put-Call Forward Parity** extends the put-call parity to include forward contracts. It is obtained by substituting the present value of the forward price, $F_0(T)$, for the underlying price in the put-call parity equation.

$$F_0(T)(1+r)^{-T} + p_0 = c_0 + X(1+r)^{-T}$$

To derive the put-call forward parity, two portfolios are considered. Portfolio A involves buying a forward contract and a risk-free bond whose face value is equal to the forward price and putting an option on the underlying asset. The cost of this strategy is:

$$F_0(T)(1+r)^{-T} + p_o$$

Portfolio A is a **Synthetic Protective Put** designed to benefit from an increase in the underlying asset's value and hedge against a decrease in value.

Portfolio B involves buying a call option and a risk-free bond redeemable at the exercise price at the time of expiration. The cost of this transaction is:

$$c_0 + X(1+r)^{-T}$$

Portfolio B is the same **Fiduciary Call** as in the put-call parity seen previously.

The put-call parity relationship can be used to define a firm's value based on equity holders' and debt holders' interests. If a firm is solvent, shareholders benefit if the company can meet its debt obligations. In contrast, if a firm is insolvent, shareholders receive nothing, and the debtholders receive the value of the firm's assets up to the face value of the debt.

The put option held by the shareholders can be seen as a form of insurance against the risk of the company becoming insolvent. If the company's value drops below the debt's face value, the shareholders' payoff from the put option can offset the losses they would incur from their equity holdings.

On the other hand, the debtholders' position $(D - p_0)$ can be seen as a form of protection against the risk of the company becoming insolvent. The put option sold by the debtholders to the shareholders provides them with a guaranteed price (the exercise price) to sell their debt in the event of the company's insolvency.

Overall, the put-call parity relationship and the options framework provide a valuable tool for understanding the interplay between a company's equity and debt holders and the implications of solvency and insolvency on their respective payoffs.

# Reading 57: Valuing a Derivative Using a One-Period Binomial Model

The law of one price states that assets with identical payoffs in all future scenarios must have the same value at a given time. Contingent claims have asymmetric payoffs, making their valuation challenging.

The **Binomial Model** can be used to model the payoffs of contingent claims by considering the spot price of an asset at maturity, which can either increase or decrease based on a random variable's outcome. The model assumes two probabilities, $q$ and $1 - q$, for an increase or decrease in asset price, respectively. The gross return for an increase or decrease in price can be calculated using the model.

The gross return when the asset price increases will be:

$$R^u = \frac{S_1^u}{S_0} > 1$$

When the asset price decreases, the gross return will be:

$$R^d = \frac{S_1^d}{S_0} < 1$$

The spread between the two possible outcomes represents the range of possible future prices.

At time t = 0, the value of the call option is $c_0$. Note that this value is unknown and needs to be calculated.

At time t = 1, if the underlying asset price rises to $S_1^u$, the call option is in the money, which means exercising the option will result in a positive payoff. On the other hand, if the underlying asset price falls to $S_1^d$, the call option is out of the money, which means that exercising the option will result in a zero payoff.

The value of an option is not affected by the real-world probabilities of underlying price movements but rather by the expected volatilities.

The value of a call or put option can be computed by discounting its expected value at expiration at the current risk-free rate using the risk-neutral probability, which equates the discounted expected values of the underlying to the option's current price.

In an equation, the value of the call option is expressed as:

$$c_0 = \frac{\pi c_1^u + (1-\pi)c_1^d}{(1+r)^T}$$

While the value of the put option is given by:

$$p_0 = \frac{p_1^u + (1-\pi)p_1^d}{(1+r)^T}$$

The **Risk-neutral Probability** (denoted by $\pi$) is calculated using the risk-free rate and the assumed up-and-down gross returns of the underlying. Risk-neutral pricing is a process that uses expected volatilities and the risk-free rate to determine the risk-neutral probability.

Risk-neutral probability is expressed as:

$$\pi = \frac{(1+r) - R^d}{R^u - R^d}$$

# Alternative Investments

## Reading 58: Categories, Characteristics, and Compensation Structures of Alternative Investments

Alternative investments include any assets that do not fall into the traditional asset class categories of equity and fixed income. It can include derivatives, real estate, commodities, short selling of stocks, etc. Common characteristics of alternative investments include limited liquidity, low correlation of returns with traditional assets, lax regulation, low transparency, and special tax and legal considerations. The primary benefit of investing in alternatives is the diversification effect due to the low correlation of returns compared to traditional investments.

**Hedge Funds** are special investment funds that aim to provide the highest possible absolute returns. They are actively managed and typically invest in aggressive strategies across many different asset classes. Many try to maximize returns through the use of leverage, arbitrage, and derivative strategies to outperform conventional funds. Investment withdrawal and lockup periods are common to prevent investors from removing their funds in the middle of an investment strategy. Hedge funds can be difficult to compare due to a lack of transparency in holdings and returns compared to other kinds of investment funds.

**Private Equity Funds** invest in non-publicly traded companies. They often use strategies including leveraged buyouts, distressed investing, and other methods of acquiring a portfolio of private companies. A frequent exit strategy involves taking a portfolio company to IPO or organizing a sale in order to recoup capital investment.

**Real Estate** is typically owned via a title, often registered by the local government in the land. Mortgages and construction financing are private debt investments, while MBS and collateralized obligations are public debts. There are **Direct Ownership**, **Partnerships**, and **Joint Ventures** as private real estate investments. Public equity investments may be REITs or shares in real estate corporations.

**Natural Resources** include commodities that may entail investment in such physical commodity products as grains, metals, and crude oil, agricultural land (or farmland), and timberland.

**Infrastructure Investments** include investments in transportation (toll roads, railways), utilities (dams, gas plants) and communication (broadcasting facilities, internet backbone).

Other investments in the alternative space include collectibles and other investments that don't fall into the traditional categories.

Methods of investing in alternative investments include:

- **Fund Investing** where investors contribute capital to a fund, and then the fund identifies, chooses, and makes investments on behalf of the investors.
- **Co-Investing** in which an investor indirectly invests in assets through a fund.
- **Direct Investing** in which an investor directly invests in an asset without using any intermediary.

It's important that due diligence is performed prior to investing in any entity. For instance, in direct investments, as much as an investor chooses the company to invest in, it's essential that due diligence is upheld before investing in a private company. As for fund investing, an investor must ensure to perform due diligence on the fund manager prior to investing in a fund. Lastly, in co-investing, the general partner assists the investor in performing direct due diligence on a portfolio company.

Different alternative investments, especially hedge and private equity funds, have a partnership structure with a **General Partner** (GP) and **Limited Partners** (LPs).

Here are some of the differences between GP and LPs:

- The GP is the fund manager, while the LPs are the accredited investors.
- LPs usually assume investment risks while the GP manages the business and bears unlimited liability in case of any unfavorable eventuality. It's equally noteworthy that the GP can manage multiple funds.
- The general partner is paid a management fee based on committed capital or the assets under their management. Besides, they earn an incentive fee based on actual profits. On the other hand, the number of LPs is limited in a given fund. LPs perform passive roles and are not part of fund management. It's the GP that makes all decisions and runs the fund.

For smooth operations between the GP and LP, **Limited Partnership Agreement (LPA)** legal document is used to govern the two. The document stipulated the rules of the partnership and institutes the structure guides for the entire fund's operations. Also, **Side Letters,** an agreement between the GP and particular number of LPs formed outside an LPA which may include potential additional reporting due to an LP's unique circumstances, such as tax requirements, first right of refusal, notice requirement in case of litigation, and related issues.

When general partners improve in performance, they are compensated through **Performance** and **Management Fees**. The management fee normally ranges between 1% and 2% of assets under management/committed funds, i.e., the funds the limited partners have invested/committed to the fund. A performance fee/incentive fee/carried interest is also awarded on funds when there is excess returns. The partnership agreement may also specify the hurdle rate. A hurdle rate is the lowest rate of return that the GP must exceed to earn a performance fee.

Common investment clauses, provisions and contingencies include:

- A **Catch-Up Clause** is intended to make the manager whole so that their incentive fee is based on the total return and not exclusively on the return in excess of the preferred return.

- A **High-Water Mark** is the highest value, net of fees, that a fund has reached in its history. It indicates the highest cumulative return used to calculate an incentive fee**.** A high-water mark clause stipulates that a GP must recover the decrease in funds value from the high-water mark prior to charging a performance fee on new profits earned.

- **Waterfall Clause** is a distribution technique that defines the order in which distributions are made to LPs and GPs. Typically, GPs get a higher proportion of profit to motivate them to make more profits. There are two major forms: **Deal-by-Deal** (also called American) and the **Whole-of-Fund** (also called European). In the deal-by-deal waterfall structure, performance fees are accumulated on per-deal methods. This makes the GP receive payments before LPs get their rates of return (hurdle rate) on the fund and initial investments. In the whole-of-fund waterfall structure, the GP does not receive any profit until LPs are paid their hurdle rate and initial investments. Clearly, the deal-by-deal waterfall is more profitable to the GP, while the whole-of-fund waterfall is more beneficial to LPs.

- **Clawback Clause** gives LPs the right to recover the performance fees from the GP. For instance, this happens if a GP pays itself an incentive fee on profit not yet fully earned.

# Reading 59: Performance Calculation and Appraisal of Alternative Investments

Alternative investments offer investors portfolio diversification opportunities and potentially high returns on a risk-adjusted basis. For that reason, they may ignore some of the obvious risks such as:

- Low transparency.
- Complex strategies and products.
- Low liquidity on a portfolio basis.
- Regular use of leverage and, sometimes, derivatives.
- Differentiation and diversification challenges.
- Compulsory/obligatory fees.
- Portfolios with specialized niche products may experience mark-to-market issues.
- Availability of redemption may be limited. At the same time, eventual redemption may mount pressure on the portfolio.

There are common approaches to performance appraisal, including:

1. **Sharpe Ratio**

The Sharpe ratio is the portfolio risk premium divided by the portfolio risk. That is,

$$Sharpe\ ratio = \frac{R_p - R_f}{\sigma}$$

Where:

$R_p$ = *Expected portfolio return*

$R_f$ = *Risk-free rate of return*

$\sigma$ = *Standard deviation of the portfolio*

Sharpe ratio is a less effective measure of alternative investment performance due to the fact that it assumes that returns are normally distributed. The return profiles of alternative investments are often asymmetrical and skewed due to underlying risks such as illiquidity and poor portfolio transparency.

2. **Sortino Ratio**

Expressed in an equation as:

$$Sortino\ ratio = \frac{R_p - R_f}{Downside\ deviation\ of\ the\ portfolio}$$

The Sortino ratio is better than the Sharpe ratio because it focuses on the performance relative to downside volatility. As such, it captures the effect of skewness and kurtosis.

3. **Calmer Ratio and MAR Ratio**

The Calmar Ratio and MAR Ratio are improved substitutes for the Sharpe ratio. They are used when evaluating the performance of alternative investments by comparing performance to a drawdown. A **Drawdown** is the net asset value's peak-to-trough (high-to-low) percentage decline (NAV). A portfolio's **Maximum Drawdown** (MDD) is the largest loss an analyst can observe between a portfolio's peak and trough before a new peak is reached.

# Reading 60: Private Capital, Real Estate, Infrastructure, Natural Resources, and Hedge Funds

Every possible investment choice is either an alternative or a traditional investment. Traditional investments are investments in long-only, publicly traded investments in stocks, long investments in publicly traded bonds and cash while alternative investments include assets such as private capital, real assets, and hedge funds.

Alternative investments are useful to an investor for reasons such as:

- They offer a range of investments that have low correlations with traditional investments.
- They allow investors to enhance the risk-return characteristics of their portfolios and increase their returns.
- They improve an investor’s return potential thanks to their higher yields.

The main characteristics of Alternative Investments include:

- Investment managers limit their specialization to alternative investments.
- They have a relatively low correlation of returns to traditional investments.

- They have less transparency and regulation compared to traditional investments.
- They have inadequate historical risk and return data.
- They have unique tax and legal issues.
- They have higher fees, usually made of performance (incentive) and management fees.
- They are typically concentrated portfolios.
- They usually have limitations on redemptions, such as lockups and gates.

Private capital is categorized into **Private Equity** and **Private Debt.** A private equity investor may directly or indirectly participate in an alternative investment through private equity funds. Private equity funds make investments in both new and established businesses. Note that these firms may not be listed on a public exchange. Some of the primary types of private equity include:

- **Leveraged Buyouts** is primarily provided to private entities. Examples of private debt include direct lending, mezzanine loans, venture debt, and distressed debt.
- **Venture Capital** refers to providing financial support to private companies with high growth potential. VC can be provided at any stage of growth, but it is usually extended to start-ups with a high-risk potential. Venture capitalists invest in companies and earn an equity interest. There are three stages of venture capital financing: **Formative-Stage Financing** (pre-seed, seed-stage, and early-stage financing), **Later-Stage Financing** (expansion VC), and **Mezzanine-Stage Financing** (financing until the IPO is completed or sold). Venture capitalists are active investors who participate in the company's decision-making processes.
- **Growth Capital** is primarily provided to private entities. Examples of private debt include direct lending, mezzanine loans, venture debt, and distressed debt.

**Exit Strategies** for private equity funds include trade sale, initial public offering (IPO), recapitalization, secondary scales, and liquidation or write-off.

**Trade Sale** involves selling a company to a strategic buyer, while **IPOs** involve selling shares to the public. **Recapitalization** is not considered a true exit strategy but can be a sign of a future exit. **Secondary Scales** involve selling a company to another private equity firm or group of investors, while **Liquidation** occurs when a private equity firm adjusts its investment value downward in a portfolio company.

Private equity funds carry a higher risk than traditional investments, and investors demand a higher return in exchange for taking on more risk, including illiquidity and leverage.

Alternatively, private equity funds may invest in public companies with the intent to privatize them. On the other hand, private debt is primarily provided to private entities. Examples of private debt include direct lending, mezzanine loans, venture debt, and distressed debt.

Private debt investments generally have higher yields compared to traditional fixed-income investments such as bonds. However, this higher yield comes with increased risk. Private debt investments have varying risk profiles depending on the type of private debt investment.

**Direct Lending** and **Mezzanine Debt** are less risky than **Venture Debt** and **Distressed Debt**. Direct lending and mezzanine debt are typically structured loans, and investors have priority over equity holders in the case of default. Venture debt and distressed debt carry higher risks since they are usually unsecured loans, and the borrower may default on their payments.

Investing in **Collateralized Loan Obligations (CLO)** and **Unitranche Debt** allows investors to diversify their investments across multiple loans, which reduces the risk associated with investing in a single borrower. However, investing in specialty loans may carry higher risks, depending on the niche borrower's creditworthiness.

Private debt investments offer higher returns than traditional fixed-income investments but come with higher risks. The risk-return characteristics of private debt depend on the type of private debt investment, and investors should carefully consider their risk appetite before investing in private debt.

Another alternative investment category is the **Real Assets,** and they include:

- **Real Estate Investments** are made in buildings or land, either directly or indirectly and they include private commercial real estate debt and private commercial real estate equity.
- **Infrastructure Assets** are long-lived, capital-intensive real estate resources such as airports and power plants. A popular method of investing in infrastructure is the **Public-Private Partnership** (PPP) strategy, in which case both public authorities and private investors have an interest.
- **Natural Resources** include commodities, agricultural land (or farmland), and timberland.

**Hedge Funds** is another alternative investment category. Hedge funds employ numerous strategies to offer investors optimal returns. These private funds are actively managed—often by "star investors"—and apply aggressive strategies across asset classes. Most hedge funds use derivatives, high leverage, arbitrage, and other investment strategies unavailable to traditional fund managers.

Other alternative investments include collectibles and other investments that don't fall into the traditional categories. Collectors have been around for centuries, and the investments are as varied as the collectors themselves. Traditional collectibles such as stamps, cars, furniture, gems, fine art, and others still dominate the market and new/emerging collectibles, include baseball cards, action figures, and almost anything you can imagine, offer investors new ways to invest their money outside traditional financial markets.

Real estate investing is the process of buying, owning, managing, and selling real estate properties to earn a profit. It can be classified into two main categories: **Equity Investing** and **Debt Investing**. Equity investing involves direct or indirect ownership of a property, while debt investing involves lending through mortgages.

Real estate investing can be further categorized into various forms, including:

- **Residential Real Estate** refers to direct equity investment in a residence, such as a single-family home or multi-family attached units. Investors can purchase a residential property either for private use, investment, or both.
- **Commercial Real Estate** includes office buildings, retail shopping centers, warehouses, hotels, and other income-generating properties. Commercial real estate is usually more complex and illiquid, requiring regular management and is popular among institutional funds and high-net-worth individuals.
- **Real Estate Investment Trusts (REITs)** are tax-advantaged investment vehicles that allow individuals to invest in real estate assets without actually owning physical property. REITs own and manage income-producing properties such as apartment buildings, office buildings, retail spaces, and industrial facilities, among others.
- **Mortgage-Backed Securities (MBS)** are investment products that are created by pooling together mortgage loans and then selling securities that represent an ownership interest in that pool. The mortgage loans that are pooled together are typically residential mortgages, such as those used to purchase homes or other properties.

Real estate ownership can also be classified based on types and sources of capital. The types of capital are debt and equity capital, while the sources of capital are private or public.

**Direct Real Estate Investing** involves an investor purchasing a property using their own funds or debt financing. The investor may manage the property themselves or hire a separate account to do so. Alternatively, they may join a joint venture with other investors. **Indirect Real Estate Investing** involves investing in real estate through public or private pooled investment vehicles. These may include **Real Estate Investment Trusts (REITs),** mortgages, or private equity funds. The risk and return characteristics of real estate depend on various factors, including economic conditions and the management of the underlying properties. While direct real estate investing allows the investor more control over their investment, it is also more capital intensive and requires significant expertise in local real estate markets.

# Portfolio Management

## Reading 61 – Portfolio Management: An Overview

Portfolio management is about creating a diversified approach to meet one's investment goals. Diversification involves avoiding too much exposure to a single asset or asset type. Diversifying the risks of a portfolio helps reduce downside risk without necessarily decreasing the expected rate of return. Portfolio risk is measured by the standard deviation of returns, and the correlations between different assets can lead to decreased overall risk when combined. The principle behind this diversification effect is based on Harry Markowitz's research and is known as Modern Portfolio Theory.

There are two broad categories of investors: individuals and institutions. **Individual Investors** are individual people or families that are investing to meet their personal goals. **Institutional Investors** are professional organizations that represent a variety of investment goals. These include groups such as pension funds, endowments, banks, investments funds, and even charities.

Employees of public and private companies have retirement savings managed in accounts that can be either defined benefit or defined contribution. **Defined Contribution** plans involve employers making specific contributions to accounts that will accrue in value for their employees. The final amounts available for retirement will depend on the performance of financial markets. In a **Defined Benefit** plan, however, the employer is responsible for providing a set number of financial benefits to their employees in retirement. This financial obligation means that the portfolio risk must be managed much more closely by the employer.

There is a series of steps that must be followed in the portfolio management process.

### 1. The Planning Step

The client's investment objectives, constraints, and portfolio benchmark need to be documented in an **Investment Policy Statement** (IPS), which is the document by which the investments will be managed.

**2. The Execution Step**

a. **Asset Allocation**: The portfolio manager needs to develop a view on the risk and return expectations of various asset classes. This analysis step can be top-down (starting from high-level macroeconomic factors) or bottom-up (starting from company-specific information). Decisions are made as to the optimal allocation of the portfolio assets among available asset classes.

b. **Security Analysis**: Specific securities are chosen for purchase that fit into the asset classes chosen in the previous step.

c. **Portfolio Construction**: The information from the previous steps is used to create an investment portfolio. Securities are purchased and trades are executed.

**3. The Feedback Step**

The portfolio is monitored and rebalanced periodically to keep its exposures in line with the **Investment Policy Statement**. Performance is also tracked and reported to the client at regular intervals.

Asset managers can either be **Active** or **Passive.** Active asset managers attempt to outperform benchmarks through fundamental and quantitative research. Passive managers simply try to replicate the returns of a market index.

There can also be either **Traditional** or **Alternative** managers. Traditional managers focus on creating diversified portfolios for their clients by using long stocks and bonds. Alternative asset managers use leverages, derivatives, etc. to create a return that is uncorrelated to the market/to outperform a pre-determined index.

In addition to building a portfolio of specific investment assets, there are also pooled investment vehicles that can be purchased. **Mutual Funds** are one example, where each investor in the mutual fund owns a pro-rata claim on the value of the securities that the mutual fund owns. In an **Open-End Fund**, investors can buy or sell positions in the fund and the portfolio manager will adjust the holdings of the fund to meet these changes in investable assets. **Closed-End Funds**, on the other hand, do not create new shares to allow for investors to enter or exit the fund. There are a fixed number of shares in the fund and any new investor that wishes to purchase must buy existing shares from a current owner.

Funds can be classified as **Load** or **No-load Funds** depending on whether there is a fee that must be paid to buy or sell shares of the fund. No-Load funds typically charge an annual management fee that is a percentage of total fund assets.

Mutual funds are categorized by the assets in which they invest. **Money Market Funds** invest in very short-term debt and are often treated as a substitute for bank deposits. **Bond Mutual Funds** buy fixed income securities of varying maturities. **Stock Mutual Funds** buy equity securities and **Hybrid/Balanced Funds** buy combinations of different asset classes. Funds can be managed either actively or passively, with passive funds trying to closely match a given index through a buy-and-hold approach and active funds doing more buying and selling to try and maximize performance results.

Other types of pooled investment vehicles include **Exchange Traded Funds** (ETFs), which are similar to mutual funds except that shares trade on the open market. A new investor in a mutual fund buys shares at the end of day Net Asset Value directly from the mutual fund provider, while a new investor in an ETF buys shares on the exchange like they would do for a regular stock. **Separately Managed Accounts** are similar to mutual funds but are created specifically for an individual or institutional client. They can be customized to fit any manner of investment or tax strategy. **Hedge Funds** are typically more complex in nature and are more loosely regulated than mutual funds. They usually require high minimum investments and have liquidity restrictions on invested assets.

## Reading 62 – Portfolio Risk and Return: Part I

The purpose of investing is to earn a return on your assets, so it's important to be able to calculate that return and the risk taken on to achieve it. There are a number of different types of returns included in the curriculum. The simplest to calculate is the **Holding Period Return**. This is simply the total investment gains as a percentage of the starting amount:

$$Return = \frac{Ending\ value - Starting\ value + Dividends}{Starting\ value}$$

This will give you the return for a single holding period, but you will also need to know how to combine multiple return periods. The first of these combination methods is to calculate the **Arithmetic Mean**:

$$Arithmetic\ Mean(5\%, 10\%, 15\%) = \frac{5 + 10 + 15}{3} = 10\%$$

A more complex approach to combining multiple return periods is the **Geometric Mean.** This utilizes the principle of compounding, so it's more accurate as a method of calculating the returns of a portfolio over time. Remember to convert all percentages to decimal values when calculating the geometric mean:

$$Geometric\ Mean(5\%, 10\%, 15\%) = \left[\left((1 + 0.05) \times (1 + 0.1) \times (1 + 0.15)\right)^{\frac{1}{3}}\right] - 1 = 9.9\%$$

One limitation of the arithmetic and geometric means is that they do not consider the money invested in the portfolio over time. For example, if a portfolio experiences large growth early when its asset base is low and then lower returns in later years after attracting large new cash flows, its performance will look much better than what is experienced by most of the investors who were not around for the earliest years. The performance "per dollar" in the fund will be much lower than what the calculated means would indicate. Money-Weighted Returns consider these flows by using the fund in- or out-flows to create a weighted average return. The IRR of a fund can also be useful in this regard. Similar to how it was used in the Quantitative Methods section, the Internal Rate of Return is the discount rate that sets the present value of the cumulative cash flows to zero.

The **Money-Weighted Rate of Return** (MWRR) is the IRR of a portfolio. It is the discount rate at which $PV\ of\ cash\ outflows\ =\ PV\ of\ cash\ inflows$. You solve an MWRR problem the same way that you solve other IRR problems using your calculator. Since this method involves finding the PV of each cash flow, it is sensitive to when funds are invested or removed from the portfolio.

On the other hand, the **Time Weighted Rate of Return** (TWRR) is the geometric mean for a series of **Holding Period Returns** (HPRs).

$$TWRR\ of\ a\ portfolio = \{(1 + HPR_1) \times (1 + HPR_2) \ldots \times (1 + HPR_n)\} - 1$$

By combining each holding period return into a geometric mean, the TWRR method gives a more robust return figure that is **not** influenced by the timing of cash flows.

You will sometimes need to compare returns for time periods of varying lengths. The **Annualized Return** calculation is used. Annualizing a return essentially compounds it by the number of times it takes for that period to equal one year. For example, to annualize a monthly return, you would compound it 12 times:

$$Annualized\ 5\%\ monthly\ Return = (1 + 0.05)^{12} - 1 = 79.6\%$$

Another important measure is the **Portfolio Return**, which is a weighted average return of all the components in a portfolio. Questions around this calculation typically give you the portfolio components and ask you to calculate the total return. All you do is multiply each component by its weight in the portfolio and add up the results.

All of the returns discussed so far do not factor in adjustments that must often be made when dealing with real investments and portfolios. You may be asked to calculate returns either **Gross** or **Net** of fees. Net returns are after all fees have been subtracted. Taxes are also relevant to most investments. Post-tax returns are what is left after taxes have been paid. The effects of inflation also impact investor returns. **Nominal** returns are what we have been calculating so far, and **Real** returns are when the effects of inflation are accounted for.

The **Money-weighted Rate of Return (MWRR)** refers to the internal rate of return on a portfolio. It is the rate of discount, r, at which:

$$PV\ of\ cash\ outflows = PV\ of\ cash\ inflows$$

When dealing with an investment portfolio, cash inflows comprise of the beginning value, dividends /interest reinvested, and withdrawals made while the cash outflows comprise of the final value of the fund, dividends or interest received and contributions.

The **Time-Weighted Rate of Return** (TWRR) measures the compound growth rate of an investment portfolio. TWRR is not sensitive to withdrawals or contributions. The time-weighted rate of return is the geometric mean of the holding period returns of the respective sub-periods involved. When working out time-weighted measurements, we break down the total investment period into many sub-periods. Each sub-period ends at the point where we have a significant withdrawal or contribution. The following are the steps to follow when calculating TWRR:

1. Establish the holding period return (HPR) for each sub-period.

2. Add 1 to each HPR.

3. Multiply all the (1 + HPR) terms.

4. Subtract 1 from the final product to get the compounded TWRR.

$$\begin{aligned} Compounded\ TWRR \\ &= \{(1 + HPR_1) \times (1 + HPR_2) \times (1 + HPR_3) \ldots \\ &\times (1 + HPR_{n-1}) \times (1 + HPR_n)\} - 1 \end{aligned}$$

$$Annual\ time\ weighted\ rate\ of\ return\ =\ (1\ +\ Compounded\ TWRR)^{\frac{1}{n}} - 1$$

It's important to understand the relative risk and return characteristics of major asset classes. For US equity securities, the smaller the company is, the higher the average return and variance of returns are. In fixed income securities, the longer the maturity of the security, the higher the average return (yield) and variance of return. Corporate bonds also return more (with higher variance) than government bonds of the same maturity. The primary concept to grasp when looking at the risk and return characteristics of different asset classes is that a higher average return with higher variance means that you cannot expect to get that increased performance consistently. A higher variance means you will see wide fluctuations in returns between different years. The primary measurement of the variation of returns is in standard deviations, which are the square root of variance.

Return characteristics between asset classes utilize what is known as their variance/covariance relationships. These are based on a set of common data points that should look familiar from the Quantitative Methods section. Each asset class will have its own **Mean** returns, which refers to the average values they have over time. They will also have their own distinct **Variance**, which is the dispersion around the mean (basically the "risk" part of the risk/return framework). The **Covariance** of asset classes measures how closely they move in relation to one another. Assets that move very differently provide diversification benefits to the portfolio when added together.

Each investor will differ in how they view risk as part of their investment portfolio. **Risk-Seeking** behavior is when investors are willing to take on additional risk, even if it doesn't seem like they are being adequately compensated for it with higher expected returns. **Risk-Neutral** investors will take on additional risk only when it's accompanied by an appropriately higher return. **Risk-Averse** investors will avoid additional risk and accept lower returns as a result. This measurement of willingness to take on risk is known as **Risk Tolerance**.

Earlier we covered the simple calculation for the portfolio return, where you multiple each portfolio component's return by its weight and then add these up. When it comes to risk in the portfolio, the calculation is a bit more complex (and you should know it, because you will probably get a question on this in your CFA level 1 exam). To calculate the standard deviation of a 2-asset portfolio, the formula is as follows:

$$Portfolio\ variance = (w_A^2 \times \sigma_A^2) + (w_B^2 \times \sigma_B^2) + (2 \times w_A \times w_B \times \sigma_A \times \sigma_B \times \rho_{AB})$$

$$Portfolio\ standard\ deviation = \sqrt{(w_A^2 \times \sigma_A^2) + (w_B^2 \times \sigma_B^2) + (2 \times w_A \times w_B \times \sigma_A \times \sigma_B \times \rho_{AB})}$$

Where:

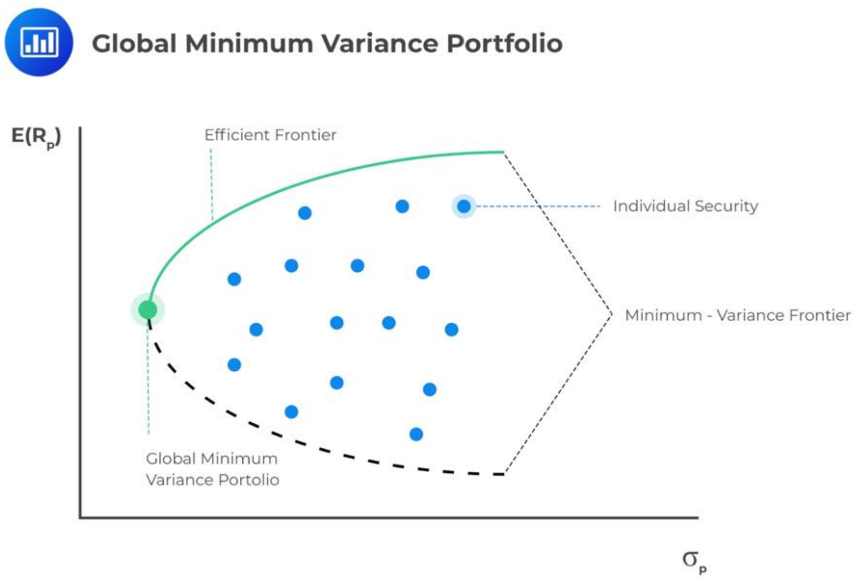
$w = Portfolio\ weight$

$\sigma = Standard\ deviation$

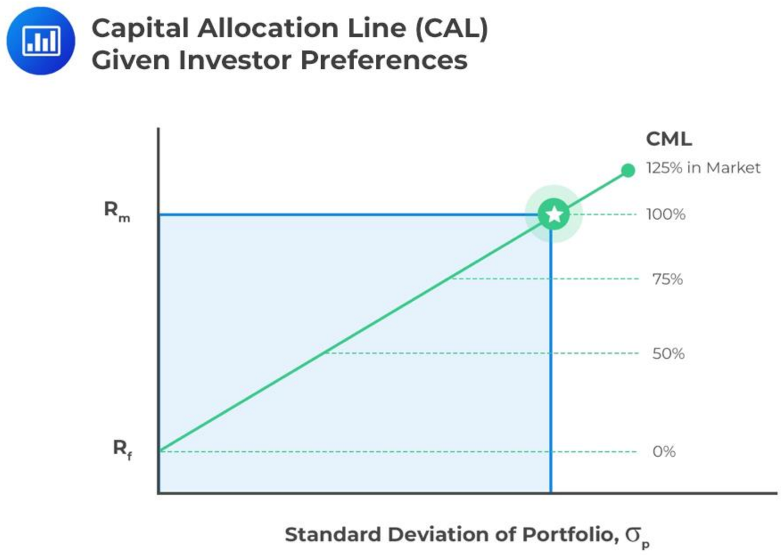
$\rho = Correlation\ coefficient$

$Note: Cov(A, B) = \sigma_A \times \sigma_B \times \rho_{AB}$

Since there is an infinite number of portfolios you could build based on different weights of asset classes, we need a way to determine what characteristics a portfolio needs to have to meet the needs of a specific investor. The **Minimum-Variance Frontier** is a way to model different portfolios along an axis of risk and return in order to find the most appropriate one. By choosing the portfolio with the least risk that meets the necessary return requirements, we ensure that an investor is maximizing their chances of meeting their investment goals. The line on the risk/return graph above which all portfolios meet the needed investment returns without undue risk is known as the **Efficient Frontier** and is represented on the graph below.




Another concept in this reading is known as the **Capital Allocation Line (CAL)**. This is based on the principle of the two-fund separation theorem, which posits that all investors will hold a portfolio that combines two assets, one risk-free portfolio and one portfolio of risky assets. Breaking the portfolio into these components allows us to graph a line showing the expected risk and returns as the portfolio weight of the two assets changes. The line represented by this is the CAL. The investor's optimal portfolio will appear along this line.




In order to determine the appropriate level of risk, it is necessary to calculate the investor's **Risk Aversion**. The formula for this is:

$$u = E(r) - \frac{1}{2}A\sigma^2$$

$U = Utility$

$E(r) = Expected\ return$

$A = Risk\ aversion\ coefficient$

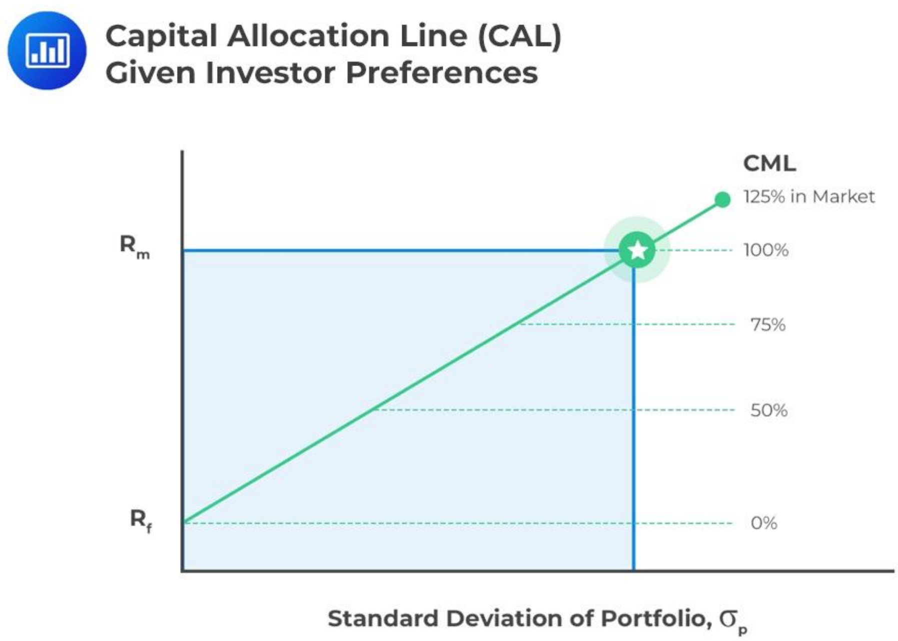
$\sigma^2 = Portfolio\ variance$

The risk aversion coefficient will be positive for risk-averse investors, zero for those risk neutral, and negative for those who are risk-seeking.

# Reading 63 – Portfolio Risk and Return: Part II

As discussed in the previous reading, an investor's risk tolerance can be quantified in portfolio construction using two assets: one totally risk-free and one composed of risky securities. The approach an investor takes can be determined by their level of risk tolerance and belief that they can achieve superior returns to the market. Investors that are more risk-averse and believe that they can only hope to match market returns will take a **Passive** investing approach, whereas more risk-tolerant investors who believe they can identify mispricing to outperform the market will take an **Active** investing approach.

This principle leads to the use of the **Capital Market Line (CML)**. This is a case of the Capital Allocation Line (CAL), where the risky asset in the equation represents the entire market portfolio. We use this approach to find the most appropriate portfolio by looking at where on a graph the risk/return line for the portfolio matches up to the efficient frontier (which is covered in the previous section as well).




The return and variance for the portfolio found using this method is calculated as:

$$R_p = wR_f + (1 - w)R_m$$
$$\sigma_p^2 = (1 - w)^2\sigma_m^2$$

$R_f = Risk\ free\ return$

$R_m = Market\ return$

$\sigma_p = Portfolio\ standard\ deviation$

$\sigma_m = Market\ standard\ deviation$

The risk an investor takes on by buying risky assets can be broken down into two categories: Systematic and Non-Systematic. **Systematic Risk** is the risk inherent to investment markets in general. It reflects the impacts of things like the business cycle or political uncertainty. This risk cannot be diversified away. **Non-Systematic Risk** is the risk inherent to specific assets or asset classes. It can be diversified away by purchasing assets with low or negative correlations. Investors should try to avoid (or diversify away if possible) any risk for which they are not compensated by higher returns.

The curriculum describes several models that investors can use to estimate the return of a portfolio based on the assets they plan to include. The most common type of these is a **Multi-Factor Model**, which allows for several inputs in order to model the expected return. These models can use macroeconomic, statistical, or fundamental factors that can be combined in order to develop the most appropriate estimate of expected return. A common calculation approach is to find a sum of the return attributed to each return factor multiplied by their respective Betas. This "weighted average" result will approximate the expected return based on the historical returns and the exposure to/correlation with the investment portfolio.

One of the most common multi-factor models is the three-factor model developed by Fama and French, which was later expanded to four factors. The original three factors they included were relative size of the company, relative book-to-market value of the company, and the market beta. The fourth factor added later was security momentum.

The simplest type of return generating model is the single factor model. A common model of this type is the market model, where returns are estimated based on the exposure to the market:

$$R = R_f(1 - \beta) + \beta R_m$$

Where:

$R_f = Risk\ free\ rate$

$R_m = Market\ return$

$\beta = Market\ beta$

Multi-factor models build on this concept by including more factors and their respective betas in order to break down the components of the expected return.

Calculating the risk of these portfolios (the exam typically will most likely ask for risk calculations for single factor models due to the complexity of multi-factor formulas) is similar to other multiple asset portfolios but simplified slightly because the risk (and therefore correlation) of the risk-free asset is zero. As a result, the calculated Beta for a single factor portfolio is:

$$\beta = \frac{Covariance(R_p, R_m)}{\sigma_m}$$

Interpreting beta is the same as correlation was used in the Quantitative Methods section. A positive market beta means the asset moves in the same direction as the market, while a negative value means they move differently. Unlike correlation, beta values can be more than 1 and less than -1.

The **Capital Asset Pricing Model (CAPM)** is used to illustrate a linear relationship between the expected return and risk of an investment asset. Since it is a simplified view, it relies on a series of assumptions about market behaviors that will not always line up with real-world results. The primary assumptions of the model include that all investors are risk-averse, utility-maximizing, and fully rational. This assumption is necessary to make the model behavior more predictable, but it's important to remember that investors in real life exhibit a wide variety of behaviors based on their own risk tolerance and preferences. The CAPM also assumes that there are no transactions costs or taxes and that investments are only held for one, uniform-length holding period.

The **Security Market Line (SML)** is the graphical representation of the CAPM using beta as the x-axis and expected return on the y-axis. The SML can be represented formulaically as:

$$E(R_p) = R_f + \beta(E(R_m) - R_f)$$

In this formula, the expected return of the portfolio is the risk-free rate plus the risk premium (market return minus risk-free rate) times the market beta. This is a crucial concept for your level I exam.

The returns calculated using the CAPM and other models can be used in a number of ways related to portfolio management. The data they provide serve as inputs for several important formulas that show up repeatedly on the exam, and also on the level 2 and 3 exams. The **Sharpe Ratio** is the slope of the CAL and is calculated as the risk premium divided by the standard deviation of the portfolio. It tells you how much return you are getting in exchange for taking on additional risk. The **Treynor Ratio** is similar to the Sharpe ratio but uses the market beta as the dividend. It tells you how much additional risk you are taking on relative to total risk. The **M-squared Ratio** tells you the weight of the risky asset you want in the portfolio in order to set portfolio risk equal to market risk. **Jensen's Alpha** finds the difference between the actual return of the portfolio and the calculated, risk-adjusted return.

$$Sharpe\ Ratio = \frac{R_p - R_f}{\sigma_p}$$

$$Treynor\ Ratio = \frac{R_p - R_f}{\beta_p}$$

$$M\ squared\ Ratio = \frac{\sigma_m}{\sigma_p}$$

$$Jensen's\ alpha = R_p - \left[R_f + \beta_p\left(R_m - R_f\right)\right]$$

# Reading 64 – Basics of Portfolio Planning and Construction

The most important part of constructing an investment portfolio is making sure that you have a solid understanding of the investment objectives that you are trying to achieve. The document that captures all of the appropriate investment goals and constraints is known as the **Investment Policy Statement (IPS)**. As outlined in the curriculum (and in the curriculum for future levels), the creation of the IPS is the first part of an advisor's job when working to determine the best approach for a client.

A typical IPS will follow the structure below:

- Introduction
  - Description of the client
- Statement of Purpose
  - Purpose of the IPS
- Statement of Duties and Responsibilities
  - Duties and responsibilities of all parties – client, advisor, custodians, etc.
- Procedures

    - The steps that will be undertaken to keep the IPS updated and procedures to respond to various contingencies.
- Investment Objectives
    - The client's investment objectives
- Investment Constraints
    - The client's investment constraints
- Investment Guidelines
    - Information on how the investment policy should be executed and includes the permissibility and exclusion of assets.
- Evaluation and Review
    - Guidance on obtaining feedback for investment results.
- Appendices
    - Strategic Asset Allocation: the policy portfolio that provides the baseline portfolio asset allocation.
    - Rebalancing policy

The primary components of the IPS that relate to the portfolio construction process are the investment objectives and constraints. These will outline the return and risk guidelines that will need to be followed throughout the portfolio creation and management process. The client's risk objectives must be clearly defined and can be stated in either absolute or relative terms. Their risk tolerance is a function of both their willingness and ability to bear risk and needs to reflect both concerns. **Willingness** represents the client's psychological comfort with investment risk, while **Ability** is a quantitative measure of the client's assets and financial ability to absorb losses and volatility. When the willingness and ability are not in alignment, it is up to the advisor to explain the implications of this to the client even though it is often not possible to change the client's mind. The prudent action is to adhere to the risk tolerance that is the lower of the two factors.

Similarly, the return objectives can be defined in either absolute or relative terms. These are typically based on an appropriate benchmark measure. For cases like a pension fund, the return objectives can be based on specific obligations that the portfolio is responsible for meeting. These returns should be stated as gross or net of fees and taxes and need to be appropriate based on the conditions of the broader market and macroeconomic factors.

There are several important investment constraints that must be outlined in the IPS that will impact how the portfolio is constructed and managed. **Liquidity** needs must be specified if they can be quantified and, if they cannot be, accounted for to create certainty that future cash flow needs can reasonably be met. The investment **Time Horizon** is an important concern that will affect which assets or asset classes can be included in the portfolio. **Tax Concerns** have a tremendous effect on investment results and so much be properly accounted for by the advisor. **Legal and Regulatory Factors** are more varied by where the client and the portfolio reside and will sometimes involve the use of outside experts such as lawyers to handle properly. **Unique Circumstances** is the final major category of limitation and covers any special concerns that do not fit in the previous sections. These concerns are often referred to as "ESG", which stands for Environmental, Social, and Governance. This section includes numerous ethical, religious, or other limitations on how the client can invest their money.

The first step in building a portfolio according to the information in the IPS is making the asset allocation decisions. The primary predictor of long-term returns is the asset allocation of the portfolio. The **Strategic Asset Allocation (SAA)** is the formal specification for how portfolio funds will be invested among available asset classes. The Capital Market Expectations for each asset class will determine the expected performance of the portfolio once it is allocated among available investment options. The level of granularity among asset classes used to specify the SAA is somewhat flexible according to client preferences. How broadly or narrowly they define each asset class is up to them as long as they adhere to a few criteria. The risk and return characteristics must be homogeneous among assets within each class, each class must be mutually exclusive, and each class must be a representation of the investable universe.

**Environmental, Social and Governance** (ESG) considerations can be implemented across all asset classes. Investors may choose to not invest in asset classes that go against ESG considerations, for example, by instructing the asset manager not to invest in military equipment stocks. There are some benchmarks that reflect excluded companies and sectors.

# Reading 65 – The Behavioral Biases of Individuals

Behavioral biases are irrational beliefs or behaviors that can influence our decision-making process. Behavioral biases may make our decisions to stray from what is prescribed or recommended, if you may, by the traditional finance theory. Behavioral biases are split into two types:

i. **Cognitive Biases** refer to erroneous ways of thinking, reasoning, and judgment that characterize how we process and interpret information. **Emotional Bias** occurs when our emotions blur our sense of judgment, reasoning, and information decoding processes.

ii. **Cognitive Biases**, also known as faulty cognitive reasoning, describe erroneous ways of thinking, reasoning, and interpreting information. They can be split into:

   a. Belief perseverance bias.

   b. Processing errors

An **Emotional Bias** is a distortion of cognition and decision-making that results from emotional factors. Emotional biases are more challenging to correct than cognitive errors since they are based on impulses or intuition rather than conscious judgments. There are six emotional biases:

i. Loss aversion.

ii. Overconfidence.

iii. Self-control.

iv. Status quo.

v. Endowment.

vi. Regret aversion.

Some persistent market patterns such as momentum, value, bubbles, and crashes impact market efficiency and are regarded as functions of behavioral finance.

**Market Anomalies** are noticeable departures from the efficient market hypothesis, as evidenced by persistently abnormal returns. **Momentum** is when future price behavior aligns with that of the recent past. **Bubbles** lead to such anomalies as overtrading, underestimation of risks, failure to diversify, and rejection of any piece of information that contradicts their assessment of the market. A **Market Crash** is a rapid and often unanticipated drop in prices that can be a side effect of a major catastrophic event, economic crisis, or the collapse of a long-term speculative bubble.

# Reading 66 – Introduction to Risk Management

Risk management is an important part of managing an investment portfolio. At an institutional level, risk management can include risks to the financial investments as well as operational risks on the enterprise side. For individuals, risk management is often more informal and unstructured.

A proper risk management framework is necessary for an institution to properly track and manage risks to its investment outcomes, regardless of the sources of risk. There are a number of important areas to be addressed by this framework. **Risk Governance** is the top-level system of structures and policies that ensures risk management supports the entire enterprise. **Risk Identification and Measurement** is the quantitative tracking and monitoring risk exposures.

**Risk Infrastructure** refers to the people and systems that carry out the risk management process. **Policies and Processes** provide constraints and guidelines for the ongoing daily operations of the firm. **Risk Monitoring, Mitigation, and Management** is the continuous evaluation of current and potential risk exposures.

**Communication** is important to keep risk information available to all affected parties in the organization. **Strategic Analysis or Integration** analysis identifies which risk activities are improving the process outcomes and which ones can be removed.

Having a risk management framework will reduce the frequency of surprises that impact the investment outcomes of the firm. It also increases the discipline around decision making and reduces operational errors. The results of having a proper framework include reputational improvements and better relationships with regulating entities.

**Effective Risk Governance** involves a visible commitment from a firm's governing body. They must be comfortable discussing risk exposures and taking preventative action even in periods of relative calm. This process needs to be comprehensive around the entire enterprise and include concerns for risks that cannot be captured through quantitative measures.

Any investor must identify the **Risk Tolerance** they have in order to set their investment expectations. There are many factors that go into this determination, including the ability to withstand losses, the competitive landscape, and the regulatory environment in which they do business. The risk tolerance should not reflect personal motivations or beliefs. The tolerance should be determined in a non-crisis period so that it can serve as a strategic guideline over a long period of time.

**Risk Budgeting** is a strategy to set useful guidelines by which day-to-day investment decisions can be made. The risk budget will quantify risk and allocate the exposures across the organization. Common risk measures used for budgeting include Standard Deviation, Beta, Value at Risk, and Scenario Losses.

Multi-dimensional approaches combine several factors in order to provide a more comprehensive view of risk.

**Financial Risks** originate from financial markets and might arise from changes in share prices or interest rates. There are three types of financial risks: market risk, credit risk and liquidity risk. **Non-Financial Risks** emanate from outside the financial market environment and could be consequences of environmental or regulatory changes or an issue with customers or suppliers. Non-financial risks include settlement risk, legal risk, compliance risk, model risk, operational risk and solvency risk.

**Value at Risk** (VaR) can be defined as the maximum amount of loss that can be incurred with a given confidence interval (under normal business conditions). A monthly VaR of $100 million at 95% confidence implies that the probability of losing $100 million or more in any given month is 5%.

# Reading 67 – Technical Analysis

Technical analysis is the practice of using price and volume data to value stocks.

The main assumptions inherent in technical analysis are:

- Market forces (supply and demand) determine the market value of assets.

- Rational factors as well as irrational ones, i.e., data and guesswork respectively, collectively drive supply and demand.

- Individual stock prices change to reflect periodic trends exhibited by the market as a whole.

- Patterns and trends are repetitive. Therefore, we can actually predict future price movements.

- Emotions play a major role in determining human behavior, which can be quite erratic.

Principles of technical analysis are:

- The market discounts everything. It is considered that a financial instrument's price reflects all of the factors that influence it.

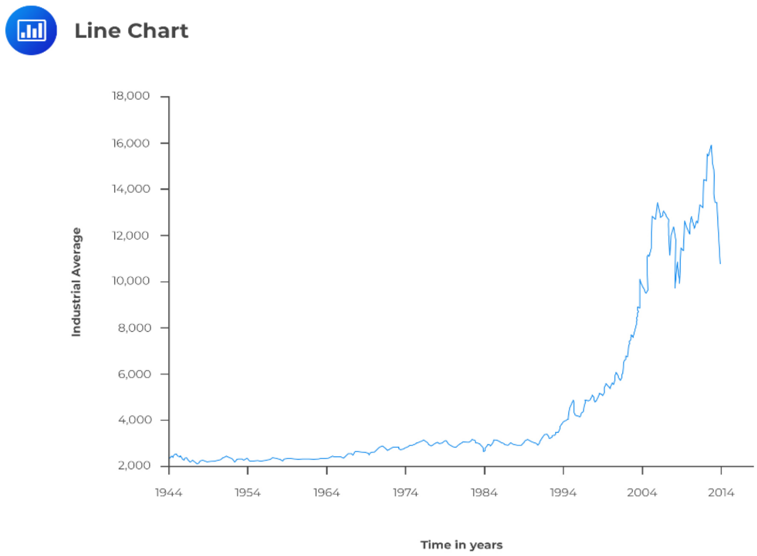
- Price moves in trends and countertrends. Prices follow trends, and if a trend is identified, the future price movement will be in line with it.

- Price action creates patterns that tend to repeat and may be cyclical. Market psychology is the reason price movements repeat themselves. Price changes can then be charted allowing technicians to spot patterns.

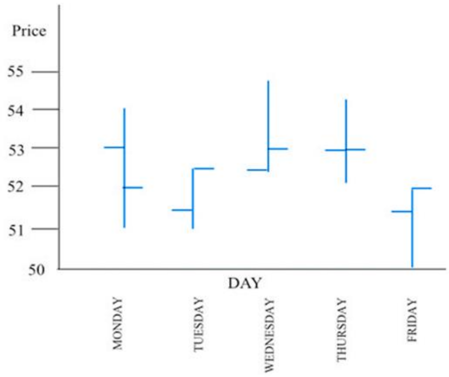
A technician's research is solely based on **Price** and **Volume Data**, and he believes that all factors are reflected in a security's price, whereas a fundamental analyst examines in detail with careful or critical attention a specific company. Fundamental analysts utilize multiple estimations and assumptions, whereas technical analysts use facts such as price and volume. Fundamental analysis is a more theoretical approach, whereas technical analysis is a more practical way.

Technical analysts are restricted to market fluctuations and do not employ other predictive analytical techniques, such as forecasting a company's future sales. Both technical and fundamental analysis can be utilized in tandem. Technical analysts often use fundamentals to back up their trades.

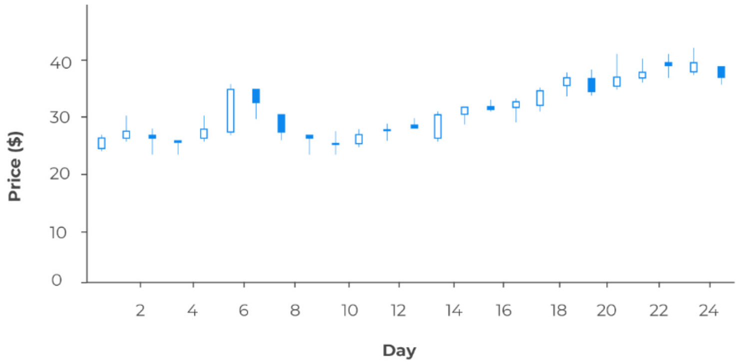
The major types of technical analysis charts are:



## Bar Chart




## Candlestick Charts




**Volume Charts** can be found below many charts. If an increase in price coincides with an increase in volume, analysts consider such a scenario as a very positive technical development. If the price increase is accompanied by dwindling volume, it means fewer and fewer buyers are willing to trade the increase in price is expected to eventually stop.

There are a number of concepts you'll be expected to understand related to technical analysis on the exam. One of these is **Trend**, which is a when a security consistently keeps moving upward or downward over time. This is the market's way of reflecting a difference between supply and demand that can persist for a while.

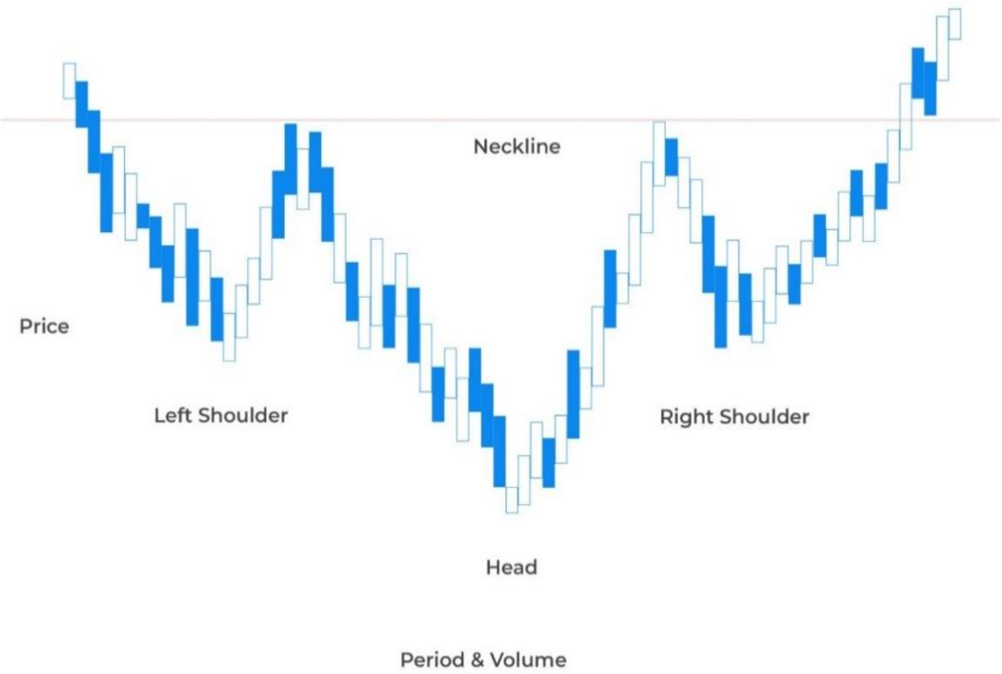
Another important concept is that is **Support** and **Resistance** levels. These are price points where market activity tends to keep a security from going below

(Support) or above (Resistance). These levels can act as floors or ceilings to the trading level of a security.

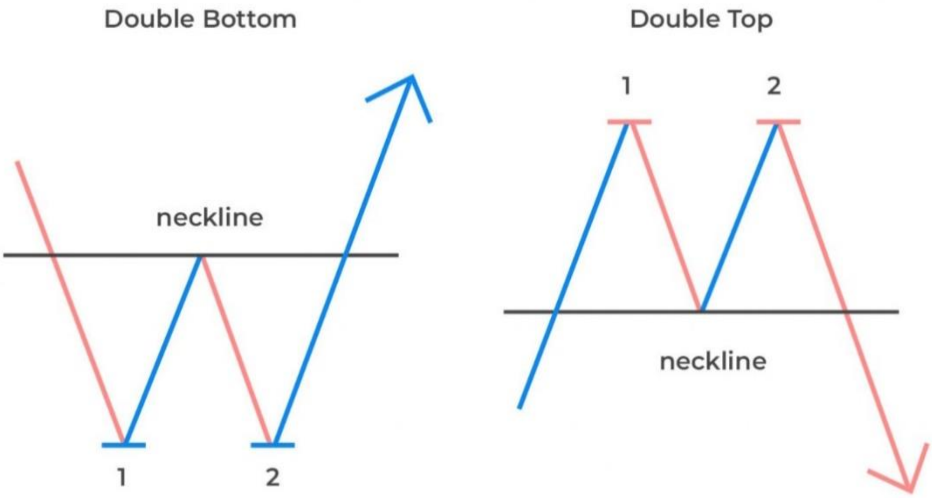
Related to the previous concept is that of the **Change in Polarity**. This is the principle that when a Support or Resistance level is breached, it then becomes its opposite. This can occur when a Resistance level is exceeded, and that price point becomes a new Support for trading activity at the new higher level.

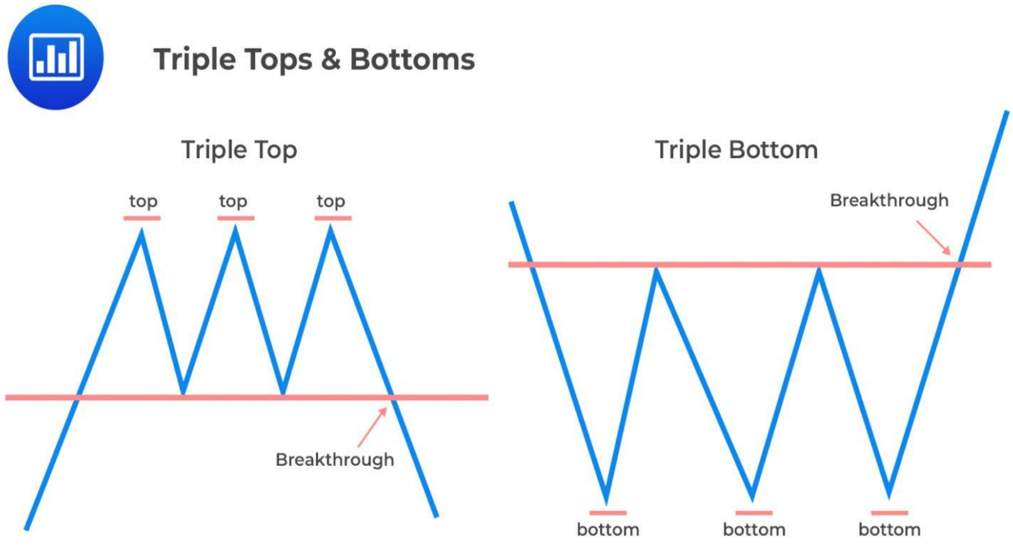
In the field of technical analysis, there are a number of patterns that appear on stock charts reflecting price movements that can be used when trying to forecast future movements in stock price. These chart patterns fall into a few categories. **Reversal Patterns** indicate that a trend is likely to reverse from the direction it has been going. A common example is the "Head and Shoulders" pattern that usually signals the end of an uptrend. Other examples of reversal patterns include inverse head and shoulders, double tops and bottoms, and triple tops and bottoms.

## Head and Shoulder Bottom

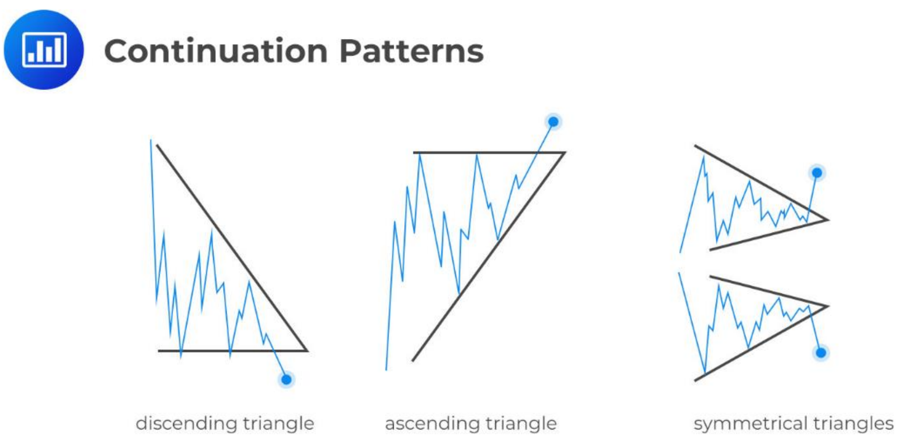


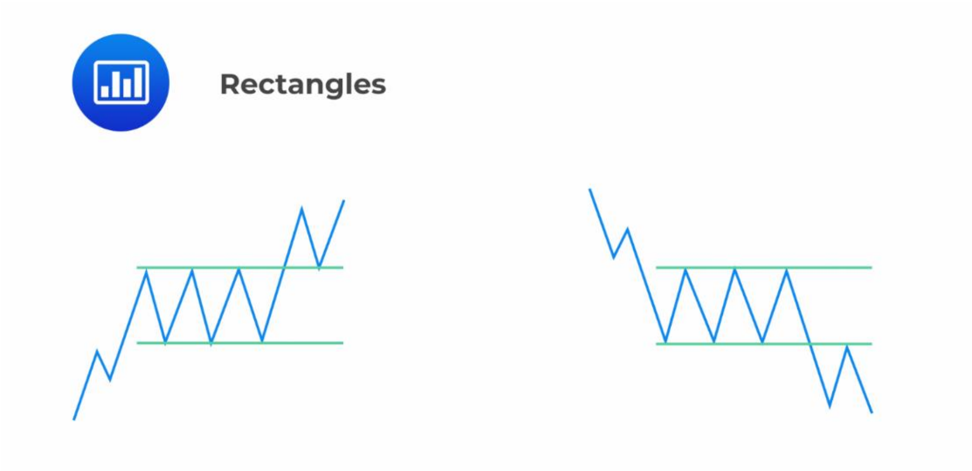

## Double Tops and Bottoms

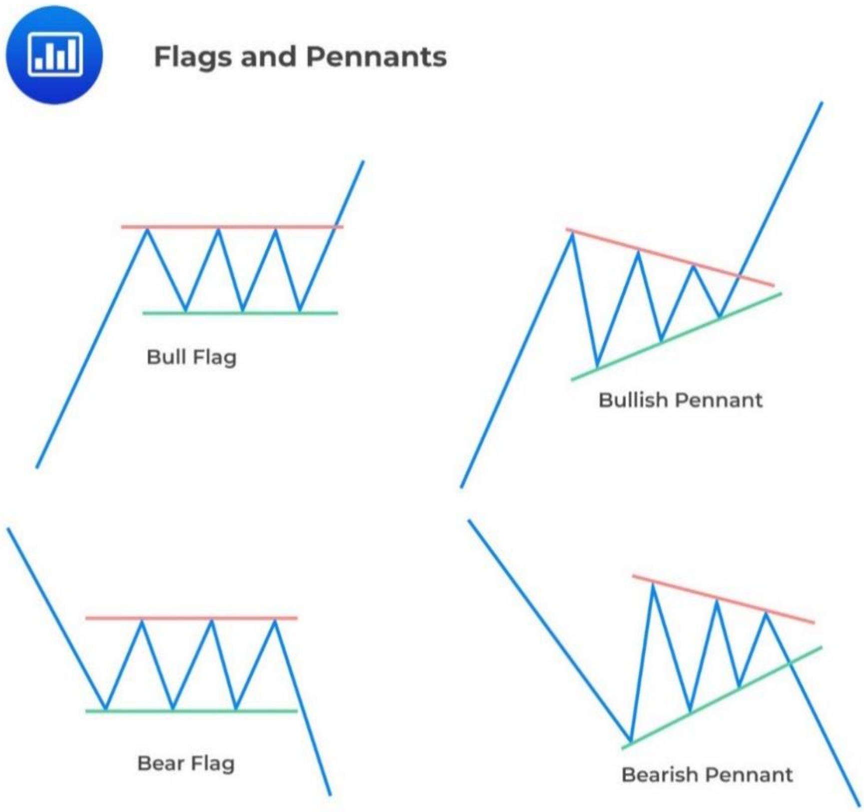





Another category of chart pattern is the **Continuation Pattern**. These patterns are used to identify trends that will continue after a pause. Many common continuation patterns are based on triangles that highlight trading movement that is converging to a given point in the future. Other continuation patterns include rectangles, flags and pennants.






In addition to chart patterns, there are also indicators used in technical analysis that are used to predict future changes in a security's price. The main categories of indicators are:

### 1. Price-Based Indicators

These indicators use information about the security's past and current price to try and predict future prices. A simple example of this is the moving average, which is the average price of a security over a number of previous time periods. This average removes the oldest observation every time a new observation is added. Bollinger Bands are another common indicator based on price. These are lines that indicate the standard deviations of prices compared to a moving average of the stock's price.

## 2. Momentum Oscillators

These are values calculated by finding the difference between the most recent price of a security and a previous price on a specified day in the past. Watching for the calculated value to switch from positive to negative is one indicator of a potential trend reversal.

$$M = (V - V_x) * 100$$

$V = Most\ recent\ closing\ price$

$V_x = Closing\ price\ at\ specified\ date\ in\ past$

Oscillators can be used to:

- Determine the strength of a trend.
- Signal an upcoming trend reversal.
- Determine short-term trading decisions.

## 3. Sentiment Indicators

These are various polls that are conducted to gauge the sentiment of individual investors or investment professionals about the state of the equity market. There are two types of sentiment indicators:

i. Investor polls. These are periodic polls conducted with the aim of gauging the sentiments of individual investors or investment professionals about the equity market.

ii. Calculated statistical indexes. These are sentiment indicators obtained from market data such as price. The commonly used ones are pull/call ratio and volatility index.

# Reading 68 – Fintech in Investment Management

The word "**Fintech**" is simply a combination of the words "financial" and "technology." Today, fintech encompasses more advanced systems that are able to analyze information and make decisions based on machine-learning logic. Machines that "learn" how to perform tasks over time have been developed. These systems can often surpass human capabilities.

Big data is a term used to refer to complex, extremely large data that may be analyzed computationally to reveal patterns, trends, and associations, especially those leading to human behavior. It encompasses both **Traditional Data Sources** such as company reports, stock exchange sources, and data gathered from governments as well as **Nontraditional (Alternative) Data** from social media, sensor networks, and electronic devices. Data can be **Structured Data** (organized in tables and are commonly stored in a database), **Unstructured Data** (cannot be presented in tabular forms, such as text messages, tweets, and emails) or **Semi-structured Data** (a mix of both).

**Artificial Intelligence** (AI) has much to do with the development of computer systems that exhibit cognitive and decision-making abilities comparable or superior to that of humans. **Machine Learning** is a current application of AI which revolves around the idea that we should really just give machines access to data and let them learn by themselves without further human intervention. Machine learning can take the form of **Supervised Learning** (supervised by humans) or **Unsupervised Learning**, where computers are only given input data and are tasked with describing the data, for instance by grouping or clustering of data points.

The ability to create computer programs that can learn on their own and improve over time creates new opportunities for investment professionals across the board, which include the **Analysis of Loads of Data** through complex algorithms developed to scour social media and sensor networks in search of consumer sentiments and product performance data that can be integrated into a manager's buy/sell decisions, **Automated Trading** to identify systematic investment strategies and automatically execute multiple trades over several financial markets worldwide, **Analytical Tools** to identify systematic investment strategies and automatically execute multiple trades over several financial markets worldwide, and **Automated Advice** through robo-advisors.

Another application of fintech is the **Distributed Ledger Technology** (DLT). A distributed ledger is a database held and updated independently by each participant (or node) in a large network. Rather than have a central authority, records are independently constructed and held by every node (computer). DLT has several features that make it a favorite among investment managers. **Cryptography** refers to algorithmic encryption of data such that it is unusable in the hands of an unauthorized party. As a result, DLT has a high level of security and integrity. **Smart Contracts** are computer programs that self-execute on the basis of pre-specified terms and conditions. If a counterparty defaults on a payment, collateral can be transferred to the relevant party instantaneously. **Blockchain** is a type of digital ledger in which information, such as changes in ownership of an asset, is recorded sequentially within blocks that are then linked together and secured using cryptography.

The most common use of the distributed ledger technology (DLT) is Bitcoin, a **Cryptocurrency** which enables payments to be sent between users without passing through a central authority. Other uses include **Tokenization** (the process of converting rights to an asset), **Post-trade Clearing and Settlement** (near-real-time trade verification, reconciliation, and settlement) and **Seamless Compliance**, which would eliminate the need for large post-trade monitoring teams and create operational efficiency.

# Ethical and Professional Standards

Ethical and Professional Standards is a major part of the CFA curriculum at all three levels, so it is very important to learn this material well. The CFA Institute places a high value on ethics, and even uses scores on this section to determine if someone passes or fails when their overall exam result is right at the edge of the Minimum Passing Score. Be ready to spend a good amount of time learning all the ethics materials.

The ethics questions on the exam will ask you to read through a situation and judge whether the actions described violate certain parts of the Code of Ethics or Standards of Professional Conduct. Since the questions only have 3 answer choices, you will only be expected to look for a small range of potential violations in an individual question.

## Reading 69: Ethics and Trust in the Investment Profession

Reading 56 is a high-level overview of the importance of ethics for investment professionals. Professions are occupational groups based on unique education, specialist knowledge, and framework of practice and behavior that establishes community trust, recognition, and respect. The primary takeaway is that ethical guidelines go beyond what is defined by legal statute. Ethics are defined as a standard of conduct that can be expressed by law and by more abstract principles that outline expected behaviors.

## Reading 70: Code of Ethics and Standards of Professional Conduct

The Code of Ethics is very straightforward. They are intended as general guidelines for appropriate behavior of an analyst. Don't expect too many tricky questions about these but become very familiar with the language of the Code.

## Code of Ethics

1. Act with integrity, competence, diligence, respect, and in an ethical manner with the public, clients, prospective clients, employers, employees, colleagues in the investment profession, and other participants in the global capital markets.

2. Place the integrity of the investment profession and the interests of clients above their own personal interests.

3. Use reasonable care and exercise independent professional judgment when conducting investment analysis, making investment recommendations, taking investment actions, and engaging in other professional activities.

4. Practice and encourage others to practice in a professional and ethical manner that will reflect credit on themselves and the profession.

5. Promote the integrity of, and uphold the rules governing, capital market.

6. Maintain and improve their professional competence and strive to maintain and improve the competence of other investment professionals.

The Standards of Professional Conduct are outlined in detail in the following reading.

# Reading 72: Guidance for Standards I-VII

The Standards of Professional Conduct are the heart of the ethics content on CFA exams. Most of the questions you see will involve the application of one or more parts of the standards applied to various real-world scenarios that a professional analyst might face.

## Standard I – Professionalism

Standard I makes it clear high ethical standards must apply even when an issue hasn't been identified in writing. It specifies that investment professionals must have a working knowledge of laws, as well as a framework for resolving ethical dilemmas.

### Standard I(A) – Knowledge of the Law

Knowledge of the law specifies that investment professionals must have a working knowledge of all applicable laws and regulations that apply where they do business. In situations where they are subject to different requirements, they are always to adhere to the strictest ones. This standard applies to rules at all levels, including national laws, local regulations, and even the CFA Code and Standards.

### Standard I(B) – Independence and Objectivity

Independence and Objectivity specifies the importance of avoiding conflicts of interest that may arise from accepting gifts, payments, or other favors from clients or business partners. It's very important for an analyst to avoid even the appearance of bias. Special note is given to travel considerations. Although a firm may view the offer to fly its investment professionals on a third-party's chartered jet as a cost-savings measure for the firm, the opportunity for conflict of interest arises.

### Standard I(C) – Misrepresentation

These standard states that CFA candidates and charterholders must not knowingly misrepresent information related to investment analysis, recommendations, or professional actions. A common example of this is guaranteeing investment results. You cannot use information in a misleading way to influence investment decisions and you absolutely cannot promise specific investment returns or superior results in any form of communication.

### Standard I(D) – Misconduct

Misconduct primarily addresses issues with honesty and professional behavior. It is more broad than Standard I(A) Knowledge of the Law because it is concerned with any behavior that could negatively impact professional integrity. This applies to activities as varied as refraining from drinking during business hours to conducting proper due diligence before making investment recommendations. Compliance with this standard means avoiding behaviors that could even affect the perception of one's integrity.

## Standard II – Integrity of Capital Markets

Standard II prohibits any activities used to manipulate investment markets. Maintaining public trust in financial markets is of paramount concern to the CFA Institute, and so this standard is very strict on actions that could distort markets through fraudulent or deceitful behavior.

### Standard II(A) – Material Non-Public Information

Standard II(A) prohibits members from acting on, or causing others to act on, information that is not available to the public and could impact market results. Doing so is a form of market manipulation, as it is providing an unfair trading advantage to some market participants over others.

### Standard II(B) – Market Manipulation

Market manipulation refers to any activity that fraudulently alters market information for the purpose of influencing investment results. Two common types of manipulation are information-based and transaction-based. Information refers to the distribution of false or misleading materials in order to manipulate the behavior of other investors. Transaction-based manipulations involve creating false impressions of market activity through misleading or fraudulent trading strategies.

## Standard III – Duties to Clients and Prospective Clients

Standard III emphasizes the importance of investment professionals placing the interests of their clients before their own or that of their firms.

### Standard III(A) – Loyalty, Prudence, and Care

This standard broadly requires CFA members to treat their clients and beneficiaries with the utmost care and concern towards meeting their financial goals. Members must practice due diligence and exercise caution when handling clients' finances.

### Standard III(B) – Fair Dealing

Fair Dealing means that members must treat all clients and partners fairly so that they are not disadvantaging some people over others. A member must not take actions that provide unfair advantages or opportunities only to certain clients.

### Standard III(C) – Suitability

This standard refers to basing investment actions and recommendations on each client's specific needs and limitations. Each client is going to be unique in terms of their abilities and willingness in regards to investment risk, and the CFA member is responsible for adjusting their strategy to best fit that client's particular situation.

### Standard III(D) – Performance Presentation

This requires members to ensure that all communication about investment performance is fair, accurate, and complete. All disclosures related to tax, fees, and prior performance should be included in the presentation of recommendations.

### Standard III(E) – Preservation of Confidentiality

Standard III(E) requires CFA members to maintain the confidentiality of current, former, and prospective clients under most circumstances. Exceptions are cases in which sharing of personal information reveals client activity of an illegal nature.

## Standard IV – Duties to Employers

Standard IV outlines basic responsibilities by investment professionals for their employers. While the loyalty to clients must be the paramount concern, a member must still also place their firm's needs ahead of their own.

### Standard IV(A) – Loyalty

As stated in Standard III(A) – Loyalty, Prudence, and Care, members must always be loyal to clients first. However, Standard IV(A) – Loyalty reinforces the fact that members have a responsibility to act in a way that sustains the integrity of their firm. CFA members must refrain from independent activity that conflicts with the best interest of their employer and must be careful not to alert current clients of separation from the employer prior to the departure or taking any documents from the employer.

### Standard IV(B) – Additional Compensation Arrangements

Additional Compensation Arrangements requires members to obtain written permission prior to accepting gifts or additional compensation by third parties.

### Standard IV(C) – Responsibilities of Supervisors

While supervisors are not completely responsible for all actions of their subordinates, this standard makes it clear that supervisors must make every effort to ensure that systems are in place to encourage compliance with all legal and ethical standards of the profession. These systems include proper incentive structures, training on ethical protocol, and the firm's policies and procedures.

## Standard V – Investment Analysis, Recommendations, and Action

Standard V outlines the responsibility of investment professionals regarding the performance of due diligence prior to making recommendations to clients.

### Standard V(A) – Diligence and Reasonable Basis

This standard requires that all investment recommendations be based on independent, verifiable information. An analyst should utilize multiple information sources and be thorough in their research before making recommendations to clients.

### Standard V(B) – Communication with Clients and Prospective Clients

Members must disclose limitations and risks associated with investment recommendations and activities. The most important part of this standard is that research based on past circumstances can be verified as fact; however, projections of future performances are the basis of opinions and should be stated in any form of communication to clients as "In my opinion, …"

### Standard V(C) – Record Retention

This standard requires members to maintain a system for maintaining records of analysis, recommendations, and investment actions that they make. The CFA Institute recommends retaining 7 years of records, but members should comply with any applicable legal requirements that require longer time periods.

## Standard VI – Conflicts of Interest

Standard VI outlines how members should mitigate the influence of conflicts of interest and how to disclose these potential risks to clients. It would be nearly impossible to avoid any conflicts of interest, so the standard focuses on how to manage that risk and avoid harm to clients through disclosure and controls.

### Standard VI(A) – Disclosure of Conflicts

Dictates that CFA members must disclose any and all circumstances which might result in a conflict of independence and objectivity or interfere with loyalty to clients or employer.

### Standard VI(B) – Priority of Transactions

This standard specifies that investment transactions for clients or employers must take precedence over investment transactions for a CFA member. Priority of Transactions clearly indicates that the order of transaction priority is (1) clients, (2) employer, and (3) member. Firms and members must also limit personal investment in IPOs and establish black-out periods.

### Standard VI(C) – Referral Fees

This requires CFA members to report to employers and clients any sums received from or paid to recommendations of products or services. This should be done in **writing**, with **both parties** signing a written agreement.

## Standard VII – Responsibilities as a CFA Institute Member or CFA Candidate

Standard VII explains the responsibilities that CFA members or Candidates should have regarding their profession and the CFA Institute.

### Standard VII(A) – Conduct as Participants in CFA Institute Programs

Requires that CFA candidates and charterholders must not engage in any behavior that could harm the integrity or reputation of the CFA Institute or its programs.

### Standard VII(B) – Reference to CFA Institute, the CFA Designation, and the CFA Program

Members must follow criteria for designating their membership or involvement in CFA programs. It prohibits the misleading use or reference to the CFA charterholder or candidate status in order to gain personal or professional favors.

# Reading 72: Introduction to the Global Investment Performance Standards (GIPS)

The **Global Investment Performance Standards** was created to establish a standardized set of ethical practices that guide practitioners in analyzing and presenting historical data as a basis for comparison of investment results. Performance reporting is an important concern to the CFA Institute, so GIPS is designed to ensure that firms are not reporting their returns in misleading ways to garner business. GIPS primarily benefits investment firms and the clients of those firms. Through compliance with GIPS, it is assured that a firm's historical results are presented consistently and fairly.

GIPS compliance is done at the firm level, and there is no partial compliance with the requirements. If a firm wants to claim GIPS compliance, they must meet every part of the standards. This includes the returns for any subadvisors the fund may utilize. Firms also cannot claim compliance for only certain portfolios or composites they manage. GIPS-compliance is voluntary and self-regulated. Due to the nature of this setup, a strong commitment to ethical integrity is required to maintain GIPS-compliance.

The GIPS Standards help asset owners, asset managers, and regulators by ensuring that firms' reports are full and fairly presented, giving investors more trust in performance presentations, and allowing regulators to better assess the performance of the firms they regulate.

A **Composite** is a collection of one or more portfolios that are all managed with the same investing mandate, purpose, or strategy. All actual, fee-paying discretionary portfolios managed in accordance with the same investing mandate, purpose, or strategy must be included in a composite.

After a firm reports compliance with the standards, it can voluntarily hire an **Independent Third Party** to verify it. It is impossible for a company to verify itself.

Non-discretionary portfolios must not be included in a firm's composites. A discretionary portfolio is one in which buy and sell decisions are made by a portfolio manager.

# Reading 73: Ethics Application

## Standard I – Professionalism

### Standard I(A) – Knowledge of the Law

Tom Peters, CFA is from Peru, a country with less strict laws than the CFA code and standards. He has just set up an investment firm in Luna. Luna's laws currently state that the client's home country laws apply. Adam Luis is one of Tom Peter's clients. Adam is from a country with minimal investment laws and no restrictions on insider trading. Which laws should Tom Peters *most likely* abide by?

A. Peru's laws
B. Luna's laws
C. CFA code and standards

**C is correct.** Tom Peters should abide by the CFA code and standards. Members and Candidates must follow the more stringent law, rule, or regulation in the event of a dispute.

**A is incorrect.** Peru's laws are less strict than the CFA code and standards. Members and candidates must refrain from engaging in activity that violates the Code and Standards, even if it is otherwise legal.

**B is incorrect.** Luna's laws are also less strict than the CFA code and standards, and therefore the CFA code and standards would apply. Even though Luna's laws state that the client's home country laws apply, by abiding by the CFA code and standards, Tom Peters would still be complying since the CFA code and standards are stricter.

### Standard I(B) – Independence and Objectivity

VVC cement Ltd is the largest publicly traded cement manufacturer in the country. VVC has taken several research analysts for a tour of its facilities and paid for their travel and accommodation expenses except one of the analysts, Martin Pablo, CFA, whose employer has a policy of paying for his travel and accommodation employees. Who among the analysts *most likely* violated standard I(B)?

A. None

B. Martin
C. The other analysts

**A is correct.** None of the analysts violated Standards I(B) of the CFA code and standards. Members and candidates must exercise their best judgment when permitting firms to pay for travel and/or accommodations. They must ensure such arrangements do not compromise or appear to compromise their independence and judgment. This particular trip was for business, and no lavish or irrelevant gifts were offered.

**B is incorrect.** Martin was not in violation of Standard I(B) by paying for his own travel expenses. He was simply following his employer's policies which are stricter than the CFA code and standards.

**C is incorrect.** The other analysts did not violate standard I(B) as the travel and accommodation expenses were for business, and no lavish gifts that would have compromised or appeared to compromise their independence and objectivity were taken.

## Standard II – Integrity of Capital Markets

### Standard II(A) – Material Nonpublic Information

John Adams is the CFO of a publicly traded bank. During a conference call with a few analysts, he mentions that his bank intends to acquire LLM Bank, one of the fastest growing in the neighboring country. Tom Michaels CFA, one of the analysts on the call, immediately places buy orders for the bank's shares. Under the CFA code and standards, Tome Michaels *most likely*?

A. violated the CFA standard II(A)
B. did not violate the CFA code and standards because the information is public
C. violated the CFA standard II(A) because he did not inform his employer before placing the orders

**A is correct.** Tom Michaels, CFA violated standard II(A) by trading on nonpublic information. A small group of analysts is not considered a public broadcast method therefore, he cannot use the information that the company has disclosed to them.

**B is incorrect.** The information disclosed to Tom Michaels, CFA is not public. A small group of analysts is not considered a public disclosure of information.

**C is incorrect.** Informing his employer before placing an order based on nonpublic information is still a violation of standard II(A).

### Standard II(B) – Market Manipulation

Philip Kinuthia, CFA, is a highly regarded independent research analyst, InstantPay Inc. has contacted him to promote their shares on his social media platforms and website in exchange for a cash compensation. This relationship is disclosed in his reports. Due to his huge audience, InstantPay Inc. share price increase drastically after Philip publishes several positive reports. Under the CFA code and standards, Philip

A. violated the CFA code and standards
B. did not violate the Code and Standards as no investors lost their investments
C. did not violate the CFA and Standard as he disclosed the relationship in his reports

**A is correct.** Phillip Kinuthia, CFA violated Standards II(B) by accepting a cash compensation in order to publish favorable reports on InstantPay Inc as this is not based on proper research.

**B is incorrect.** Whether investors lost their investments or not, Philips violated The CFA Code and Standards by accepting a cash compensation for issuing favorable reports. Accepting a compensation to offer a research report is acceptable, but a favorable research should not be guaranteed, the report should be independent, objective, and well researched.

**C is incorrect.** Disclosing the nature of the relationship is necessary, however, accepting compensation in order to positive reports is a violation.

## Standard III – Duties to Clients

### Standard III(A) – Loyalty, Prudence and Care

JLL Investment Ltd is an investment manager. Most of the trades that JLL performs are done by BTS Securities mainly because of the close relationship between the CEOs of the two firms. BTS Securities' commission structure is higher than other stockbrokers. However, they provide average research reports to JLL investments in exchange for this relationship. Under the CFA Code and Standards, JLL investments *most likely*,

A. violated the standards III(A)
B. did not violate the Code and Standards as JLL can choose who to use for trading purposes
C. did not violate the Code and Standards as they receive research reports for directing trades to BTS Securities

**A is correct.** JLL investments violated Standard III(A) by directing their trades to BTS Securities mainly because of the relationship between the firm's two CEOs. JLL should obtain the best price and best execution for their clients. JLL Investments are breaking Standard III(A) by failing to honor their duty of loyalty, Prudence, and Care.

**B is incorrect.** JLL Investments' choice of who to direct their trades should be guided on their clients' best price and execution. BTS Securities charges higher than average for the trades performed negatively impacting JLL investments' clients.

**C is incorrect.** The research provided by BTS Securities is average, and JLL Investments would incur more costs to obtain good research. Additionally, JLL Investments pays more for this research.

## Standard III(B) – Fair Dealing

Tom Hanks, CFA, is the investment manager at PKN Investments. BGT Bank's recent IPO was oversubscribed, resulting in investors receiving fewer shares than they had initially subscribed. PKN Investments allocated its share of BGT Bank's shares to its High-Net-Worth Investors without allocating the shares to its other clients. Under the CFA code and standards, Tom *most likely*:

A. violated the CFA Standard III(B)
B. did not violate the Code and Standards by giving their largest accounts BGT Bank's shares
C. did not violate the Code and Standards because did not keep the shares for themselves

**A is correct.** Tom Hanks, CFA violated the CFA Code and Standards by distributing their allocated shares to their high-net-worth- investors. These shares should have been prorated to all the clients that had bought.

**B is incorrect.** By allocating BGT Bank's shares to their high-net-worth investors, Tom Hanks discriminated against their other clients and violated Standard III(B).

**C is correct.** Anything done after allocating BGT Bank's shares to the high-net-worth investors does not matter, as the manner in which the allocation was done is a violation.

## Standard IV – Duties to Employers

### Standard IV(A) – Loyalty

John Adams, CFA just starting his new job as a portfolio manager at Heading Fund Manager. He is currently marketing to prospective clients. While going through his laptop, he comes across a list of top clients at his former place of work and decides to reach out to them. Under the CFA Codes and Standards, John Adams most likely:

A. violated the Code and Standards.
B. did not violate the Code and Standards because he did not intentionally take these clients contact information.
C. did not violate the Code and Standards because he had already left his previous employment.

**A is correct.** John Adams, CFA violated the CFA Code and Standards by using his previous employer's confidential material to contact prospective clients. All the material developed while working at his previous employer is the property of the previous employer and should not be used without their permission.

**B is incorrect.** Whether it was intentional or not John Adams, CFA should not use the property of his previous employer to solicit business from prospective clients without their consent.

**C is incorrect.** John Adams, CFA violated the Code and Standards by using his previous employers material without their consent. John Adams, CFA still owes loyalty to his former employee even though he does not work with them.

**Standard IV(B) – Additional Compensation Arrangements**

Stephen Lee, CFA is a research analyst at a leading investment bank. He has recently published a strong buy recommendation for Cobra Inc., a leading a shipping company after a thorough analysis of the company. After reading his report, Trevor Adam, Cobra Inc.'s CEO arranges to meet Stephen. Trevor offers to pay for Stephen's travel and accommodation expenses but after seeking his supervisor's approval to meet with Trevor, Stephen accepts to meet with Trevor but declines the travel and accommodation arrangement. After the meeting, Stephen provides his supervisor with a report of the meeting. Under the CFA Code and Standards, Stephen *most likely*:

A. did not violate the CFA Code and Standards.
B. violated Standard IV(B) by accepting to meet Trevor Adam.
C. violated the CFA Code and Standards by issuing a favorable report on Cobra Inc.

**A is correct.** Stephen did not violate the CFA Code and Standards by agreeing to meet Cobra Inc.'s CEO. Additionally, he sought approval and disclosed the discussions at the meeting to his supervisor. Declining the travel and accommodation arrangements offered by Trevor ensured there is no conflict or appearance of a conflict of interest.

**B is incorrect.** Meeting with Cobra Inc.'s CEO by itself is not a violation of the CFA Code and Standards. Stephen took the necessary steps to ensure there is no conflict of interest or the appearance of a conflict of interest.

**C is incorrect.** Stephen did not violate the CFA Code and Standards by issuing a favorable report on Cobra Inc. as he had done a thorough analysis of the company and his report was not externally influenced.

# Standard V – Investment Analysis, Recommendations, and Actions

## Standard V(A) – Diligence and Reasonable Basis

Abel Jones, CFA is the research director at Pharaoh Inc. an investment management firm that publishes share recommendations for the top twenty largest public companies trading on the stock market. Due to time constraints and limited manpower, Abel Jones uses the research report of a well-known research company to issue a recommendation in its monthly reports. Pharaoh Inc. buys these third-party research reports every month from the research company. Under the CFA Code and Standards, Abel Jones, *most likely*:

A. did not violate the CFA Code and Standards
B. violated the CFA Code and Standards by using third party research reports
C. violated the CFA Code and Standards by not making reasonable and diligence efforts on the third-party research report

**C is correct.** Abel Jones, CFA violated Standard V(A) by not making reasonable and diligence effort on the third-party research reports before using it in his publication. Using third-party reports is not a violation but one must ensure the report is objective and reasonably based.

**B is incorrect.** Abel Jones can use third-party research reports, but he must ensure that the research is objective and reasonably based.

**A is incorrect.** Abel Jones violated the CFA Code and Standards by using third-party research reports without performing proper diligence on it.

## Standard V(B) – Communication with Clients and Prospective Clients

Sofia Emma, CFA is a research analyst at Green Investment Bank covering oil drilling companies. RMK Drilling Inc. has just won an oil exploration license in the Southern African country of Persia. The area to be explored is similar in size to another that was discovered to have 200,000 barrels of oil. Using this information, Sofia concludes that RMK will extract 200,000 barrels and states in her report, “RMK Drilling Inc. recent license has given it access to 200,000 barrels of oil, and I recommend a strong buy”. Under the CFA Code and Standards, Sofia Emma most likely:

A. violated the CFA Standard V(B).
B. did not violate Standard V(B) by issuing a buy recommendation on estimated oil reserves.
C. did not violate Standard V(B) as she estimated RMK Drilling Inc.'s oil using a similar license

**A is correct.** Sofia violated the CFA Standard V(B), phrasing opinions as facts in her report. Her estimation of RMK Drilling Inc's oil reserves from a similar license awarded is an opinion and this should be distinguished from facts.

**B is incorrect.** Estimating the oil reserves awarded to RMK Drilling Inc. is not a violation of the CFA Code and Standards. Sofia can estimate RMK Drilling Inc's oil reserves and issue a recommendation based on that information, but she must distinguish between facts and opinions.

**C is incorrect.** Estimations are not a violation of the CFA Code and Standards, but facts must be clearly distinguished from opinions.

## Standard VI – Conflicts of Interest

### Standard VI(A) – Disclosure of Conflicts

Susan Shell, CFA is a research analyst at True Capital, who owns 1,000 shares of Unicorn Inc., a payment service provider. The director of research at True Capital has asked Susan to write a research report on Unicorn Inc. that will be published at the end of the quarter. To abide by the CFA Code and Standards, Susan Shell, CFA should *most likely*:

A. sell her shares at Unicorn Inc. before writing her report
B. disclose her shareholding to her employer and in her report
C. write and publish her report without disclosing her shareholding in Unicorn Inc.

**B is correct.** Susan Shell, CFA must disclose her shareholding in Unicorn Inc. to her employer, and if allowed to proceed with writing the report she must disclose this shareholding in her report in order to be in compliance with the CFA Code and Standards.

**A is incorrect.** Susan Shell, CFA does not have to sell her shares in Unicorn Inc. She should disclose her shareholding to her employer, and if her employer allows her to proceed with the report, she must disclose it in her report.

**C is incorrect.** Writing the report and publishing it without disclosing her shareholding and informing her employer prior to writing the report would be a violation of Standard VI(B).

### Standard VI(B) – Priority of Transactions

Michael Thomas, CFA is an investment manager at Thomson Capital. Thomas Capital bought shares in a recent IPO that was oversubscribed for their clients whom the investment was appropriate. Michael Thomas distributes the shares to his clients during share allocation and then allocates a disproportionately small amount of shares to brother's account. He does this to dispel the perception of favoring his brother. Under the CFA Code and Standards, Michael Thomas *most likely*:

A. violated Standard VI(B).
B. did not violate Standard VI(B) by not favoring his brother.
C. did not violate Standards VI(B) as no clients were discriminated against.

**A is correct.** Michael Thomas, CFA violated the CFA Code and Standards on priority of transactions. He violated his duty to his brother by treating them differently than the other clients. Michael's brother is entitled to the same level of service as other clients. However, if Michael has beneficial ownership in his brother's account, transactions may need to be cleared.

**B is incorrect.** Michael Thomas, CFA has violated his duty to his brother by not treating his account in a similar manner as the other accounts.

**C is incorrect.** Michael Thomas, CFA discriminated against his brother by allocating a disproportionately small number of shares of the IPO compared to other accounts.

## Standard VII – Responsibilities as a CFA Institute Member of CFA Candidate

### Standard VII(A) – Conduct as Participants in CFA Programs

John Phillips is a CFA level II candidate who just sat his level II exam. He is a member of an online CFA Candidates Forum. After finishing his exam, he enters the online forum to see other candidates' views on the exam. During the discussion, John mentioned that he found residual income model particularly difficult. Under the CFA Code and Standards, John Phillips *most likely*:

A. violated the CFA Code and Standards.
B. did not violate the CFA Code and Standards as he was expressing his opinion.
C. did not violate the CFA Code and Standard because he did not divulge specific details.

**A is correct.** John Phillips violated the CFA Code and Standards by discussing specific exam questions on the online forum. As per the Standard VII(A), candidates should not disclose specific details of questions or broad topical areas and formulas tested or not tested on the exam.

**B is incorrect.** John Phillips went beyond expressing his opinion by divulging specific details about topics tested in the exam. Candidates and members are allowed to express opinions on policies and procedures but prohibited from disclosing content specific information including actual exam questions.

**C is incorrect.** John Phillips disclosed specific content tested in the exam by stating that he found the residual income model particularly difficult and therefore violated Standard VII(A).

**Standard VII(B) - Reference to CFA Institute, the CFA Designation, and the CFA program**

Taya Jones, CFA is the chief investment officer at Grey Capital. On the company's website and marketing material, she states that she completed the CFA exams first attempt, and that the completion of the CFA Program has enhanced her portfolio management skills. Under the CFA Code and Standards, Taya Jones *most likely*:

A. did not violate the CFA Code and Standards.
B. violated the CFA Code and Standards by stating that she completed the CFA program on her first attempt.
C. violated the CFA Code and Standards by claiming the CFA program enhanced her portfolio management skills.

**A is correct.** Taya Jones did not violate the CFA Code and Standards by stating that she completed the program on her first attempt as these are facts. Additionally, her claim that the program enhanced her portfolio management skills is also acceptable. It becomes a violation when candidates claim to achieve better performance results because of the CFA Program.

**B is incorrect.** Stating that you completed the CFA program on your first attempt is not a violation of the CFA Code and Standards as this is a fact.

**C is incorrect.** Claiming to have gained portfolio management skills from the CFA program is not a violation. However, claiming to achieve better performance results from the CFA program is a violation.